

LOCAL EXCLUSION OF RESIDUAL TYPE I NAVIER–STOKES SEREGIN LIMITS

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ABSTRACT. This paper gives the refined residual analysis for Type I Navier–Stokes Seregin limits. A finite-energy local pointwise Type I singularity for a suitable weak solution produces, through the endpoint weak Serrin local Type I bounds, terminal no-escape, and the Seregin blow-up procedure, a nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation carrying positive local velocity L^3 -mass. After the previously treated small, stationary L^3 , uniformly L^3 -tight, and closed structured/decay alternatives have been removed, the remaining residual class is divided into four analytic classes: axisymmetric bounded-circulation profiles, rotational relative equilibria, degenerate stationary-hull profiles with retained local concentration, and the generic terminal residual. The terminal case is then refined into the affine/parasitic quotient stratum, log-diffuse and Young critical-tail alternatives, coherent homogeneous/log-periodic/aperiodic critical tails, and the final generic residual class.

The first class is excluded by a centered angular-circulation equation and an axis-absorption Liouville theorem. The second class is closed by a local Coriolis-flux identity and a local concentration decomposition: a superthreshold rotational flux gives a nontrivial recentered local profile, while a subthreshold flux remains inside the compact local limiting equation. The third class is reduced, by scale reselection for time-translation hulls, to a stationary profile in a degenerate stationary family which occurs as a Seregin limit and still carries local CKN concentration; it is then excluded by the terminal local concentration theorem. The final class is reduced by covariant tail normalization, concentration-set extraction, diffuse-defect compactness, critical-tail compactification, recurrence for chains of concentration profiles, exclusion of finite separated concentration profiles, indecomposable-profile sequence- L^3 completeness, parasitic-free mildness inheritance, and the Albritton–Barker endpoint Liouville theorem. The paper distinguishes statements inherited from the setup paper, local compactness estimates and reduction results proved here, and the minimal mesoscopic-scale reduction theorem which excludes the critical-tail boundary defects. The affine stratum and the coherent critical-tail alternatives are closed here; the log-diffuse and Young alternatives are both reduced to the same minimal mesoscopic-scale theorem, which is then proved by pressure compactness and a bounded ancient variance Liouville argument. No global Coriolis–Leray Liouville theorem for bounded profiles or global critical-norm estimate is used in the residual closure.

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1. INTRODUCTION AND MAIN THEOREM

The incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$ are written as

$$(1.1) \quad \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0.$$

The residual analysis in this paper begins after the standard Type I blow-up construction of Seregin has been performed [8]. Thus the basic solution is a smooth bounded ancient solution of the centered equation

$$(1.2) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0,$$

on $\mathbb{R}_y^3 \times \mathbb{R}_\tau$. The equation is obtained from the physical ancient representative (U, Q) by

$$(1.3) \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t}U(x, t), \quad P(y, \tau) = (-t)Q(x, t),$$

where $t < 0$ and pressures are always understood modulo addition of a function of time. Conversely,

$$(1.4) \quad U(x, t) = (-t)^{-1/2}V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0.$$

The purpose of this paper is to present the refined residual decomposition as a single standard PDE argument, using only the reductions already established in the setup paper. In this two-paper presentation, the current paper depends only on the setup paper `proof_setup.tex` and on cited peer-reviewed Navier–Stokes regularity and Liouville literature [4, 8]. The rotational-flux estimates, terminal local concentration decompositions, exclusions of local compactness failure

and diffuse exterior concentration, and finite separated-profile exclusions are proved in this paper from local estimates rather than assumed as global estimates.

This paper does not reconstruct the Seregin extraction or the endpoint mildness theory. Those are imported from the setup paper. The sole new task is to prove that the residual alternative left open there is empty, namely

$$\text{NoResidual} : \quad \mathcal{R}^\#(\mathcal{S}; I, J) = \emptyset.$$

Definition 1.1 (Type I convention used in the chain). In this paper and in the final two-paper chain, a terminal singular point $z_* = (x_*, T)$ is called *local pointwise Type I* if there are $\rho > 0$ and $M < \infty$ such that

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

This is the local pointwise Type I hypothesis used in the setup paper and in [Section 1.2](#). For suitable weak solutions this envelope lies in the endpoint weak Serrin class $L_t^{2,\infty} L_x^\infty$. The Albritton–Barker weak-Serrin local Type I estimate, recorded below as [Theorem 1.23](#), upgrades it on smaller cylinders to the scale-invariant local energy bounds needed for terminal velocity no-escape. Thus the terminal no-escape condition used by the setup paper is automatic for the local pointwise Type I singularities considered here. The paper does not use the phrase “Type I” to mean an unrelated global L^∞ , local CKN, weak- L^3 , or Serrin scale condition unless that condition is explicitly stated.

Remark 1.2 (Translation table for standard Type I hypotheses). The global pointwise condition

$$\sup_{t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty$$

implies [Theorem 1.1](#) at every terminal point. Any stronger local criterion that yields the displayed L^∞ envelope on a neighborhood of x_* enters the same argument, provided the solution is suitable in the corresponding cylinder. The endpoint weak Serrin estimate then gives the local scale-invariant energy bounds and terminal no-escape. Purely CKN-type, weak- L^3 , or other critical criteria are not silently identified with this hypothesis; they require their own implication into the local pointwise envelope, or directly into the local Type I estimates, before the present exclusion applies.

1.1. Imported setup results and remaining hypothesis.

Theorem 1.3 (Imported setup theorem). *The setup paper `proof_setup.tex` proves the following facts.*

- (1) *Positive terminal local concentration at every local pointwise Type I singular point.*
`p0:thm:singular-positive`
`p0:prop:concentration-package`
- (2) *Local pointwise Type I control gives an admissible Seregin extraction sequence with pressure gauges, no-escape, and retained unit-scale concentration.*
`p0:thm:local-weak-serrin-typeI`
`p0:cor:paper0-local-typeI-admissible`
`p0:lem:local-seregin-extraction`
- (3) *Every normalized Seregin limit produced by this extraction is a bounded ancient suitable weak solution, admits the pressure atlas of the setup paper, and has the mild ancient representative required by the endpoint Liouville theorem.*
`p1:thm:extraction`
`p1:prop:ancient-limit`
`p1:prop:nonzero`
`p1:lem:pressure`

- p1:lem:gauge-compatibility
- p1:lem:pressure-atlas
- p1:lem:mild-representation
- p2:lem:mild-limit
- p2:lem:physical-mild
- (4) *The previously classified small, stationary L^3 , uniformly tight, and known structured/decay alternatives are empty.*
 - p1:thm:classical-classes
 - p1:thm:known-structure-decay-liouville
 - p2:thm:no-tight-normalized-collection
 - p2:thm:tight-liouville
- (5) *The endpoint and mildness results include the Albritton–Barker input and the setup paper proves local pointwise Type I exclusion conditionally on the residual-class hypothesis.*
 - p2:thm:AB-endpoint
 - p1:hyp:no-remainder
 - p1:thm:final-assembly

The purpose of the present paper is exactly to prove **NoResidual**. This verifies the setup residual hypothesis

p1:hyp:no-remainder.

Remark 1.4 (Use of imported labels). Below, the phrase “by the imported setup theorem” means that the terminal profile is bounded, locally suitable, pressure-normalized by the setup pressure atlas, parasitic-free modulo the affine class, and mild after finite terminal shift, with the exact imported labels listed in [Theorem 1.3](#). Paper IV proves only the residual closure **NoResidual**.

- (1) Local concentration and terminal no-escape entry: positive local concentration depends on CKN theory and the setup paper; terminal no-escape follows from the endpoint weak Serrin local Type I bounds in [Theorems 1.23](#) and [1.24](#).
- (2) Seregin extraction and parasitic-free normalization: follows from the setup paper and the mildness and compactness arguments in this paper.
- (3) Small and stationary L^3 strata: handled in the setup paper.
- (4) Uniformly tight stratum: handled in the setup paper.
- (5) Closed structured/decay strata: handled in the setup paper.
- (6) Axisymmetric and rotational residual strata: proved in this paper.
- (7) Terminal concentration-set decomposition, recurrence for concentration chains, and diffuse-defect compactness: proved in this paper.
- (8) Critical-tail compactification and realized critical-tail decomposition: proved in this paper.
- (9) Residual indecomposable-profile closure: proved in this paper, with the final sequence- L^3 endpoint supplied by Albritton–Barker.

Theorem 1.5 (Admissible Type I Seregin extraction theorem). *An admissible finite-energy Type I singular point produces a normalized Seregin ancient limit (V, P) satisfying (1.2), with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M < \infty,$$

local suitability and local pressure-gauge compactness, and a compact parabolic cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ for which

$$(1.5) \quad \iint_K |V|^3 dy d\tau \geq \eta_0 > 0.$$

The collection of all normalized Seregin limits generated by the original solution is denoted by $\mathcal{S}$.

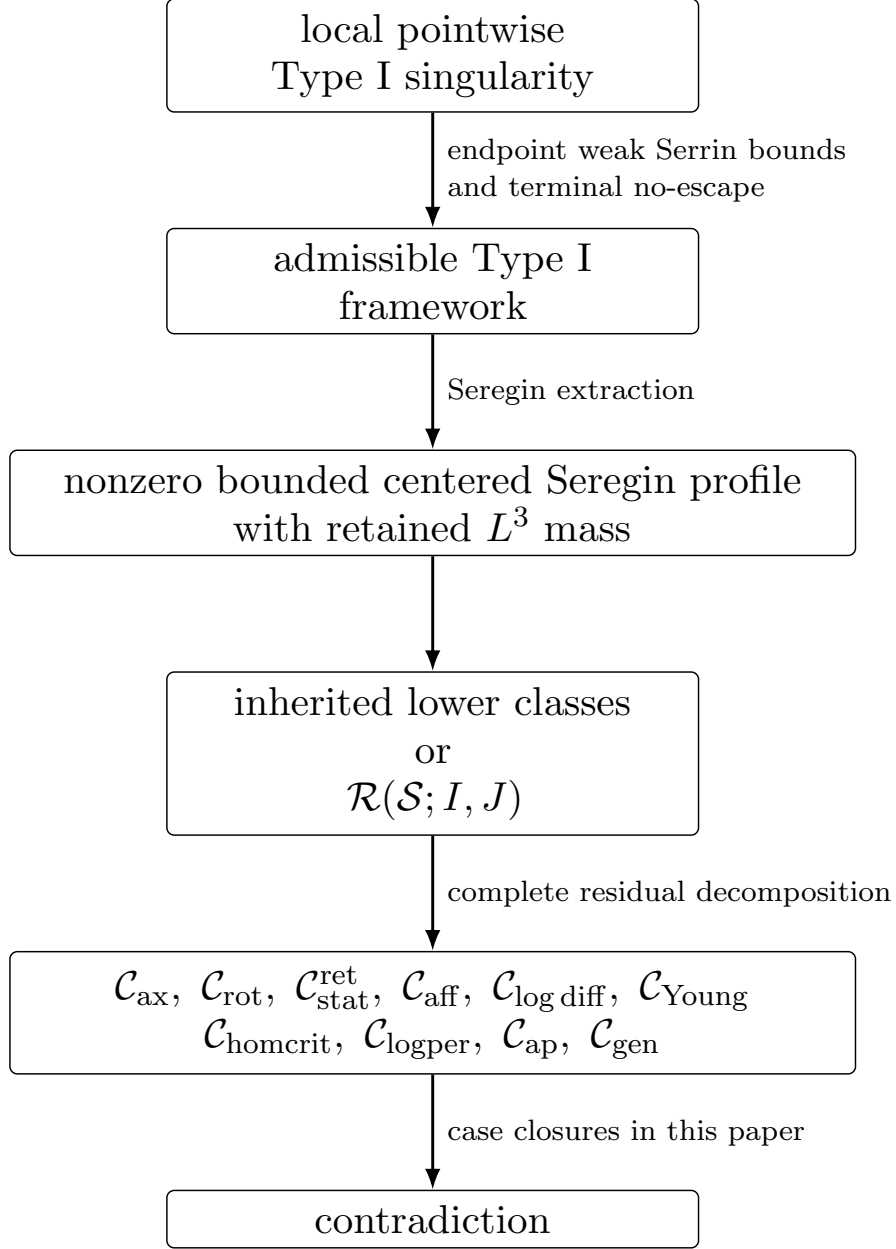


FIGURE 1. Decision diagram for the Type I profile decomposition. The first arrow uses the local pointwise Type I to terminal no-escape bridge supplied by the endpoint weak Serrin local Type I estimate. The inherited lower classes in the split node are already closed in the setup paper. The residual alternatives in the boxed list are excluded by the case-exclusion theorems cited in the proof of [Theorem 1.13](#).

Proof. This is the base admissible Seregin extraction theorem from `proof_setup.tex`, including velocity nonvanishing, local suitability, and local pressure compactness. The underlying positive local concentration input is also provided there. The present paper uses this as an inherited reduction and does not reprove the Type I blow-up compactness theorem. $\square$

Definition 1.6 (Earlier alternatives inherited from the setup paper). Within a fixed normalized Seregin collection $\mathcal{S}$, the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes are the non-residual alternatives already established in the setup paper. The small-amplitude and stationary L^3 alternatives are closed there; the uniformly tight class is excluded there; and the closed structured/decay class is precisely the part proved empty there. Any structured profile not belonging to that closed class is residual by definition and is handled by the present paper. The present paper does not reprove those earlier alternatives and does not add any hypothesis to their statements.

Definition 1.7 (Coarse residual class). Let I, J be the index sets used in the base Type I residual criterion. The pre-quotient coarse residual class

$$\mathcal{R}_{\text{pre}}(\mathcal{S}; I, J)$$

is the complement in $\mathcal{S}$ of the already removed alternatives in [Theorem 1.6](#). The retained normalized residual class

$$\mathcal{R}^\#(\mathcal{S}; I, J)$$

is obtained from $\mathcal{R}_{\text{pre}}(\mathcal{S}; I, J)$ by passing to the canonical affine/parasitic representative and by removing any representative assigned to the affine/parasitic lower stratum. Unless the word unreduced is used explicitly, $\mathcal{R}(\mathcal{S}; I, J)$ denotes this retained normalized residual class. Thus an “empty affine/parasitic stratum” below always means that no retained normalized residual representative remains in that quotient stratum.

Definition 1.8 (Refined residual decomposition). The refined residual decomposition is an ordered definition, not a separate compactness theorem. Starting with the pre-quotient coarse residual class $\mathcal{R}_{\text{pre}}(\mathcal{S}; I, J)$, one removes the following alternatives in order; the affine/parasitic alternative is then recorded as a quotient stratum with no retained normalized representative in $\mathcal{R}^\#(\mathcal{S}; I, J)$. After the first nine alternatives have been removed, whatever remains is by definition the generic terminal residual $\mathcal{C}_{\text{gen}}$:

$$\begin{array}{ccccccc} \mathcal{C}_{\text{ax}}, & \mathcal{C}_{\text{rot}}, & \mathcal{C}_{\text{stat}}^{\text{ret}}, & \mathcal{C}_{\text{aff}}, & & & \\ \mathcal{C}_{\text{log diff}}, & \mathcal{C}_{\text{Young}}, & \mathcal{C}_{\text{homcrit}}, & \mathcal{C}_{\text{logper}}, & \mathcal{C}_{\text{ap}}, & \mathcal{C}_{\text{gen}}. \end{array}$$

Here $\mathcal{C}_{\text{ax}}$ is the axisymmetric bounded-circulation class, $\mathcal{C}_{\text{rot}}$ is the rotational relative-equilibrium class, $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is the degenerate stationary-hull class with retained local concentration, $\mathcal{C}_{\text{aff}}$ is the affine/parasitic lower stratum selected by affine-normalized oscillatory entry, $\mathcal{C}_{\text{log diff}}$ and $\mathcal{C}_{\text{Young}}$ are respectively the log-diffuse and Young critical-tail alternatives, $\mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}$ are the coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives, and $\mathcal{C}_{\text{gen}}$ is the generic terminal residual class. The inclusion below is part of this ordered definition, written for the retained normalized residual class; the appearance of $\mathcal{C}_{\text{aff}}$ records the quotient alternative before retained normalization, and $\mathcal{C}_{\text{aff}} \cap \mathcal{R}^\#(\mathcal{S}; I, J) = \emptyset$ is proved in [Theorem 9.61](#). The substantive content is the closure of each non-quotient alternative, especially $\mathcal{C}_{\text{gen}}$. Equivalently,

$$(1.6) \quad \mathcal{R}(\mathcal{S}; I, J) \subset \mathcal{C}_{\text{ax}} \cup \mathcal{C}_{\text{rot}} \cup \mathcal{C}_{\text{stat}}^{\text{ret}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{log diff}} \cup \mathcal{C}_{\text{Young}} \cup \mathcal{C}_{\text{homcrit}} \cup \mathcal{C}_{\text{logper}} \cup \mathcal{C}_{\text{ap}} \cup \mathcal{C}_{\text{gen}}.$$

Definition 1.9 (Profiles realized from the original blow-up sequence). Fix a normalized Seregin collection $\mathcal{S}$ and a retained residual candidate $(V, P) \in \mathcal{R}(\mathcal{S}; I, J)$. An auxiliary profile or limiting measure used in the residual proof is *realized from the original blow-up sequence* if it is obtained from (V, P) , or from the terminal sequence which generated (V, P) , by a finite or diagonal sequence of the following operations:

- (i) exact centered symmetries T_a , Θ_s , and the covariant observer maps $\mathcal{R}_z$;
- (ii) unreduced parabolic recenterings before passing to centered variables, with the local pressure gauges supplied by Seregin compactness;

- (iii) retained tail limits or concentrating successors with $\mathcal{E}_1 \geq \varepsilon_*$ after the fixed admissible pressure gauges;
- (iv) normalized diffuse or critical-tail defect limits obtained by restricting the associated defect measure to a positive-measure window and renormalizing it;
- (v) minimal mesoscopic blow-downs selected from such retained or normalized defect windows.

When the shorter phrase “realized from the original sequence” is used below, it refers to this definition and to the original singularity-generated candidate (V, P) , not merely to the immediately preceding normalized profile.

Definition 1.10 (Retained descendants and normalized support profiles). *A retained descendant* is a descendant with a genuine inherited positive unit CKN concentration bound

$$\liminf_n \mathcal{E}_1(\cdot; 0) \geq \eta_* > 0.$$

A *normalized support profile* is a normalized profile in the support of a defect measure or normalized window measure associated with the original blow-up sequence. It may arise from unnormalized mass tending to zero and therefore need not carry the original retained threshold. Log-window and Young/variance normalizations are called retained only when the displayed unit CKN concentration bound is explicitly proved; otherwise they are normalized support profiles.

Table 1: Stability of the equation, gauges, and concentration bounds under limiting operations.

Operation	Equation	Boundedness	Pressure gauge	Mildness	Threshold
Centered time shift	yes	yes after finite shift	shifted gauge	setup mildness	yes
Spatial recentering	covariant	yes	translated gauge	yes	yes
Parabolic rescaling	yes	normalized scale	scaled gauge	yes	yes
Tail recentering	quotient covariant	locally yes	tail gauge	setup closure	support profile or retained
Log-window normalization	normalized defect equation	normalized	quotient gauge	not automatic	support profile
Affine pressure quotient	modulo $\mathcal{C}_{\text{aff}}$	yes	affine removed	yes outside $\mathcal{C}_{\text{aff}}$	specified
Critical-tail blow-down	limiting equation	locally yes	scaled/quotient gauge	after bounded-origin realization	support profile

Theorem 1.11 (Stability under realized limiting operations). *Every descendant, tail limit, concentrating successor, log-window normalization, critical-tail profile, and minimal-scale blow-down used in the proof of Theorem 1.13 is realized from the original blow-up sequence in the sense of Theorem 1.9. More precisely, each such limit satisfies one of the following alternatives.*

- (a) *It is a retained Seregin descendant of the original candidate, with the fixed descendant lower bound $\mathcal{E}_1 \geq \varepsilon_*$.*
- (b) *It is a normalized defect profile whose unnormalized defect measure has positive total mass on the selected window. If every normalized window descendant is excluded, then the original defect is excluded.*
- (c) *It lies in a previously closed lower stratum, and the retained descendant heredity theorem applies.*

- (d) *It generates a retained descendant or a normalized defect descendant satisfying (a), (b), or (c).*

Consequently a contradiction obtained for a realized limiting profile is a contradiction for the original retained residual candidate, not merely for an auxiliary compactification limit.

Proof. It suffices to verify the assertion for the operations listed in [Theorem 1.9](#). Exact centered symmetries preserve the centered equation, local suitability, pressure gauges, and retained local concentration; this is recorded in [Theorems 8.2](#) and [8.6](#). Unreduced terminal recenterings before centering are the recenterings in the Seregín extraction theorem of the setup paper and are compact after the gauges in [Theorem 1.18](#). Retained tail limits and concentrating successors are realized from the original sequence by diagonal lifting: [Theorems 10.10](#) and [10.11](#) show that the limit remains in the normalized compactness class and keeps the fixed retained CKN threshold, up to the once-for-all threshold loss in [Theorem 3.4](#).

Normalized defect profiles are obtained by restricting a positive defect measure to a window and dividing by its mass. The mass of the chosen window may depend on the index and may tend to zero; this does not detach the normalized profile from the original sequence. It records a point in the support of the defect compactification. The renormalized-window heredity lemma [Theorem 9.28](#) below states explicitly that if all such support profiles are assigned to closed lower alternatives, then the defect measure itself has no surviving mass outside those alternatives.

Minimal-scale blow-downs are selected from retained or normalized defect windows; their centers and scales therefore come with the same defect measure or retained CKN lower bound. The pressure/gauge normalization for those blow-downs is treated in [Theorems 9.42](#), [9.45](#) and [9.48](#). Thus every contradiction later in the proof is applied either to a retained descendant of the original profile or to a normalized support profile whose closure removes the corresponding defect. This proves the stated principle. $\square$

Lemma 1.12 (Transfer of contradictions to the original blow-up sequence). *Let $(V, P) \in \mathcal{R}(\mathcal{S}; I, J)$ be a retained residual candidate, and let $\mathcal{A}$ be a closed union of lower alternatives which is stable under the descendant operations used in this paper. Suppose that every limit realized from (V, P) by the relevant construction has one of the following two properties:*

- (i) *it is a retained Seregín descendant lying in $\mathcal{A}$;*
- (ii) *it is a normalized support profile of a defect measure associated with the original sequence and lies in $\mathcal{A}$.*

If $\mathcal{A}$ has already been excluded in the ordered residual decomposition, then the original candidate (V, P) cannot exist.

Proof. In case (i), retained descendant heredity [Theorem 10.11](#) identifies the descendant as an admissible successor of the original residual candidate with the retained threshold fixed in [Theorem 3.4](#). Since $\mathcal{A}$ is closed under the allowed limiting operations and is excluded, the retained descendant cannot occur.

In case (ii), the normalized support profile is obtained by restricting a positive defect measure to a window and renormalizing. If all such normalized support profiles lie in the closed excluded set $\mathcal{A}$, then regularity of the compactified defect measure implies that the defect measure has no mass outside $\mathcal{A}$; this is exactly the mechanism made explicit for logarithmic windows in [Theorem 9.28](#). Hence the defect has no surviving support in the residual class after $\mathcal{A}$ is removed. In both cases the construction cannot produce a retained residual profile from (V, P) , contradicting the defining retained concentration of the residual candidate. $\square$

Theorem 1.13 (Refined residual closure for the complete ordered decomposition). *Relative to the inherited alternatives in Theorems 1.5 and 1.6, one has*

$$\mathcal{C}_{\text{ax}} = \mathcal{C}_{\text{rot}} = \mathcal{C}_{\text{stat}}^{\text{ret}} = \mathcal{C}_{\text{gen}} = \emptyset,$$

$$\mathcal{C}_{\text{log diff}} = \mathcal{C}_{\text{Young}} = \mathcal{C}_{\text{homcrit}} = \mathcal{C}_{\text{logper}} = \mathcal{C}_{\text{ap}} = \emptyset.$$

Moreover,

$$\mathcal{C}_{\text{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \emptyset,$$

that is, no retained normalized residual representative remains in the affine/parasitic quotient stratum. Consequently

$$\mathcal{R}^{\#}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Deferred. The proof is given in Section 12, after the terminal decomposition theorem and the exclusion of all reduced cases. The dependency order is recorded in Table 2. $\square$

Table 2: Dependency table for the residual closure proof.

Result	Uses	Does not use
Imported setup theorem	Setup paper only: local concentration, local Type I bridge, normalized Seregin extraction, pressure atlas, lower classes, mildness, and conditional final assembly.	Paper IV residual closure.
R1 axisymmetric closure	Centered circulation equation, axis regularity, killed-strip comparison, and setup lower classes.	Terminal concentration theorem.
R2 reduction	Rotational flux identities, finite unit-cylinder concentration cover, and terminal concentration extraction.	R2 closure.
R3 reduction	Hull realization, centered scale reselection, and setup heredity.	R3 closure.
Concentration-set extraction	Local compactness, local defect measures, compact observer windows, and pressure atlas.	Endpoint Albritton–Barker theorem.
Diffuse-defect compactification	Concentration-set residual measure, observer compactification, and amenability of $\mathbb{R}^3 \rtimes \mathbb{R}$.	Generic closure.
Critical-tail closure	Bounded-origin realization, affine quotient, hidden-scale compactness, and coherent-tail alternatives.	Final residual exclusion.
Finite separated-profile exclusion	Terminal stratification, diagonal lifting, finite directed graph recurrence, and recurrence for concentration chains.	Global L^3 bound.
Atomic sequence- L^3	Indecomposability and absence of diffuse, tail, separated, pressure-loss, and critical-tail alternatives.	Albritton–Barker endpoint conclusion.
Endpoint atomic contradiction	Bounded mild finite terminal shifts, sequence- L^3 , and retained nonzero CKN concentration.	Residual decomposition.
Final assembly	All previous rows and the ordered decomposition.	Itself.

Corollary 1.14 (Residual input from the setup paper). *For every normalized Seregin collection $\mathcal{S}$ generated by an admissible finite-energy Type I singularity in the setup paper, the setup remaining class is empty.*

Proof. The setup paper defines its remaining class as the complement, inside $\mathcal{S}$, of the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes. This is the pre-quotient residual class $\mathcal{R}_{\text{pre}}(\mathcal{S}; I, J)$. Passing to the canonical affine/parasitic representative gives the retained normalized class $\mathcal{R}^\#(\mathcal{S}; I, J)$; if the representative is assigned to $\mathcal{C}_{\text{aff}}$, it is already in a closed lower quotient stratum. The indices I, J add no analytic assumption. By [Theorem 1.13](#), no retained normalized residual representative remains, and hence the setup remaining class is empty. $\square$

Theorem 1.15 (Local pointwise Type I exclusion after residual closure). *No finite-energy local pointwise Type I singularity can generate a normalized Seregin collection $\mathcal{S}$.*

Proof. Suppose that such a singular point exists. The setup paper supplies a positive local velocity concentration sequence. By [Theorems 1.24](#) and [1.26](#), the local pointwise Type I envelope gives the terminal velocity no-escape condition for this sequence. Hence [Theorem 1.5](#) applies, and the Seregin blow-up procedure produces a normalized ancient limit (V, P) satisfying [\(1.2\)](#) and the local velocity nonvanishing condition [\(1.5\)](#). By the inherited class exhaustion, V is either in one of the non-residual classes recorded in [Theorem 1.6](#) or in $\mathcal{R}(\mathcal{S}; I, J)$. The first alternative is impossible by the corresponding closed inputs: the small and stationary L^3 classes are excluded in the setup paper, the uniformly L^3 -tight class is also excluded there, and the closed structured/decay class is excluded there. The second alternative is impossible by [Theorem 1.13](#). This contradiction proves the theorem. $\square$

1.2. Local Type I singularities enter the terminal concentration framework. The purpose of this subsection is to make explicit the local reduction that is implicit in the two-paper proof architecture here. There are two logically different issues. The local pointwise Type I envelope gives Seregin compactness on every compact negative-time cylinder

$$B_R \times [-S, -\sigma], \quad \sigma > 0,$$

but this compactness is strictly away from the terminal slice $t = 0$. It does not alone imply that the positive L^3 -mass in $B_1 \times (-1, 0)$ survives away from the terminal layer $(-\sigma, 0)$. The latter is exactly the terminal velocity no-escape condition used in the setup paper.

The bridge below first records the admissible form used by the unified setup paper. It then uses the endpoint weak Serrin local Type I estimate of Albritton–Barker to derive the required scale-invariant L^2 bound on the terminal interval. Interpolation with the Type I envelope gives terminal velocity no-escape.

For this subsection, if $z_* = (x_*, T)$ and $r_n \downarrow 0$, write

$$u_n(x, t) := r_n u(x_* + r_n x, T + r_n^2 t), \quad p_n(x, t) := r_n^2 p(x_* + r_n x, T + r_n^2 t).$$

Also write $Q_R := B_R(0) \times (-R^2, 0)$ and

$$Q_1(0, 0) = B_1(0) \times (-1, 0).$$

Definition 1.16 (Admissible local Type I concentration sequence). Let $z_* = (x_*, T)$ satisfy the local Type I bound [\(1.10\)](#), and let $r_n \downarrow 0$. The corresponding rescaled velocities u_n form an admissible local Type I concentration sequence if there are $\eta_v > 0$ such that

$$(1.7) \quad \iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v \quad \text{for every } n,$$

and the terminal velocity no-escape condition holds:

$$(1.8) \quad \limsup_{\sigma \downarrow 0} \sup_n \iint_{B_1 \times (-\sigma, 0)} |u_n|^3 dx dt = 0.$$

Equivalently, after lowering the mass threshold, there are $\sigma_0 \in (0, 1)$ and $\eta_1 > 0$ such that

$$(1.9) \quad \iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 dx dt \geq \eta_1 \quad \text{for all } n.$$

Lemma 1.17 (No-escape leaves positive mass away from the terminal layer). *If u_n is an admissible local Type I concentration sequence in the sense of Theorem 1.16, then (1.9) holds for some $\sigma_0 \in (0, 1)$ and some $\eta_1 > 0$.*

Proof. Choose $\sigma_0 \in (0, 1)$ so small that

$$\sup_n \iint_{B_1 \times (-\sigma_0, 0)} |u_n|^3 dx dt \leq \frac{1}{2} \eta_v,$$

which is possible by (1.8). Subtracting this terminal-layer contribution from (1.7) gives

$$\iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 dx dt \geq \frac{1}{2} \eta_v.$$

Take $\eta_1 = \eta_v/2$. □

Proposition 1.18 (Local Type I rescalings have Seregin compactness). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that there exist $\rho > 0$ and $M < \infty$ such that*

$$(1.10) \quad \operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

Let $r_n \downarrow 0$, and let (u_n, p_n) be the rescaled sequence above. Then the following hold.

(i) *For every compact cylinder*

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \quad R, S < \infty, \quad \sigma > 0,$$

there exists $n_0(K)$ such that

$$(1.11) \quad \|u_n\|_{L^\infty(K)} \leq M \sigma^{-1/2}, \quad n \geq n_0(K).$$

(ii) *On every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$ there are time-dependent pressure gauges $a_{n,K}(t)$ and a constant $C_K < \infty$ such that, for $n \geq n_0(K)$,*

$$(1.12) \quad \|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \leq C_K.$$

The constant C_K may depend on the fixed Leray energy level of the original solution, but no concentration or no-escape assumption enters it.

(iii) *After subtracting these local pressure gauges, the family (u_n, p_n) is locally precompact in the standard Seregin topology on $\mathbb{R}^3 \times (-\infty, 0)$. Along every subsequence there is a further subsequence and an ancient suitable weak solution (U, Q) such that, on every compact $K \Subset \mathbb{R}^3 \times (-\infty, 0)$,*

$$u_n \rightarrow U \quad \text{strongly in } L^q(K), \quad 1 \leq q < \infty,$$

and

$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}(K).$$

The limit is bounded and satisfies

$$|U(x, t)| \leq M(-t)^{-1/2}, \quad t < 0.$$

- (iv) *More generally, let $\mathbf{c} = (z_n)$, $z_n = (y_n, \tau_n)$, be a parabolic recentering sequence for (u_n, p_n) with $\tau_n \leq 0$. Assume that $\mathbf{c}$ remains in the local Type I window in the following precise sense: for every $R, S < \infty$, all sufficiently large n satisfy*

$$r_n(|y_n| + R) < \rho, \quad T + r_n^2(\tau_n - S) > T - \rho^2.$$

If $\mathbf{c}$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness in the sense of [Theorem 9.3](#), and its recentered fields converge locally to a bounded suitable terminal profile. Recenterings which do not satisfy this local-window condition are, by definition, assigned to the exterior or diffuse terminal alternatives rather than to the compact local profile class.

Proof. Fix

$$K = B_R \times [-S, -\sigma] \in \mathbb{R}^3 \times (-\infty, 0).$$

For n sufficiently large,

$$r_n R < \rho/2, \quad r_n^2 S < \rho^2/2.$$

Hence, for a.e. $(x, t) \in K$,

$$x_* + r_n x \in B_\rho(x_*), \quad T + r_n^2 t \in (T - \rho^2, T).$$

The local Type I bound [\(1.10\)](#) gives

$$|u_n(x, t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \leq r_n M (T - (T + r_n^2 t))^{-1/2} = M(-t)^{-1/2} \leq M\sigma^{-1/2}.$$

This proves [\(1.11\)](#). In particular,

$$\operatorname{ess\,sup}_{-S < t < -\sigma} \int_{B_R} |u_n(x, t)|^2 dx \leq |B_R| M^2 \sigma^{-1},$$

and

$$\iint_K |u_n|^3 dx dt \leq |K| M^3 \sigma^{-3/2}.$$

We next record the pressure estimate, where the nonlocal pressure contribution is localized in the local Type I compactness argument. We use the whole-space Leray pressure representative, so at each rescaled time

$$p_n(\cdot, t) = \mathcal{R}_i \mathcal{R}_j (u_{n,i} u_{n,j})(\cdot, t)$$

modulo a function of time. Fix a cutoff $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R \equiv 1$ on B_{4R} , and write in B_{4R}

$$p_n = q_n + h_n, \quad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$

The local term satisfies

$$\|q_n\|_{L^{3/2}(B_{4R} \times [-S, -\sigma])} \leq C_{R,S,\sigma,M}.$$

For the harmonic part choose the gauge $a_{n,K}(t) = h_n(0, t)$. The standard kernel difference estimate for the Riesz pressure kernel is as follows. If

$$K_{ij}(x) = c_3 \text{ p.v. } \frac{3x_i x_j - \delta_{ij} |x|^2}{|x|^5}$$

is the Calderón–Zygmund pressure kernel, then for $|x| \leq 2R$ and $|z| \geq 4R$,

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq C_R |z|^{-4}.$$

Decomposing the exterior region into dyadic annuli gives, for $x \in B_{2R}$,

$$|h_n(x, t) - a_{n,K}(t)| \leq C_R \sum_{m \geq 2} 2^{-4m} \int_{2^m R < |z| < 2^{m+1} R} |u_n(z, t)|^2 dz.$$

For annuli whose physical image remains in $B_\rho(x_*)$, namely while $2^{m+1}Rr_n \leq \rho$, the local Type I bound and $-t \geq \sigma$ on K give

$$\int_{2^m R < |z| < 2^{m+1} R} |u_n(z, t)|^2 dz \leq CM^2 \sigma^{-1} (2^m R)^3.$$

Multiplication by 2^{-4m} leaves a summable 2^{-m} factor, so the corresponding part of the series is bounded by a geometric sum depending only on R, M, σ . For the remaining outer annuli one uses only the finite Leray energy of the original solution:

$$\int_{\mathbb{R}^3} |u_n(z, t)|^2 dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 dy.$$

If m_n is the first index for which $2^{m_n} Rr_n \gtrsim \rho$, then

$$\sum_{m \geq m_n} 2^{-4m} \int_{\mathbb{R}^3} |u_n(z, t)|^2 dz \leq C_{\rho, R} r_n^3 \|u\|_{L^\infty(0, T; L^2)}^2 \leq C_{\rho, R, u}.$$

Thus $h_n - a_{n, K}$ is uniformly bounded on $B_{2R} \times [-S, -\sigma]$. Since this cylinder has finite measure,

$$\|h_n - a_{n, K}\|_{L^{3/2}(B_{2R} \times [-S, -\sigma])} \leq C_K.$$

Together with the Calderón–Zygmund bound for q_n , this gives

$$\|p_n - a_{n, K}(t)\|_{L^{3/2}(B_{2R} \times [-S, -\sigma])} \leq C_K.$$

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder $K' \Subset \mathbb{R}^3 \times (-\infty, 0)$. The already obtained local L^2 , L^3 , and pressure-gauge bounds control every term on the right-hand side, and therefore

$$\iint_K |\nabla u_n|^2 dx dt \leq C_K.$$

This proves (1.12).

The Navier–Stokes equation now bounds $\partial_t u_n$ locally in a negative Sobolev space, for instance in $L_t^{3/2} H_x^{-2}(K)$. Aubin–Lions compactness and the uniform L^∞ bound give strong local convergence of the velocity in every finite L^q , after passing to a subsequence. The pressure bound gives weak $L^{3/2}$ convergence after gauges. Passing to the limit in the distributional equations and in the local energy inequality gives an ancient suitable weak solution (U, Q) . The pointwise Type I bound passes to the limit, so U is bounded on every compact negative-time cylinder. This proves (iii).

Finally, let $\mathfrak{c} = (y_n, \tau_n)$ be as in (iv), and set

$$u_n^{\mathfrak{c}}(y, s) = u_n(y + y_n, s + \tau_n), \quad p_n^{\mathfrak{c}}(y, s) = p_n(y + y_n, s + \tau_n).$$

The local-window condition, together with $\tau_n \leq 0$, is exactly what is needed to repeat the preceding estimates on each fixed backward compact cylinder in the (y, s) -variables: the physical cylinders remain in $B_\rho(x_*) \times (T - \rho^2, T)$, the local Type I envelope gives the same L^∞ bound on compact windows, and the pressure decomposition has the same local and finite-energy tail estimates. Thus the two failure modes in Theorem 9.3, blow-up of the local dissipation measure or failure of local pressure-gauge control, cannot occur for such a retained local recentering sequence after subsequence extraction. The recentered fields therefore converge locally to a bounded suitable terminal profile. If the local-window condition fails, then the recentering leaves every bounded terminal window attached to the singular point, or its critical mass persists only through expanding regions; those are precisely the exterior and diffuse terminal alternatives rather than loss of local compactness inside the local Type I bridge. $\square$

1.3. Pressure-projected terminal energy estimate. The preceding compactness result is stated away from the terminal slice. The next estimate supplies the terminal-layer bound required for velocity no-escape. The local pressure projection removes the nonlocal pressure flux from the local energy inequality; the remaining estimate is an interior Caccioppoli propagation bound on balls contained in the local Type I window.

Let $v = u^{(r)}$, $\pi = p^{(r)}$, and $\Lambda = \rho/r$. After translating z_* to $(0, 0)$, the local Type I bound gives

$$|v(x, s)| \leq M(-s)^{-1/2} \quad (x, s) \in B_\Lambda \times (-4, 0),$$

for all sufficiently small r . For $a \in \mathbb{R}^3$, $L \geq 1$, and $-4 < s < t < 0$, write

$$A_L(a, s) := \int_{B_L(a)} |v(x, s)|^2 dx, \quad D_L(a; s, t) := \int_s^t \int_{B_L(a)} |\nabla v|^2 dx d\sigma.$$

Lemma 1.19 (Local pressure-projected Caccioppoli inequality). *There is a universal constant C with the following property. Let (v, π) be a suitable weak solution in $B_{8L}(a) \times (s_-, 0)$, $L \geq 1$, and assume*

$$|v(x, s)| \leq M(-s)^{-1/2} \quad \text{on } B_{8L}(a) \times (s_-, 0).$$

If $s_0 \in (s_-, 0)$ is a Lebesgue time for v and ∇v , then for a.e. $t \in (s_0, 0)$,

$$(1.13) \quad \int_{B_L(a)} |w(x, t)|^2 dx + D_L(a; s_0, t) \leq C A_{4L}(a, s_0) + C \int_{s_0}^t \left(L^{-2} + M L^{-1} (-s)^{-1/2} \right) A_{4L}(a, s) ds.$$

The estimate is entirely local on $B_{8L}(a)$; no pressure tail from outside $B_{8L}(a)$ appears. Here $w = v + \nabla p_h$ is the local projected velocity defined in the proof below.

Proof. The proof uses the standard local pressure-projection argument in the scale L normalization. Let $B = B_{8L}(a)$. Denote by E_B^* the Stokes pressure projection on B : for $F \in W^{-1,q}(B)$, $1 < q < \infty$, $\nabla E_B^*(F)$ is the gradient part of the stationary Stokes solution in B with zero boundary velocity. It satisfies the scale-invariant estimates

$$\|\nabla E_B^*(\nabla \cdot F)\|_{L^q(B)} \leq C_q \|F\|_{L^q(B)}, \quad \|\nabla E_B^*(\Delta w)\|_{L^2(B)} \leq C \|\nabla w\|_{L^2(B)}.$$

Set

$$\nabla p_h = -\nabla E_B^*(v), \quad w := v + \nabla p_h.$$

The harmonic pressure p_h is harmonic in B , and the projection estimates give

$$\|w\|_{L^2(B)} + \|\nabla p_h\|_{L^2(B)} \leq C \|v\|_{L^2(B)}, \quad \|\nabla w\|_{L^2(B_{6L}(a))} \leq C \|\nabla v\|_{L^2(B)}.$$

The projected local energy inequality for w , obtained by applying the Stokes projection to the distributional Navier–Stokes equations and testing with $\zeta^2 w$, reads

$$(1.14) \quad \begin{aligned} & \int |w(t)|^2 \zeta^2 + \int_{s_0}^t \int |\nabla v|^2 \zeta^2 \\ & \leq C \int |w(s_0)|^2 \zeta^2 + C \int_{s_0}^t \int |w|^2 (|\partial_s \zeta| + |\Delta \zeta| + |\nabla \zeta|^2) \\ & \quad + C \int_{s_0}^t \int |v| |w|^2 |\nabla \zeta| + C \int_{s_0}^t \int |v| |\nabla p_h|^2 |\nabla \zeta|. \end{aligned}$$

Here $\zeta \in C_c^\infty(B_{4L}(a))$, $\zeta \equiv 1$ on $B_L(a)$, and

$$|\nabla \zeta| \leq C L^{-1}, \quad |\Delta \zeta| + |\partial_s \zeta| \leq C L^{-2}.$$

The pressure-projection estimates allow w and ∇p_h to be bounded by v in $L^2(B_{4L}(a))$. Therefore the cutoff terms in (1.14) are bounded by

$$CL^{-2} \int_{s_0}^t A_{4L}(a, s) ds.$$

The two convective terms are bounded, using the Type I envelope, by

$$CL^{-1} \int_{s_0}^t \|v(s)\|_{L^\infty(B_{4L}(a))} A_{4L}(a, s) ds \leq CML^{-1} \int_{s_0}^t (-s)^{-1/2} A_{4L}(a, s) ds.$$

The initial projected energy is controlled by

$$\int |w(s_0)|^2 \zeta^2 dx \leq CA_{4L}(a, s_0).$$

Combining the preceding bounds gives (1.13). $\square$

Definition 1.20 (Interior terminal energy estimate). The local pointwise Type I window at z_* is said to satisfy the interior terminal energy estimate if, for every fixed $R < \infty$, there are

$$r_R > 0, \quad B_R < \infty,$$

such that every rescaling $v = u^{(r)}$, $0 < r < r_R$, with $\Lambda = \rho/r$, satisfies

$$(1.15) \quad \text{ess sup}_{-1 < s < 0} \int_{B_{8R}} |v(x, s)|^2 dx + \int_{-1}^0 \int_{B_{8R}} |\nabla v|^2 dx ds \leq B_R.$$

Equivalently, the dependence on A_{4L} in Theorem 1.19 can be stopped inside a fixed multiple of the target ball, uniformly as the boundary $\partial B_{\rho/r}$ of the local Type I window recedes to infinity.

Proposition 1.21 (Pressure-projected reduction to an interior energy bound). *Assume the local pointwise Type I bound (1.10) at z_* . If the interior energy estimate (1.15) holds with $R = 1$, then the velocity no-escape condition (1.8) holds for every positive local Type I concentration sequence at z_* . Consequently such a sequence is admissible in the sense of Theorem 1.16.*

Proof. Let $u_n = u^{(r_n)}$ be any positive local Type I concentration sequence. By (1.15) with $R = 1$, after discarding finitely many indices,

$$\text{ess sup}_{-1 < s < 0} \int_{B_1} |u_n(x, s)|^2 dx \leq B_1.$$

The local Type I envelope gives

$$\|u_n(s)\|_{L^\infty(B_1)} \leq M(-s)^{-1/2}.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_n|^3 dx ds &\leq \int_{-\sigma}^0 \|u_n(s)\|_{L^\infty(B_1)} \int_{B_1} |u_n(x, s)|^2 dx ds \\ &\leq 2MB_1\sigma^{1/2}. \end{aligned}$$

The finitely many discarded indices have absolutely continuous $|u_n|^3$ -integrals on $B_1 \times (-1, 0)$, so taking the supremum over the whole sequence and then letting $\sigma \downarrow 0$ gives (1.8). The positive Q_1 mass hypothesis is the other part of Theorem 1.16. $\square$

Remark 1.22 (Pressure-projected reduction). Theorem 1.19 shows, independently of the Albritton–Barker endpoint input used below, that terminal no-escape reduces to an interior terminal energy estimate: after local pressure projection, the local energy inequality controls the projected velocity and the dissipation up to the larger-scale quantity A_{4L} . The estimate (1.15) prevents

terminal-layer concentration entering from the boundary of the local Type I window. In the present paper this bound is supplied instead by the endpoint weak Serrin local Type I bounds in [Theorem 1.23](#).

Theorem 1.23 (Albritton–Barker weak Serrin local Type I bounds). *Let (v, q) be a suitable weak solution in a parabolic cylinder*

$$Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0).$$

Assume

$$v \in L_t^{q,\infty} L_x^{p,\infty}(Q_\rho(z_0)), \quad 3 \leq p \leq \infty, \quad 2 \leq q \leq \infty, \quad \frac{3}{p} + \frac{2}{q} = 1.$$

Then for every $0 < \theta < 1$, the scale-invariant local Type I bounds are finite on $Q_{\theta\rho}(z_0)$: there is

$$K_\theta < \infty$$

such that every cylinder $Q_\lambda(y, \tau) \subset Q_{\theta\rho}(z_0)$ satisfies

$$A(Q_\lambda(y, \tau)) + C(Q_\lambda(y, \tau)) + D(Q_\lambda(y, \tau)) + E(Q_\lambda(y, \tau)) \leq K_\theta.$$

Here

$$A(Q_\lambda(y, \tau)) := \lambda^{-1} \operatorname{ess\,sup}_{\tau - \lambda^2 < s < \tau} \int_{B_\lambda(y)} |v(x, s)|^2 dx,$$

and C, D, E denote the usual scale-invariant L^3 -velocity, $L^{3/2}$ -pressure, and dissipation quantities of Albritton–Barker. In the endpoint $p = \infty, q = 2$, the constant is allowed to depend on the local suitable-solution background quantities on $Q_\rho(z_0)$, including the local C - and D -bounds.

Proof. After translating and scaling to the unit cylinder, this is Lemma 2.5 of Albritton–Barker [1]. Their proof combines the weak Serrin bound with critical Morrey estimates for suitable weak solutions. The endpoint $p = \infty, q = 2$ is included in the weak Serrin lemma; as emphasized there, the resulting local Type I bound is a suitable-solution estimate and its constant may depend on the background local energy quantities on the larger cylinder. $\square$

Theorem 1.24 (Local pointwise Type I implies terminal no-escape). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$, and assume that*

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)).$$

Assume that z_* satisfies the local pointwise Type I bound: there are $\rho > 0$ and $M < \infty$ such that

$$(1.16) \quad \operatorname{ess\,sup}_{T - \rho^2 < t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

For $0 < r < \rho$, set

$$u^{(r)}(x, s) = r u(x_* + rx, T + r^2 s), \quad p^{(r)}(x, s) = r^2 p(x_* + rx, T + r^2 s).$$

Then for every $0 < R < \infty$ there are constants

$$r_R \in (0, \rho/R), \quad B_R < \infty,$$

and time-dependent pressure gauges $a_{r,R}(s)$, depending on R, M, ρ, T and the local suitable-solution background quantities of (u, p) in $Q_\rho(z_*)$, such that the following terminal local energy estimate holds for every $0 < r < r_R$:

$$(1.17) \quad \begin{aligned} & \operatorname{ess\,sup}_{-1 < s < 0} \int_{B_R} |u^{(r)}(x, s)|^2 dx + \int_{-1}^0 \int_{B_R} |\nabla u^{(r)}|^2 dx ds \\ & + \int_{-1}^0 \int_{B_R} |p^{(r)}(x, s) - a_{r,R}(s)|^{3/2} dx ds \leq B_R. \end{aligned}$$

Consequently, for every $0 < \sigma < 1$ and every $0 < r < r_R$,

$$(1.18) \quad \int_{-\sigma}^0 \int_{B_R} |u^{(r)}(x, s)|^3 dx ds \leq 2MB_R\sigma^{1/2},$$

and in particular

$$(1.19) \quad \lim_{\sigma \downarrow 0} \sup_{0 < r < r_R} \int_{-\sigma}^0 \int_{B_R} |u^{(r)}(x, s)|^3 dx ds = 0.$$

Proof. Fix once and for all $0 < \theta < 1$, for instance $\theta = 1/2$, and put $L_R := \max\{R, 1\}$. By decreasing r_R , we may assume

$$B_{L_R r}(x_*) \times (T - L_R^2 r^2, T) \subset Q_{\theta\rho}(z_*) \quad (0 < r < r_R).$$

The pointwise Type I bound implies the endpoint weak Serrin estimate on $Q_{\theta\rho}(z_*)$. Indeed, after writing

$$f(t) = \|u(t)\|_{L^\infty(B_\rho(x_*))},$$

one has

$$|\{t \in (T - \rho^2, T) : f(t) > \alpha\}| \leq M^2 \alpha^{-2}, \quad \alpha > 0,$$

and therefore $u \in L_t^{2,\infty} L_x^\infty(Q_{\theta\rho}(z_*))$. Applying [Theorem 1.23](#) gives a finite constant K_θ such that $A, C, D, E \leq K_\theta$ on all cylinders contained in $Q_{\theta\rho}(z_*)$.

Taking the cylinder $Q_{L_R r}(z_*)$ gives

$$(L_R r)^{-1} \operatorname{ess\,sup}_{T - L_R^2 r^2 < t < T} \int_{B_{L_R r}(x_*)} |u(x, t)|^2 dx \leq K_\theta,$$

and similarly the D - and E -bounds give

$$(L_R r)^{-1} \int_{T - L_R^2 r^2}^T \int_{B_{L_R r}(x_*)} |\nabla u|^2 dx dt \leq K_\theta,$$

$$(L_R r)^{-2} \int_{T - L_R^2 r^2}^T \int_{B_{L_R r}(x_*)} |p - a_{r,R}^{\text{phys}}(t)|^{3/2} dx dt \leq K_\theta$$

for suitable time-dependent pressure gauges $a_{r,R}^{\text{phys}}(t)$. Scaling these three estimates and restricting to

$$B_R \times (-1, 0) \subset B_{L_R} \times (-L_R^2, 0)$$

gives [\(1.17\)](#) with

$$a_{r,R}(s) := r^2 a_{r,R}^{\text{phys}}(T + r^2 s),$$

after enlarging the constant B_R by a harmless factor depending on R and K_θ .

After translating (x_*, T) to $(0, 0)$, the rescaled fields also satisfy

$$|u^{(r)}(x, s)| \leq M(-s)^{-1/2} \quad (x, s) \in B_R \times (-1, 0)$$

for all $r < r_R$, after decreasing r_R if necessary. The terminal local energy estimate established above gives

$$\operatorname{ess\,sup}_{-1 < s < 0} \int_{B_R} |u^{(r)}(x, s)|^2 dx \leq B_R.$$

For $0 < \sigma < 1$, the local Type I bound above and the kinetic bound give

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_R} |u^{(r)}|^3 dx ds &\leq \int_{-\sigma}^0 \|u^{(r)}(s)\|_{L^\infty(B_R)} \int_{B_R} |u^{(r)}(x, s)|^2 dx ds \\ &\leq MB_R \int_{-\sigma}^0 (-s)^{-1/2} ds = 2MB_R\sigma^{1/2}. \end{aligned}$$

This proves (1.18) and (1.19). $\square$

Corollary 1.25 (No diffuse terminal-layer mass). *Under the hypotheses of Theorem 1.24, for every fixed $R < \infty$ there do not exist $r_n \downarrow 0$, $\sigma_n \downarrow 0$, and $\varepsilon_0 > 0$ such that*

$$r_n^{-2} \int_{T-\sigma_n r_n^2}^T \int_{B_{Rr_n}(x_*)} |u(x, t)|^3 dx dt \geq \varepsilon_0 \quad \text{for all } n.$$

Proof. For $u_n = u^{(r_n)}$, the left hand side is

$$\int_{-\sigma_n}^0 \int_{B_R} |u_n(x, s)|^3 dx ds.$$

By (1.18), this is bounded by

$$2MB_R\sigma_n^{1/2} \rightarrow 0,$$

which contradicts any fixed positive lower bound. $\square$

Corollary 1.26 (Local pointwise Type I concentration sequences are admissible). *Let (u, p) and $z_* = (x_*, T)$ satisfy the hypotheses of Theorem 1.24. Let $r_n \downarrow 0$ be any velocity concentration sequence satisfying*

$$\iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v > 0, \quad u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

Then u_n is an admissible local Type I concentration sequence in the sense of Theorem 1.16.

Proof. Apply Theorem 1.24 with $R = 1$. After discarding finitely many n , all $r_n < r_1$, and

$$\limsup_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n|^3 dx dt = 0.$$

The finitely many discarded indices do not affect the limit as $\sigma \downarrow 0$, because each fixed suitable weak solution has $|u_n|^3 \in L^1(B_1 \times (-1, 0))$. The positive mass hypothesis is the other condition in Theorem 1.16. $\square$

Lemma 1.27 (Local Type I concentration enters the terminal dichotomy). *Under the hypotheses of Theorem 1.18, let $r_n \downarrow 0$ be a velocity concentration sequence at z_* satisfying*

$$\iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v > 0.$$

Then r_n is admissible by Theorem 1.26. Let σ_0, η_1 be given by Theorem 1.17. For the associated terminal extraction sequence (V_N, P_N) generated by a Seregin limit of this sequence, choose the small-data threshold $\varepsilon_{sd} \leq \eta_1/2$. Then the origin extraction produces a bounded ancient Seregin profile with retained compact local velocity mass away from the terminal layer. In particular the retained compact concentration generated from the origin recentering has no loss of local compactness. Any retained local recentering which stays in the local Type I window has the same compactness property by Theorem 1.18; any concentration leaving that window is assigned to the exterior or diffuse terminal remainder. The paired terminal decomposition of Theorem 9.11, together with the mixed-profile exclusions and concentration-chain exclusions Theorems 9.69 and 10.20, reduces the retained possibilities to a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles.

Proof. By Theorem 1.17,

$$\iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 dx dt \geq \eta_1 \quad \text{for all } n.$$

The cylinder $B_1 \times [-1, -\sigma_0]$ is compactly contained in $\mathbb{R}^3 \times (-\infty, 0)$. Therefore [Theorem 1.18](#) gives, after passing to a subsequence, strong local L^3 convergence on this cylinder:

$$u_n \rightarrow U \quad \text{in } L^3(B_1 \times (-1, -\sigma_0)).$$

Consequently

$$\iint_{B_1 \times (-1, -\sigma_0)} |U|^3 dx dt \geq \eta_1.$$

Cover $B_1 \times (-1, -\sigma_0)$ by finitely many unit terminal cylinders strictly away from $t = 0$. At least one such cylinder carries a fixed positive velocity lower bound, and since the local defect measure dominates the velocity part, choosing $\varepsilon_{\text{sd}} \leq \eta_1/2$ after adjusting by this finite cover makes the corresponding origin-frame recentering ε_{sd} -concentrating. This argument is precisely where no-escape is used: without [\(1.8\)](#), the lower bound could remain entirely in $B_1 \times (-\sigma, 0)$ for every fixed $\sigma > 0$.

The distinguished recentering satisfies the local-window condition in [Theorem 1.18](#), so it is not in the local-compactness-failure class. Other retained local recenterings inside the same Type I window have the same compactness. Recenterings leaving the window are exterior or diffuse terminal remainders. The paired decomposition [Theorem 9.11](#) records these remainders alongside the concentration set rather than treating them as mutually exclusive alternatives. The sole-source radiative cases are excluded by [Theorem 9.14](#), mixed compact–diffuse or compact–noncompact coexistence is excluded by [Theorem 9.69](#), and infinite chains of retained concentrating descendants are excluded by [Theorem 10.20](#). Hence [Theorem 9.12](#) leaves only a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles. $\square$

Theorem 1.28 (Local Type I singularities enter the exclusion argument). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that z_* satisfies the local Type I bound [\(1.10\)](#). Then there exists a bounded centered Type I Seregin profile with retained local concentration generated from (u, p, z_*) . Consequently the local terminal concentration framework of [Theorem 10.24](#) applies. In particular, no finite-energy local pointwise Type I singular point exists.*

Proof. By the local velocity concentration theorem in the setup paper, choose $r_n \downarrow 0$ and $\eta_v > 0$ with

$$\iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v, \quad u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

[Theorems 1.24](#) and [1.26](#) show that this concentration sequence satisfies the terminal velocity no-escape condition required by the setup paper. Apply [Theorem 1.27](#) to this sequence. The local Type I rescalings therefore produce at least one bounded terminal profile carrying retained local concentration. After passing to the centered variables, the bound $|U(x, t)| \leq M(-t)^{-1/2}$ becomes the global centered Type I bound

$$|V(y, \tau)| \leq M,$$

so each retained profile generated by the bridge is a bounded centered Type I Seregin profile.

If every such retained profile belongs to one of the previously closed lower classes, then the contradiction is supplied by the setup paper: small-amplitude and stationary L^3 profiles are excluded there, the uniformly tight class is excluded there, and the closed structured/decay class is excluded there. Otherwise choose a retained profile outside those lower classes. It is then a refined residual candidate, so [Theorem 10.24](#) applies. In that residual setting, a finite separated family of retained concentrating terminal profiles is excluded by [Theorem 10.23](#), infinite chains of retained concentrating descendants are excluded by [Theorem 10.20](#), and mixed compact–diffuse tails are

excluded by [Theorem 9.69](#). The remaining single retained profile is terminally indecomposable, has the backward finite-critical-norm sequence, has parasitic-free finite-shift mildness, and is then excluded by the Albritton–Barker endpoint Liouville input used in this paper. Both alternatives are impossible. Hence the assumed finite-energy local Type I singular point does not exist. $\square$

Corollary 1.29 (Local pointwise Type I exclusion after setup plus residual closure). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$. Combining the residual closure [Theorem 1.13](#) of this paper with the setup paper, the following holds: if $z_* = (x_*, T)$ is a singular point satisfying the local Type I bound (1.10), then the completed case closures give a contradiction. Equivalently, in the local Type I/local Type II dichotomy of the setup paper, every finite-energy singular point belongs to the local Type II alternative. The conclusion is the joint output of the two-paper chain; it is not asserted as a hidden hypothesis inside Paper IV alone.*

Proof. The first assertion is exactly [Theorem 1.28](#). The equivalence is the local Type I/local Type II dichotomy of the setup paper. $\square$

2. DECISION TREE FOR THE RESIDUAL PROFILE DECOMPOSITION

The following rendered decision tree records the ordered partition used in the paper. Each diamond is a stratification criterion. Each terminal node is a possible singularity or candidate blow-up profile type selected by the criteria. The local Type II terminal alternative is the remaining regime after the local pointwise Type I case is excluded; every Type I terminal alternative is excluded either by the inherited alternatives or by the case-closure theorems proved below.

The loop from the last criterion back to the residual class is only a graphical way to indicate the ordered nature of the decomposition: if a lower stratum has not actually been removed, it is assigned to its corresponding terminal alternative rather than to $\mathcal{C}_{\text{gen}}$. After the ordered removals are performed, $\mathcal{C}_{\text{gen}}$ is exactly the remaining residual class by definition. For the table, write

$$\mathfrak{T}_\rho(u; z_*) := \operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))}.$$

Table 3: Classification and closure table for the decision tree. Each row gives the criterion selecting the terminal alternative, the mechanism that excludes it when it is a Type I alternative, and the statement used at that point. For the inherited alternatives, the displayed condition is a representative form of the inherited definition; the precise definitions and closures are those of `proof_setup.tex`.

Terminal alternative	Criterion, exclusion mechanism, and reference
Local Type II singularity	Criterion. For every $\rho > 0$, $\mathfrak{T}_\rho(u; z_*) = \infty$. Role. This is the complementary alternative left by the local Type I/Type II dichotomy after local pointwise Type I is excluded. Link. Theorem 1.29 .
Small Type I profile	Criterion. Inherited small-amplitude class; schematically $\ V\ _{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_{\text{sm}}$. Exclusion. Small-data/regularity closure recorded in the setup paper. Link. Theorem 1.6 .
Stationary L^3 Type I profile	Criterion. Inherited stationary class; representative condition $\partial_\tau V = 0$ and $V \in L^3(\mathbb{R}^3)$. Exclusion. Stationary L^3 Liouville/regularity closure recorded in the setup paper. Link. Theorem 1.6 .
Tight Type I profile	Criterion. Inherited tight class; representative condition $\lim_{R \rightarrow \infty} \sup_\tau \int_{ y > R} V(y, \tau) ^3 dy = 0$. Exclusion. Endpoint L^3 rigidity for uniformly tight ancient dynamics. Link. Theorem 1.6 .

Terminal alternative	Criterion, exclusion mechanism, and reference
Structured Type I profile	Criterion. Inherited closed structured/decay class, $V \in \mathcal{C}_{\text{struct/decay}}^{\text{setup}}$. Exclusion. Closed structured/decay alternatives recorded in the setup paper. Link. Theorem 1.6 .
$\mathcal{C}_{\text{ax}}$ residual profile	Criterion. Axisymmetric residual profile with bounded centered circulation $\Gamma(\rho, Z, \tau) = \rho V_\theta$ and $\ \Gamma\ _{L^\infty} < \infty$. Exclusion. Centered circulation equation plus axis-absorption Liouville theorem; the profile becomes no-swirl and falls into an earlier structured class. Link. Theorem 4.3 .
$\mathcal{C}_{\text{rot}}$ residual profile	Criterion. Rotational relative equilibrium $V(y, \tau) = Q(\tau)W(Q(\tau)^T y)$, $Q(\tau) = e^{\tau\Omega}$. Exclusion. Localized Coriolis-flux reduction into the terminal local concentration decomposition. Link. Theorem 10.26 .
$\mathcal{C}_{\text{stat}}^{\text{ret}}$ residual profile	Criterion. Degenerate stationary-hull retained profile: along time translations $T_{s_k} V \rightarrow W$, where W is stationary and $\mathcal{E}_1(W, \Pi; 0) \geq \varepsilon_*$. Exclusion. Scale reselection reduces to a stationary retained concentration profile, then terminal concentration excludes it. Link. Theorem 10.27 .
$\mathcal{C}_{\text{aff}}$ lower stratum	Criterion. The affine-normalized oscillation vanishes: $\Phi(S_c U) = 0$ for an admissible affine symmetry S_c . Exclusion. Oscillatory entry normalization forces retained residual profiles to keep positive non-affine oscillation after affine normalization. Link. Theorem 9.61 .
$\mathcal{C}_{\text{log diff}}$ critical tail	Criterion. Logarithmic escape component in the realized critical-tail compactification: $\Lambda \neq 0$. Exclusion. Log-window descendant selection plus minimal mesoscopic-scale reduction and hidden-scale compactness. Link. Theorem 9.30 .
$\mathcal{C}_{\text{Young}}$ tail	Criterion. Tight annular critical-tail profile with non-Dirac Young measure: $\Lambda = 0$ and $\mathcal{Y}_{x,s} \neq \delta_{\tilde{W}(x,s)}$ on a set of positive measure. Exclusion. No hidden-scale variance gives strong L_{loc}^3 compactness, hence a Dirac Young measure. Link. Theorem 9.62 .
$\mathcal{C}_{\text{homcrit}}$ coherent tail	Criterion. Coherent (-1) -homogeneous tail: $W(r\omega, s) = r^{-1}A(\omega, s)$. Exclusion. Bounded-origin realization gives $W \in L_{\text{loc}}^\infty$ near $r = 0$, forcing $A = 0$. Link. Theorem 9.67 .
$\mathcal{C}_{\text{logper}}$ coherent tail	Criterion. Nonzero log-periodic coherent tail: $Z(\rho + L, \omega, s) = Z(\rho, \omega, s)$, $L \neq 0$, where $Z = e^\rho W(e^\rho \omega, s)$. Exclusion. Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \rightarrow 0$ as $\rho \rightarrow -\infty$; periodicity forces $Z \equiv 0$. Link. Theorem 9.67 .
$\mathcal{C}_{\text{ap}}$ coherent tail	Criterion. Nonzero aperiodic minimal logarithmic hull: $\mathcal{H}(Z)$ is minimal, nonperiodic, and nonzero. Exclusion. Bounded-origin realization puts 0 in the compact log-translation hull; minimality forces the hull to be $\{0\}$. Link. Theorem 9.67 .
$\mathcal{C}_{\text{gen}}$ generic terminal residual	Criterion. Ordered complement $V \in \mathcal{R}(\mathcal{S}; I, J) \setminus \bigcup_{\mathcal{B} \in \mathfrak{B}_{\text{lower}}} \mathcal{B},$ where $\mathfrak{B}_{\text{lower}}$ is the list of all preceding Type I alternatives. Exclusion. Paired concentration-set decomposition, finite separated-family exclusion, terminal indecomposability, sequence-L^3, parasitic-free mildness, and the Albritton–Barker endpoint Liouville theorem. Link. Theorem 11.2.

3. SHARED LOCAL NOTATION

For $z_0 = (y_0, \tau_0)$ and $r > 0$, write

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0).$$

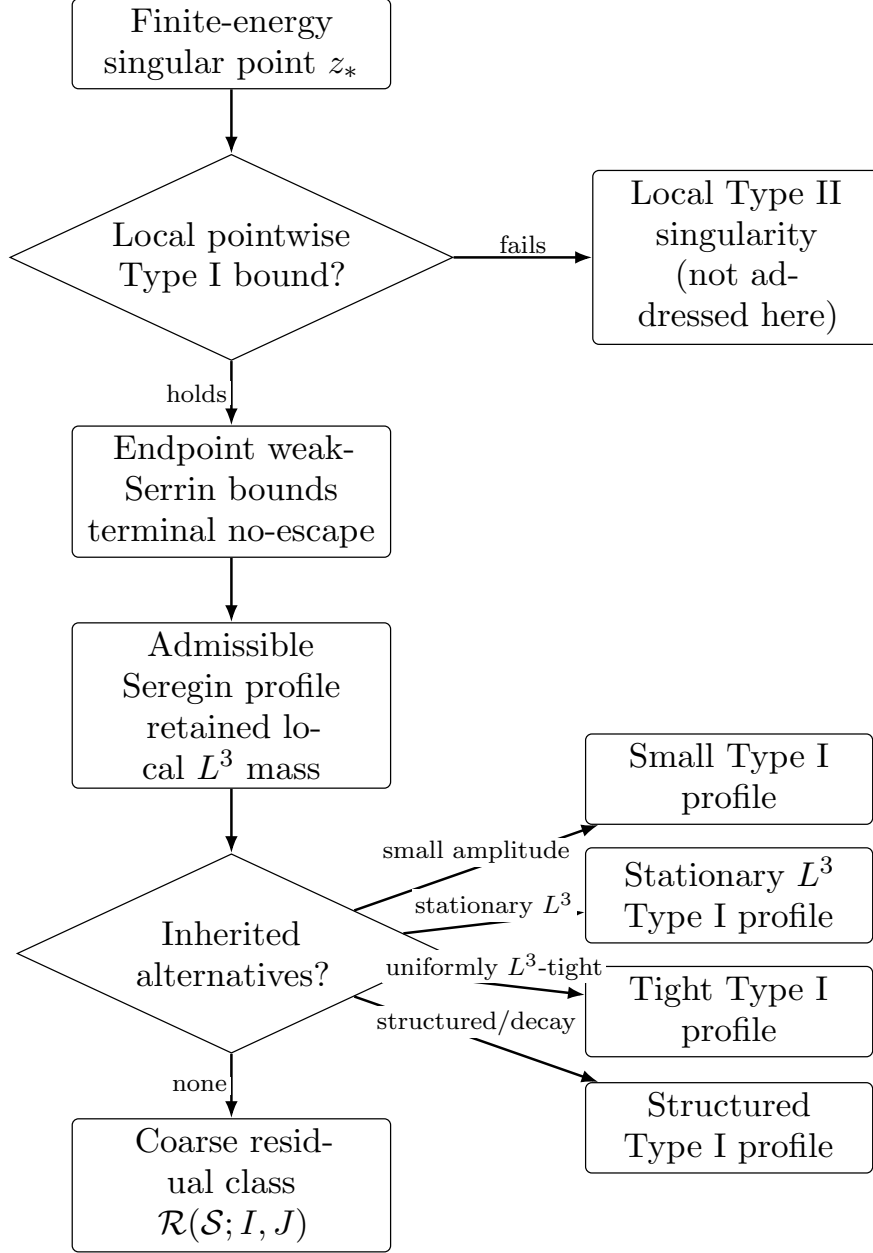


FIGURE 2. Rendered entry tree. Local pointwise Type I enters the admissible Seregin framework through endpoint weak Serrin estimates and terminal no-escape. If the bound fails, the point belongs to the local Type II alternative, which is not addressed here.

Definition 3.1 (Scale-invariant CKN concentration and defect measures). For a centered profile (V, P) , define the velocity concentration

$$(3.1) \quad C_r(V; z_0) := r^{-2} \iint_{Q_r(z_0)} |V|^3 dy d\tau,$$

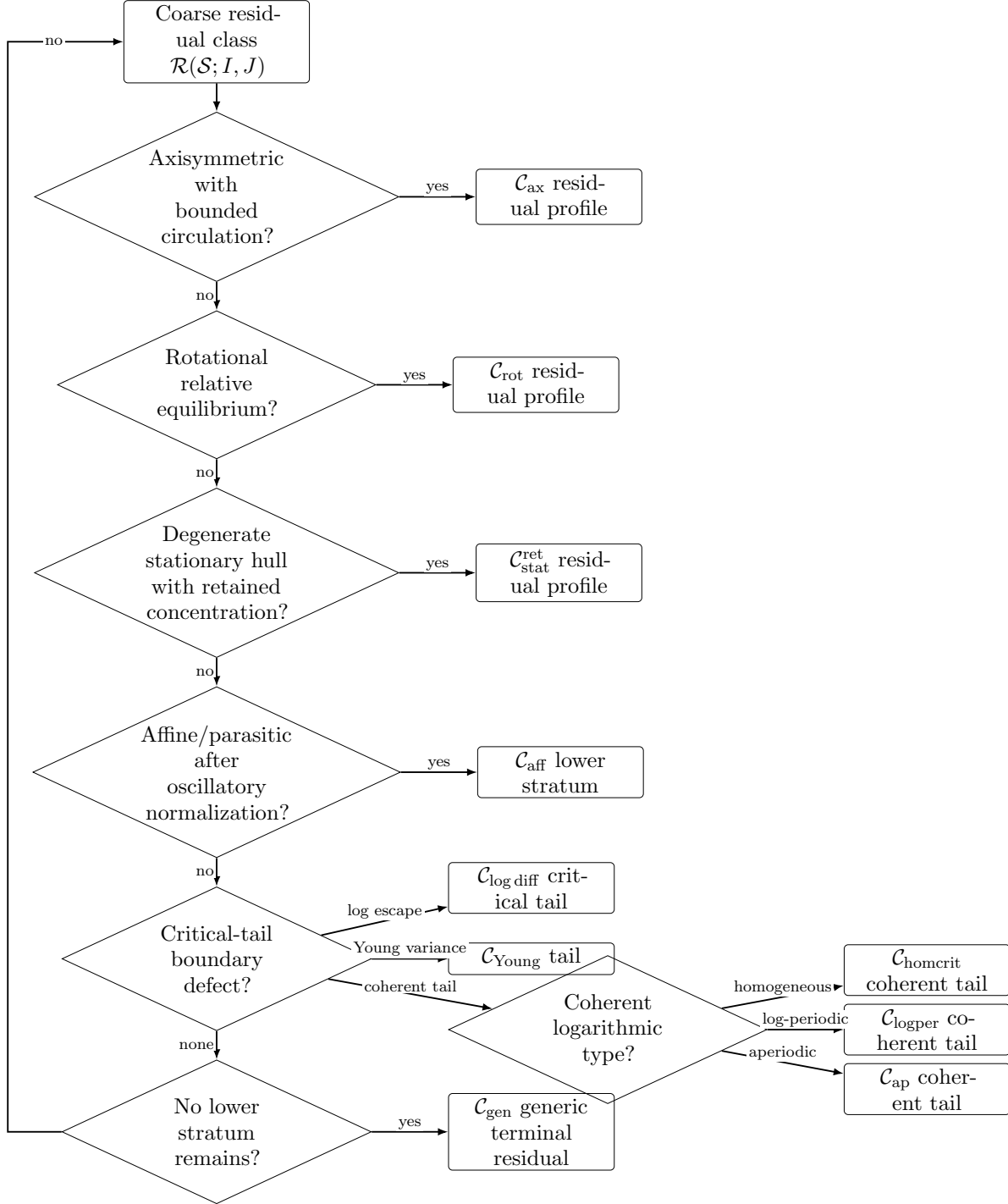


FIGURE 3. Rendered residual stratification tree. The final return arrow only records the ordered nature of the decomposition: if a lower stratum has not been removed, the profile is assigned to that terminal alternative; after all lower alternatives are removed, the remaining residual class is $\mathcal{C}_{gen}$.

and the quotient pressure concentration

$$(3.2) \quad D_r(P; z_0) := r^{-2} \inf_{c \in L^{3/2}(\tau_0 - r^2, \tau_0)} \iint_{Q_r(z_0)} |P(y, \tau) - c(\tau)|^{3/2} dy d\tau.$$

The local CKN concentration is

$$(3.3) \quad \mathcal{E}_r(V, P; z_0) := C_r(V; z_0) + D_r(P; z_0).$$

Thus $\mathcal{E}_r$ is parabolically scale-invariant under the Navier–Stokes scaling.

If an explicit pressure atlas $\mathbf{a}$ has been chosen, with local representative $a_{z_0, r}(\tau)$, we also write

$$(3.4) \quad \mathcal{E}_r^{\mathbf{a}}(V, P; z_0) := r^{-2} \iint_{Q_r(z_0)} (|V|^3 + |P - a_{z_0, r}(\tau)|^{3/2}) dy d\tau.$$

Then

$$\mathcal{E}_r(V, P; z_0) \leq \mathcal{E}_r^{\mathbf{a}}(V, P; z_0).$$

Thresholds and regularity statements use the quotient quantity $\mathcal{E}_r$; compactness statements use a fixed setup pressure atlas when an explicit representative is needed.

For a terminal sequence (V_n, P_n) with chosen pressure gauges $a_n(\tau)$, the atlas-dependent Radon defect measure is

$$(3.5) \quad d\mu_n := (|V_n|^3 + |P_n - a_n(\tau)|^{3/2}) dy d\tau.$$

The measure μ_n is not denoted by $\mathcal{E}_r$. On unit cylinders the factor 1^{-2} is invisible, so unit-scale atlas mass and unit-scale CKN concentration are compatible, with the atlas quantity dominating the quotient quantity.

Remark 3.2 (Two-scale dictionary). We use two different normalizations. The quantity $\mathcal{E}_r$ is parabolic and CKN-critical, hence carries the factor r^{-2} . By contrast, the tail quantities $\mathcal{A}_n(z, r)$ are spatial/logarithmic averaged oscillations on fixed centered time windows; their natural normalization is r^{-3} . These two quantities are never identified.

Definition 3.3 (Retained local concentration profile). Fix $\varepsilon_* > 0$. A terminal profile (U, Π) has retained local concentration if, after the terminal normalization under consideration,

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Definition 3.4 (Threshold hierarchy). The constants used in the terminal concentration argument are chosen in the following order. The compact velocity lower bounds η_0 and η_v are supplied by the setup paper `proof_setup.tex` and the local concentration input therein, respectively. The local small-data threshold ε_{sd} is then fixed with

$$0 < \varepsilon_{\text{sd}} \leq \min\{\eta_0, \eta_v, \varepsilon_{\text{CKN}}\},$$

where ε_{CKN} denotes the universal CKN regularity threshold whenever that threshold is invoked. The retained descendant threshold ε_* is chosen after the pressure gauges and compactness topology are fixed, with

$$0 < \varepsilon_* \leq \frac{1}{2} \varepsilon_{\text{sd}}.$$

Whenever a compactness or lower-semicontinuity passage loses a harmless fixed factor, ε_* is replaced once and for all by a smaller positive value. No later argument chooses a new concentration threshold depending on a previously extracted profile tree.

Definition 3.5 (Refined residual candidate). A refined residual candidate is a bounded normalized Seregin ancient limit obtained from the admissible Seregin extraction theorem in `proof_setup.tex`, which carries the positive compact local velocity mass in (1.5), and which is not in any alternative already closed before the refined residual decomposition is entered. The definition is local: the only persistent lower bound attached to the candidate by the setup paper is the fixed compact velocity L^3 lower bound.

Remark 3.6 (No hidden global L^3 assertion). The terminal arguments below do not claim that a generic residual profile is globally L^3 -tight or globally bounded in $L^\infty_\tau L^3_y$. The only nonlocal critical-norm conclusion used at the endpoint is the weaker sequence- L^3 completeness of retained indecomposable concentration profiles, and that conclusion is derived from local terminal alternatives rather than from an a priori global estimate on the original residual class.

3.1. Notation, threshold, and gauge directories. The following compact directories collect the persistent symbols, thresholds, and gauges used in the residual closure. They are not new definitions; each entry refers to the location where the symbol is first defined.

Velocity, pressure, and profile-space symbols.

- (V, P) : centered Seregin ancient profile in (1.2); first used in Section 1.
- (U, Q) : physical ancient representative when (V, P) is treated under physical pullback (1.4); also used in $\mathcal{R}_z$ -recentered form throughout Sections 8 and 9. The intended meaning is fixed locally in each section by the explicit definition of the immediately preceding rescaling.
- (W, Q_W) : blown-down, co-rotating, or tail profile, depending on the section; defined locally in Sections 5 and 9.3.
- $\mathcal{E}_r(V, P; z_0)$: quotient, scale-invariant parabolic CKN concentration, see Theorem 3.1.
- $\mathcal{E}_1$: pressure-normalized unit CKN concentration, fixed at $r = 1$; used throughout the terminal concentration arguments and in the retained-concentration definition Theorem 3.3.
- μ_n : atlas-dependent Radon defect measure in (3.5); used for concentration-set and defect compactness, not as a scale-invariant quantity.
- $\mathcal{A}_n(z, r)$: spatial/logarithmic averaged affine-quotient oscillation with r^{-3} normalization, distinct from $\mathcal{E}_r$.
- $\mathfrak{T}_M$: terminal profile class, Theorem 8.7.
- $\mathfrak{T}_M^\#$: compact normalized descendant profile class, Theorem 8.10.
- $\mathcal{C}_{\text{aff}}$: affine/parasitic reduction/quotient stratum, Theorem 8.16; not a usual nonempty/empty PDE class but an exclusion label, see Theorems 8.17 and 9.61.
- $\text{osc}_{3,\text{sp}}$: spatial oscillation modulo time-dependent homogeneous gauges, Theorem 9.33.
- $\text{osc}_{3,\text{ct}}$ and Φ : constant-mode/non-affine retained oscillation, Theorems 8.8 and 8.19.

Threshold ordering.

- η_0 : compact velocity L^3 lower bound supplied by the base Seregin extraction Theorem 1.5.
- η_v : compact velocity lower bound from the local concentration input of the setup paper, used as initial mass for terminal extraction; see Theorem 3.4.
- ε_{sd} : local small-data threshold, $0 < \varepsilon_{\text{sd}} \leq \min\{\eta_0, \eta_v, \varepsilon_{\text{CKN}}\}$, Theorem 3.4.
- ε_* : retained descendant CKN threshold, $0 < \varepsilon_* \leq \varepsilon_{\text{sd}}/2$, used in Theorem 3.3 and in concentration-set extraction.
- ε_0 : minimal mesoscopic-scale oscillation threshold; depends only on the hidden-scale defect lower bound and on the finite-overlap constants of Theorem 9.35; first appears in Theorem 9.34.
- η_K : compact oscillation lower bound on the recurrent concentration core, first introduced in Theorem 10.16.

The thresholds are fixed in the order $\eta_0, \eta_v \Rightarrow \varepsilon_{\text{sd}} \Rightarrow \varepsilon_* \Rightarrow \varepsilon_0, \eta_K$, and no later argument chooses a new concentration threshold depending on a previously extracted profile tree. Phrases of the

form “after lowering the threshold” refer either to the once-and-for-all reduction of ε_* recorded in [Theorem 3.4](#) or to a slight reduction of the mesoscopic threshold ε_0 recorded in [Theorem 9.38](#); in either case the reduction does not disturb earlier choices.

Pressure and velocity gauges.

- $c(\tau)$ or $a_{z_0,r}(\tau)$: local pressure time gauge in (3.2) and (3.4); measurable in τ ; used because pressure is defined modulo functions of time.
- $c_n(s)$: homogeneous velocity gauge at the minimal mesoscopic scale, [Theorem 9.39](#); bounded; either compactifies or assigns the descendant to $\mathcal{C}_{\text{aff}}$ by [Theorem 9.49](#); no pointwise derivative $\partial_s c_n$ is used unless that compactness has already been obtained.
- $\beta(s) \cdot y + \alpha(s)$: affine parasitic pressure paired with a spatially homogeneous velocity mode; quotient notation introduced in [Theorem 8.16](#) and used only in the gauged formulation of the affine/parasitic stratum.
- $L_n(x, s) = b_n(s) \cdot x + a_n(s)$: scaled affine pressure at the minimal mesoscopic scale, [Theorem 9.46](#); affine in x , measurable in s ; discarded in the non-affine quotient and assigned to $\mathcal{C}_{\text{aff}}$ when it does not vanish.
- $a_{n,K}(t)$: local Seregin pressure time gauge on a compact cylinder K , used by the local pressure-gauge compactness theorem inherited from the setup paper and recorded in [Theorem 1.18](#).

Table 4: Glossary of residual symbols.

Symbol	Meaning
$\mathcal{E}_r$	Scale-invariant parabolic CKN concentration.
μ_n	Local defect measure for concentration-set and compactness arguments.
$\mathcal{A}_n(z, r)$	Spatial/log-tail averaged affine-quotient oscillation.
$\mathcal{C}_{\text{ax}}$	Axisymmetric bounded-circulation residual class.
$\mathcal{C}_{\text{rot}}$	Rotational relative-equilibrium residual class.
$\mathcal{C}_{\text{stat}}^{\text{ret}}$	Stationary-hull residual class with retained local concentration.
$\mathcal{C}_{\text{aff}}$	Affine/parasitic pressure residual quotient.
$\mathcal{C}_{\text{log diff}}$	Log-diffuse critical-tail residual alternative.
$\mathcal{C}_{\text{Young}}$	Young/variance critical-tail residual alternative.
$\mathcal{C}_{\text{gen}}$	Generic terminal residual class.
Retained descendant	Descendant with inherited positive unit CKN concentration.
Normalized support profile	Normalized defect support profile, not necessarily retained at the original threshold.

4. EXCLUSION OF THE AXISYMMETRIC BOUNDED-CIRCULATION RESIDUAL CLASS

In this section we prove the rigidity statement needed for the first refined residual class. The argument uses the scale-invariant angular circulation in centered variables. The essential point is that the centered Type I equation contains the Ornstein–Uhlenbeck drift. Thus the angular-circulation equation has an absorbing boundary at the symmetry axis together with a confining drift at spatial infinity. This gives a scalar Liouville theorem for bounded ancient solutions of the circulation equation. Related axisymmetric Liouville theorems for Navier–Stokes, in different settings, are proved in [\[6, 5\]](#).

Throughout the section we write cylindrical coordinates as (ρ, θ, Z) , and an axisymmetric vector field as

$$V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The centered Navier–Stokes equation is

$$(4.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For an axisymmetric solution define the centered angular circulation

$$(4.2) \quad \Gamma(\rho, Z, \tau) := \rho V_\theta(\rho, Z, \tau).$$

This is the same scale-invariant quantity as the physical circulation rU_θ . If the physical representative is

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)),$$

then

$$rU_\theta(r, z, t) = \rho V_\theta(\rho, Z, \tau), \quad r = \sqrt{-t} \rho, \quad z = \sqrt{-t} Z.$$

Lemma 4.1 (Centered circulation equation). *Let (V, P) be a smooth bounded axisymmetric solution of (4.1). Then $\Gamma = \rho V_\theta$ satisfies*

$$(4.3) \quad \partial_\tau \Gamma = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma - \left(V_\rho + \frac{\rho}{2}\right) \partial_\rho \Gamma - \left(V_Z + \frac{Z}{2}\right) \partial_Z \Gamma$$

on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Moreover

$$(4.4) \quad \Gamma(0, Z, \tau) = 0 \quad \text{for all } (Z, \tau) \in \mathbb{R} \times \mathbb{R}.$$

Proof. The angular component of (4.1) is

$$\partial_\tau V_\theta - \left(\Delta - \frac{1}{\rho^2}\right) V_\theta + \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta + V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} = 0.$$

Multiplying by ρ , using

$$\begin{aligned} \rho \left(\Delta - \frac{1}{\rho^2}\right) V_\theta &= \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma, \\ \rho \left[\frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta\right] &= \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) \Gamma, \end{aligned}$$

and

$$\rho \left(V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho}\right) = V_\rho \partial_\rho \Gamma + V_Z \partial_Z \Gamma,$$

gives (4.3). Finally, smoothness of a bounded axisymmetric vector field at the axis implies $V_\theta(\rho, Z, \tau) = O(\rho)$ as $\rho \downarrow 0$, locally uniformly in (Z, τ) . Hence $\Gamma = \rho V_\theta = O(\rho^2)$, which gives (4.4). $\square$

We next record the scalar Liouville theorem used below. It is stated in a form that isolates precisely the PDE mechanism: bounded drift perturbations of the centered circulation operator cannot sustain a bounded ancient solution which vanishes on the symmetry axis.

Lemma 4.2 (Axis-absorption Liouville theorem). *Let $B = (B_\rho, B_Z)$ be a smooth vector field on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$ satisfying*

$$\|B\|_{L^\infty} < \infty.$$

Let $G \in C^{2,1}((0, \infty) \times \mathbb{R} \times \mathbb{R}) \cap L^\infty$ solve

$$(4.5) \quad \partial_\tau G = \partial_\rho^2 G - \frac{1}{\rho} \partial_\rho G + \partial_Z^2 G - \left(B_\rho + \frac{\rho}{2}\right) \partial_\rho G - \left(B_Z + \frac{Z}{2}\right) \partial_Z G$$

for all $\tau \in \mathbb{R}$, and assume

$$(4.6) \quad \lim_{\rho \downarrow 0} G(\rho, Z, \tau) = 0 \quad \text{locally uniformly in } (Z, \tau).$$

Then $G \equiv 0$.

Proof. Set $M_B = \|B\|_{L^\infty}$. For $R > 1$ define

$$(4.7) \quad h_R(\rho) := \frac{\int_0^{\min\{\rho, R\}} s \exp\left(\frac{s^2}{4} - M_B s\right) ds}{\int_0^R s \exp\left(\frac{s^2}{4} - M_B s\right) ds}.$$

Then $0 \leq h_R \leq 1$, $h_R(0) = 0$, $h_R(R) = 1$, and, on $0 < \rho < R$,

$$(4.8) \quad h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0, \quad h_R' > 0.$$

Let

$$\mathcal{L}_\tau f := \partial_\rho^2 f - \frac{1}{\rho} \partial_\rho f + \partial_Z^2 f - \left(B_\rho + \frac{\rho}{2}\right) \partial_\rho f - \left(B_Z + \frac{Z}{2}\right) \partial_Z f.$$

Since $B_\rho \geq -M_B$, (4.8) gives

$$(4.9) \quad \mathcal{L}_\tau h_R = h_R'' - \left(\frac{1}{\rho} + \frac{\rho}{2} + B_\rho\right) h_R' \leq h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0.$$

Thus $(\partial_\tau - \mathcal{L}_\tau)h_R \geq 0$.

We now compare G to this barrier on the strip $\Omega_R = (0, R) \times \mathbb{R}$. Let $M_G = \|G\|_{L^\infty}$. We first record the Dirichlet strip decay estimate used by the comparison argument. If $q_{s,R}$ solves

$$\partial_\tau q_{s,R} = \mathcal{L}_\tau q_{s,R}, \quad q_{s,R} = 0 \text{ on } \rho = 0, R, \quad q_{s,R}(\rho, Z, s) = 1,$$

then, for every compact $K \Subset \Omega_R$ and every fixed t ,

$$(4.10) \quad \lim_{s \rightarrow -\infty} \sup_{(\rho, Z) \in K} q_{s,R}(\rho, Z, t) = 0.$$

To prove (4.10), fix $K \Subset \Omega_R$. Choose $\delta > 0$, $Z_K > 0$, and $Z_1 > Z_K + 1$ so that

$$K \subset [\delta, R - \delta] \times [-Z_K, Z_K] \subset (0, R) \times (-Z_1, Z_1).$$

On the finite cylinder

$$\Omega_{R,Z_1} := (0, R) \times (-Z_1, Z_1)$$

with Dirichlet data on the full parabolic boundary, the operator $\mathcal{L}_\tau$ is uniformly parabolic on $[\delta/2, R - \delta/2] \times [-Z_1, Z_1]$, has bounded first-order coefficients there, and has absorbing boundary at $\rho = 0, R$. The boundary parabolic Harnack inequality and the strong maximum principle give a time $T_R > 0$ and a number $\vartheta_R \in (0, 1)$, depending only on R, M_B, δ, Z_K and not on Z_1 , such that every nonnegative solution with initial data bounded by 1 and zero lateral data satisfies

$$0 \leq q(\cdot, s + T_R) \leq 1 - \vartheta_R \quad \text{on } [\delta, R - \delta] \times [-Z_K, Z_K].$$

Applying the same estimate after each time translation and using the maximum principle between the compact set and the boundary gives

$$\sup_K q_{s,R}(\cdot, s + mT_R) \leq (1 - \vartheta_R)^m, \quad m = 1, 2, \dots$$

For a fixed $t > s$, choose m with $s + mT_R \leq t < s + (m+1)T_R$. The maximum principle and interior Harnack estimate on the final interval $[s + mT_R, t]$ preserve the preceding bound up to a constant depending only on R, M_B, δ, Z_K, T_R . Letting $s \rightarrow -\infty$ forces $m \rightarrow \infty$, proving decay on K for the truncated problem. The solutions on Ω_{R,Z_1} increase monotonically to the Dirichlet strip solution as $Z_1 \rightarrow \infty$; passing to this limit gives (4.10).

Fix $R > 1$, $s < t$, and set

$$W_R^+ := \frac{G}{M_G} - h_R, \quad W_R^- := -\frac{G}{M_G} - h_R,$$

with the convention that the conclusion is immediate if $M_G = 0$. By (4.5) and (4.9),

$$(\partial_\tau - \mathcal{L}_\tau)W_R^\pm \leq 0 \quad \text{in } \Omega_R \times (s, t).$$

On $\rho = 0$, (4.6) and $h_R(0) = 0$ give $W_R^\pm \leq 0$. On $\rho = R$, $h_R(R) = 1$ and $|G| \leq M_G$ give again $W_R^\pm \leq 0$. The parabolic comparison principle with the Dirichlet strip kernel therefore yields, for $\tau = t$,

$$(W_R^\pm)_+(\rho, Z, t) \leq q_{s,R}(\rho, Z, t) \quad ((\rho, Z) \in \Omega_R).$$

Letting $s \rightarrow -\infty$ and using (4.10) gives $W_R^\pm \leq 0$ on compact subsets of Ω_R . Since the compact subset is arbitrary,

$$(4.11) \quad |G(\rho, Z, t)| \leq M_G h_R(\rho) \quad (0 < \rho < R, Z \in \mathbb{R}, t \in \mathbb{R}).$$

Finally, for each fixed $\rho < \infty$, the denominator in (4.7) tends to infinity super-exponentially as $R \rightarrow \infty$, while the numerator remains fixed. Hence $h_R(\rho) \rightarrow 0$. Letting $R \rightarrow \infty$ in (4.11) gives $G(\rho, Z, t) = 0$ for every $\rho > 0$, $Z \in \mathbb{R}$, and $t \in \mathbb{R}$. The boundary value at $\rho = 0$ is already zero by assumption. Therefore $G \equiv 0$. $\square$

Theorem 4.3 (Exclusion of the axisymmetric bounded-circulation residual class). *Let $\mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$ be the axisymmetric bounded-circulation residual class in the refined residual decomposition. Then*

$$(4.12) \quad \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Assume, for contradiction, that $V \in \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$. By definition of $\mathcal{C}_{\text{ax}}$, the associated physical ancient representative is axisymmetric and has bounded angular circulation. Equivalently, in centered variables,

$$\Gamma(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau)$$

is bounded on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Since V is a Seregin ancient limit, it is smooth and bounded in centered variables. Hence $B = (V_\rho, V_Z)$ is bounded.

By Theorem 4.1, Γ satisfies (4.3) and vanishes at the axis. Applying Theorem 4.2 with $G = \Gamma$ and $B = (V_\rho, V_Z)$ gives

$$\Gamma \equiv 0.$$

Therefore $V_\theta \equiv 0$ for $\rho > 0$, and by smoothness also on the axis. Thus the associated ancient solution is axisymmetric without swirl. But the axisymmetric no-swirl class is one of the structured alternatives removed before the residual class is defined. This contradicts $V \in \mathcal{R}(\mathcal{S}; I, J)$, and therefore no such V exists. $\square$

Remark 4.4 (Why this does not assert the full bounded-swirl Liouville theorem). The proof uses the centered Type I equation, specifically the Ornstein–Uhlenbeck drift terms $-\rho\partial_\rho/2$ and $-Z\partial_Z/2$ in the scalar equation for Γ . It is therefore a Type I Seregin limit argument. It does not prove that every bounded ancient axisymmetric solution of the unrenormalized Navier–Stokes equations with bounded physical circulation is swirl-free. The latter lacks the confining centered drift and remains a different shell-transport problem.

5. THE ROTATIONAL RELATIVE-EQUILIBRIUM RESIDUAL CLASS

We next isolate the second refined residual class. This class consists of centered Type I ancient limits whose time dependence is a rigid rotation of a fixed spatial profile. Passing to the co-rotating frame converts the centered Navier–Stokes equation into a stationary Leray profile equation with a Coriolis transport term. A whole-space Coriolis–Leray Liouville theorem for all bounded profiles is not used here. Instead, the Coriolis term is localized as an annular flux and is then treated by the local concentration analysis proved in Sections 9 and 10.

Throughout this section the centered equation is

$$(5.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For $\Omega \in \mathfrak{so}(3)$, set

$$(5.2) \quad Q(\tau) := \exp(\tau\Omega) \in SO(3).$$

Definition 5.1 (Rotational relative-equilibrium profile). A bounded smooth ancient solution (V, P) of (5.1) is a rotational relative equilibrium if there exist $\Omega \in \mathfrak{so}(3)$, a bounded smooth divergence-free vector field W , and a pressure profile Π , such that

$$(5.3) \quad V(y, \tau) = Q(\tau)W(Q(\tau)^T y), \quad P(y, \tau) = \Pi(Q(\tau)^T y)$$

up to the usual time-dependent pressure gauge. The class $\mathcal{C}_{\text{rot}}$ consists of such residual Seregin profiles, after the axisymmetric bounded-circulation class and the previously closed alternatives in the setup paper have been removed.

Lemma 5.2 (Co-rotating stationary equation). *Let (V, P) be a rotational relative equilibrium. Then the co-rotating profile (W, Π) satisfies*

$$(5.4) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Equivalently,

$$(5.5) \quad \mathbb{P} \left[-\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W \right] = 0,$$

where $\mathbb{P}$ is the whole-space Leray projector.

Proof. Let $z = Q(\tau)^T y$. Since $Q'(\tau) = \Omega Q(\tau)$ and $Q(\tau)^T \Omega Q(\tau) = \Omega$, one has

$$\partial_\tau z = -\Omega z.$$

Therefore

$$\partial_\tau V(y, \tau) = Q(\tau) \left(\Omega W(z) - (\Omega z) \cdot \nabla W(z) \right).$$

The Laplacian, divergence, convection term, and pressure gradient are invariant under the orthogonal change of variables. Substitution into (5.1), followed by multiplication by $Q(\tau)^T$, gives (5.4). Applying the Leray projector gives (5.5). $\square$

5.1. The Bernoulli identity and the local Coriolis flux. Write

$$(5.6) \quad F(y) := \frac{1}{2}y - \Omega y, \quad A(y) := W(y) + F(y), \quad \omega := \nabla \times W.$$

Since Ω is skew-symmetric, there is a unique vector $\varpi \in \mathbb{R}^3$ such that

$$(5.7) \quad \Omega y = \varpi \times y.$$

Then

$$(5.8) \quad \nabla \times F = -2\varpi.$$

Lemma 5.3 (Coriolis Bernoulli identity). *Let (W, Π) be a smooth solution of (5.4). Define*

$$(5.9) \quad B := \frac{1}{2}|W|^2 + \Pi + F \cdot W.$$

Then

$$(5.10) \quad \nabla B + \nabla \times \omega - A \times \omega = 0,$$

and consequently

$$(5.11) \quad -\Delta B + A \cdot \nabla B = -|\omega|^2 - \omega \cdot (\nabla \times F) = -|\omega|^2 + 2\varpi \cdot \omega.$$

Proof. The vector identity

$$(W \cdot \nabla)W = \nabla \left(\frac{1}{2}|W|^2 \right) - W \times \omega$$

and the identity

$$(F \cdot \nabla)W + (\nabla F)^T W = \nabla(F \cdot W) - F \times \omega$$

apply because $(\nabla F)^T W = \frac{1}{2}W + \Omega W$. Thus (5.4) can be rewritten as

$$-\Delta W + \nabla \left(\frac{1}{2}|W|^2 + \Pi + F \cdot W \right) - (W + F) \times \omega = 0.$$

Since $\nabla \cdot W = 0$, one has $-\Delta W = \nabla \times \omega$, which gives (5.10).

Taking divergence in (5.10) gives

$$\Delta B = \nabla \cdot (A \times \omega),$$

because the divergence of a curl vanishes. Using

$$\nabla \cdot (A \times \omega) = \omega \cdot (\nabla \times A) - A \cdot (\nabla \times \omega), \quad \nabla \times A = \omega + \nabla \times F,$$

we obtain

$$\Delta B = |\omega|^2 + \omega \cdot (\nabla \times F) - A \cdot (\nabla \times \omega).$$

On the other hand, by (5.10),

$$A \cdot \nabla B = A \cdot (A \times \omega) - A \cdot (\nabla \times \omega) = -A \cdot (\nabla \times \omega).$$

Combining the last two identities gives (5.11). Finally, (5.8) follows from (5.7). $\square$

Lemma 5.4 (Localized Coriolis flux identity). *Let (W, Π) be as in Theorem 5.3, and let $\chi \in C_c^\infty(\mathbb{R}^3)$. Then*

$$(5.12) \quad \int_{\mathbb{R}^3} \chi (-\Delta B + A \cdot \nabla B + |\omega|^2) dy = -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy.$$

In particular the Coriolis contribution is supported only where the cutoff varies.

Proof. By (5.11),

$$-\Delta B + A \cdot \nabla B + |\omega|^2 = 2\varpi \cdot \omega.$$

Because ϖ is constant and $\omega = \nabla \times W$,

$$\varpi \cdot \omega = \nabla \cdot (W \times \varpi).$$

Multiplying by χ , integrating over $\mathbb{R}^3$, and integrating by parts gives

$$2 \int \chi \varpi \cdot \omega = 2 \int \chi \nabla \cdot (W \times \varpi) = -2 \int (W \times \varpi) \cdot \nabla \chi,$$

which proves (5.12). $\square$

Lemma 5.5 (Non-small Coriolis flux creates a retained local concentration). *Let $\chi \in C_c^\infty(\mathbb{R}^3)$, let $A_\chi := \text{supp } \nabla \chi$, and put*

$$C_\chi := \int_{A_\chi} |\nabla \chi|^{3/2} dy.$$

Fix $T \in (0, 1]$, and let

$$\mathcal{T}_{\chi, T} := \{(y, \tau) : -T < \tau < 0, Q(\tau)^T y \in A_\chi\}$$

be the swept annular tube in the original centered variables. Cover $\mathcal{T}_{\chi,T}$ by finitely many unit terminal cylinders

$$\mathcal{T}_{\chi,T} \subset \bigcup_{m=1}^{N_{\chi,T}} Q_1(z_m).$$

The number $N_{\chi,T}$ depends only on the compact annulus A_χ , the time length T , and Ω , and the cover is chosen with bounded overlap inside the normalized terminal domain under consideration. Let $\varepsilon_{\text{sd}} > 0$ be the local concentration threshold used in [Theorem 9.3](#). If $\varpi \neq 0$ and

$$(5.13) \quad \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right| \geq 2|\varpi| C_\chi^{2/3} \left(\frac{N_{\chi,T} \varepsilon_{\text{sd}}}{T} \right)^{1/3},$$

then at least one cylinder $Q_1(z_m)$ is ε_{sd} -concentrating for the original centered solution V .

Proof. Write

$$\delta_\chi := \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy \right|.$$

Hölder's inequality gives

$$\begin{aligned} \delta_\chi &\leq 2|\varpi| \int_{A_\chi} |W| |\nabla \chi| \, dy \\ &\leq 2|\varpi| \left(\int_{A_\chi} |W|^3 \, dy \right)^{1/3} C_\chi^{2/3}. \end{aligned}$$

Thus (5.13) implies

$$\int_{A_\chi} |W|^3 \, dy \geq \frac{N_{\chi,T} \varepsilon_{\text{sd}}}{T}.$$

Since $V(y, \tau) = Q(\tau)W(Q(\tau)^T y)$ and $Q(\tau)$ is an isometry,

$$\begin{aligned} \iint_{\mathcal{T}_{\chi,T}} |V(y, \tau)|^3 \, dy \, d\tau &= \int_{-T}^0 \int_{Q(\tau)A_\chi} |W(Q(\tau)^T y)|^3 \, dy \, d\tau \\ &= T \int_{A_\chi} |W(z)|^3 \, dz \geq N_{\chi,T} \varepsilon_{\text{sd}}. \end{aligned}$$

Because the tube is covered by $N_{\chi,T}$ unit terminal cylinders with bounded overlap, one of them satisfies

$$\iint_{Q_1(z_m)} |V|^3 \, dy \, d\tau \geq \varepsilon_{\text{sd}}.$$

The pressure contribution to the local unit defect measure is nonnegative after the allowed pressure gauge. Hence

$$\mu(Q_1(z_m)) \geq \varepsilon_{\text{sd}},$$

so the corresponding recentering sequence is ε_{sd} -concentrating in the sense of [Theorem 9.3](#). Since the estimate is on unit cylinders, the corrected normalization $\mathcal{E}_1$ agrees with unit atlas mass up to the fixed pressure atlas versus quotient gauge convention in [Theorem 3.1](#). $\square$

Lemma 5.6 (Local Coriolis-flux dichotomy). *Fix $\chi \in C_c^\infty(\mathbb{R}^3)$ and $T \in (0, 1]$. Along every terminal extraction of a rotational relative equilibrium, exactly one of the following local alternatives occurs:*

- (i) $\varpi \neq 0$, the flux across $A_\chi = \text{supp } \nabla \chi$ satisfies the threshold (5.13), and the swept annular tube contains a unit-scale concentration;

(ii) either $\varpi = 0$, or the threshold (5.13) fails; then the flux is a bounded compactly supported functional of W on A_χ , passes to terminal limits by local L^3 convergence, and creates no terminal alternative other than the single-profile, finite separated-profile, exterior-escape, diffuse-exterior-concentration, local-vanishing, or local-compactness-failure alternatives of Theorem 9.11.

Proof. Alternative (i) is precisely Theorem 5.5. If $\varpi = 0$, then the flux vanishes identically. If $\varpi \neq 0$ and (5.13) fails, the flux is bounded by the right hand side of that threshold. In both cases the functional

$$W \mapsto -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi \, dy$$

depends only on W on the compact annular set A_χ . Under the local Seregin compactness supplied by the admissible setup framework, velocities converge strongly in L^3_{loc} ; since $\nabla \chi \in L^{3/2}$ and ϖ is fixed, the functional is continuous along every terminal subsequence:

$$\int (W_n \times \varpi) \cdot \nabla \chi \longrightarrow \int (W_\infty \times \varpi) \cdot \nabla \chi.$$

Thus a subthreshold flux is retained only as a local term in the limiting localized Bernoulli identity (5.12). It is not a new location of local velocity mass. Every actual location of compact velocity L^3 -mass is still one of the terminal local concentration alternatives listed in Theorem 9.11. $\square$

Remark 5.7 (Why this is not a global Liouville argument). For $\Omega = 0$, the Coriolis flux in Theorem 5.4 vanishes. For $\Omega \neq 0$, the same lemma does not give a sign-definite maximum principle for B . It gives only the local dichotomy in Theorem 5.6: either a chosen annulus already contains a recentered local concentration, or the annular flux is a subthreshold compact functional which passes to the local terminal limit. No tail decay, whole-space L^p bound, Gaussian energy estimate, or global pressure normalization is used.

5.2. Reduction of the rotational relative-equilibrium class.

Proposition 5.8 (Rotational residual reduction). *Every rotational relative-equilibrium residual either belongs to a previously closed lower class or produces a terminal concentration profile realized from the original blow-up sequence satisfying the hypotheses of the terminal stratification theorem.*

Proof. Let $V \in C_{\text{rot}}(\mathcal{S}; I, J)$. By Theorem 5.1 and Theorem 5.2, there are $\Omega \in \mathfrak{so}(3)$ and a bounded smooth divergence-free profile (W, Π) satisfying the Coriolis–Leray equation (5.4). The centered representative V is a normalized Seregin ancient limit, so by Theorem 1.5 it carries a fixed positive compact velocity L^3 -mass.

If $\Omega = 0$, then V is stationary in the centered frame. It is reduced to the stationary setup class if it lies in the already closed stationary alternative, and otherwise to the stationary-hull residual branch handled by Theorem 6.12. We therefore assume $\Omega \neq 0$.

Choose a countable family of smooth cutoffs which is cofinal among compact annuli in the terminal variables. For each cutoff and each rational $T \in (0, 1]$, apply the local dichotomy Theorem 5.6. A superthreshold Coriolis flux produces a unit-scale concentration and is therefore already part of the local concentration decomposition. A subthreshold Coriolis flux is a compactly supported functional of W , continuous under the local convergence supplied by the admissible setup compactness framework, and hence remains inside the localized limiting equation; it carries no independent CKN mass and creates no additional terminal alternative.

We now apply the terminal local concentration decomposition to the inherited compact density. Loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration cannot be the sole locations of this density by Theorem 9.14. By the preceding paragraph, the

Coriolis flux has already led either to a local concentration or to a compact local term in the selected terminal equation. Thus, after local concentration extraction, the retained density is located either in a finite separated family of retained concentration profiles or in a single retained concentration terminal profile, together with any residual diffuse, noncompact, or critical-tail remainder selected by the terminal extraction procedure. This is exactly a terminal input realized from the original blow-up sequence for the terminal stratification theorem; no R2 closure is used in this reduction step. $\square$

Proposition 5.9 (Already-closed integrable subcase). *Let V be a rotational relative equilibrium with co-rotating profile W . If*

$$(5.14) \quad W \in L^3(\mathbb{R}^3),$$

then $V \equiv 0$. More generally, the same conclusion holds under any condition which places the physical representative in $L_t^\infty L_x^3$ on its ancient time interval.

Proof. Rotations and the Navier–Stokes critical scaling preserve the L^3 norm. Thus $W \in L^3$ implies

$$\sup_{\tau \in \mathbb{R}} \|V(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Equivalently, the physical representative lies in the critical class $L_t^\infty L_x^3$. The endpoint L^3 class is one of the already closed alternatives used in the setup paper, by the Escauriaza–Seregin–Šverák endpoint regularity theorem [3] and the stationary Nečas–Růžička–Šverák Liouville theorem [7]. Therefore $V \equiv 0$. A zero profile has zero local velocity mass, so no such profile can belong to the residual nonvanishing class. $\square$

6. THE DEGENERATE STATIONARY-HULL RESIDUAL CLASS

The third refined residual class consists of Type I ancient limits whose time-translation hull contains a stationary profile with retained local concentration lying on a non-isolated family of stationary centered profiles. The role of the stationary-family condition is to separate genuinely degenerate stationary profile families from isolated stationary profiles and from the symmetry-generated classes already removed in $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$.

The retained-concentration condition is essential. The mere presence of a stationary-family element in the hull of a residual ancient solution is insufficient. A stationary element may occur as a limit without retained local concentration and therefore need not retain the local CKN concentration inherited from the original singular point. Such a hull element is not placed in a class that is to be closed by a stationary-profile exclusion. Therefore the class below is defined using *retained-concentration occurrence*: the stationary-family profile must itself occur as a Seregin limit carrying a positive local CKN concentration.

Throughout this section the centered equation is

$$(6.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

Its projected form is

$$(6.2) \quad \partial_\tau V = \mathcal{F}(V), \quad \mathcal{F}(W) := \mathbb{P} \left[\Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W \right],$$

where $\mathbb{P}$ is the whole-space Leray projector.

6.1. Stationary families and retained-concentration hull limits.

Definition 6.1 (Admissible stationary phase space). An admissible stationary phase space is a Banach space $\mathfrak{B}$ of solenoidal vector fields on $\mathbb{R}^3$ such that:

- (i) $\mathfrak{B} \hookrightarrow C_{\text{loc}}^{2,\alpha}(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$ continuously for some $\alpha \in (0, 1)$;
- (ii) the map $\mathcal{F}$ in (6.2) is well-defined on an open subset $\mathcal{U} \subset \mathfrak{B}$ and maps $\mathcal{U}$ into a Banach space $\mathfrak{Y}$;
- (iii) $\mathcal{F} : \mathcal{U} \rightarrow \mathfrak{Y}$ is C^1 .

Definition 6.2 (Nontrivial stationary family). Let $W \in \mathcal{U} \subset \mathfrak{B}$ satisfy $\mathcal{F}(W) = 0$. We say that W lies on an admissible nontrivial stationary family if there exist $\varepsilon > 0$ and a C^1 curve

$$a \mapsto W_a \in \mathcal{U}, \quad |a| < \varepsilon,$$

such that

$$(6.3) \quad W_0 = W, \quad \mathcal{F}(W_a) = 0 \quad (|a| < \varepsilon),$$

and such that the family is not locally generated only by the action of the rigid rotation group. Equivalently, after possibly reparametrizing the curve, one may assume that

$$(6.4) \quad \dot{W}_0 \notin \{\Omega W - (\Omega y) \cdot \nabla W : \Omega \in \mathfrak{so}(3)\}$$

whenever the tangent $\dot{W}_0$ exists and is nonzero.

Definition 6.3 (Retained local CKN concentration). Let (W, Π) be a stationary centered profile. We say that (W, Π) is ε_* -concentrating if it has retained concentration in the normalized unit sense: there is a recentering/rescaling realized from the original blow-up sequence $\mathcal{T}$ such that

$$(6.5) \quad \mathcal{E}_1(\mathcal{T}(W, \Pi); 0) \geq \varepsilon_*.$$

Equivalently, before the final unit normalization one may require $\mathcal{E}_r(W, \Pi; z_0) \geq \varepsilon_*$ for some parabolic cylinder $Q_r(z_0)$. Unnormalized large-cylinder mass is not a retained concentration criterion.

Definition 6.4 (Degenerate stationary-hull residual class with retained concentration). Fix an admissible stationary phase space $\mathfrak{B}$. The degenerate stationary-hull residual class with retained local concentration

$$\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$$

consists of those $V \in \mathcal{R}(\mathcal{S}; I, J)$, after the ordered removal of $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$, for which there exists a sequence $s_k \in \mathbb{R}$ and a stationary profile (W, Π) such that

$$(6.6) \quad (V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \longrightarrow (W, \Pi)$$

locally smoothly for the velocity and locally in $L^{3/2}$ for the pressure modulo admissible gauges, and such that:

- (i) $W \in \mathfrak{B}$ and $\mathcal{F}(W) = 0$;
- (ii) W lies on an admissible nontrivial stationary family in the sense of [Theorem 6.2](#);
- (iii) (W, Π) is ε_* -concentrating in the sense of [Theorem 6.3](#);
- (iv) (W, Π) is not in any previously closed stationary, structured, axisymmetric bounded-circulation, or rotational relative-equilibrium class.

In the ordered residual decomposition one keeps this class of stationary-hull limits with retained concentration as the degenerate stationary-hull alternative. The hull cases without retained local concentration are assigned to the generic terminal residual class, where they are handled by the terminal local concentration decomposition rather than by a stationary-profile exclusion.

Remark 6.5 (Why retained concentration is necessary). The condition (6.5) prevents the following logical error. A nonstationary ancient solution may have a stationary profile in its hull with zero local CKN mass on the relevant compact cylinders. Excluding that stationary profile would not exclude the original ancient solution. The stationary-profile exclusion used below applies only to stationary profiles that themselves occur as nontrivial Seregin limits with positive local concentration.

6.2. Hull limits are again Seregin limits. The next lemma is the basic compatibility between time-translation hulls in centered variables and rescaling sequences in physical variables. It is the reason that a stationary profile in the hull which retains local concentration may be treated as a blow-up profile of the original solution, not merely as a limit inside the already extracted ancient solution.

Lemma 6.6 (Scale reselection for centered hull limits). *Let (V, P) be a centered Type I Seregin limit obtained from a finite-energy suitable weak solution (u, p) at a singular point $z_* = (x_*, t_*)$. Suppose that, for some sequence $s_k \in \mathbb{R}$,*

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \rightarrow (W, \Pi)$$

locally in the Seregin topology. Then, after passing to a diagonal subsequence and reselecting the physical blow-up scales, (W, Π) is itself a centered Type I Seregin limit of (u, p) at z_ . If the convergence to (W, Π) retains a positive local CKN lower bound, then (W, Π) is a nontrivial Seregin limit with the same type of inherited nonvanishing condition.*

Proof. Let $\lambda_n \downarrow 0$ be a sequence of scales producing (V, P) . In centered variables, a time translation by s is exactly a change of the physical blow-up scale by a factor depending only on s . With the convention used in (6.1), write this adjusted scale as

$$(6.7) \quad \lambda_{n,s} := e^{-s/2} \lambda_n.$$

Indeed, writing $t = -e^{-\tau}$, a shift $\tau \mapsto \tau + s$ corresponds to multiplying the physical length scale by $e^{-s/2}$, so that

$$V_{\lambda e^{-s/2}}(y, \tau) = V_\lambda(y, \tau + s)$$

with the corresponding pressure gauge. Thus the rescaled fields built from (u, p) at scale $\lambda_{n,s}$ are the same as the fields at scale λ_n , translated by s in centered time.

Fix an exhaustion of compact cylinders $K_m \subset \mathbb{R}^3 \times \mathbb{R}$. For each k , first require

$$\lambda_{n(k)} \leq k^{-1} e^{s_k/2}$$

when $s_k < 0$, and otherwise require simply $\lambda_{n(k)} \leq k^{-1}$. Then $\lambda_{n(k), s_k} = e^{-s_k/2} \lambda_{n(k)} \rightarrow 0$. Increasing $n(k)$ if needed, arrange that the rescaled fields at scale $\lambda_{n(k)}$, shifted by s_k , are within $1/k$ of $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ on K_k in the local Seregin topology. This is possible because the original sequence converges locally to (V, P) .

By the assumed hull convergence, the shifted limit $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ converges to (W, Π) on each fixed compact cylinder. The diagonal sequence of physical rescalings with scales $\lambda_{n(k), s_k}$ therefore converges locally to (W, Π) . Hence (W, Π) is again a centered Type I Seregin limit at z_* . The assertion about nonvanishing follows from lower semicontinuity and local convergence of the CKN quantities after the prescribed pressure gauges. $\square$

6.3. Elementary consequences of stationary-family degeneracy. The stationary-family condition has two immediate PDE consequences. First, a differentiable nontrivial family produces a nontrivial element of the kernel of the linearized stationary Leray operator. Second, if an ancient trajectory is exactly constrained to a stationary family, then it is stationary.

Lemma 6.7 (Stationary family implies linearized degeneracy). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family satisfying (6.3). Then*

$$(6.8) \quad D\mathcal{F}(W_0)\dot{W}_0 = 0.$$

In particular, if $\dot{W}_0 \neq 0$ modulo infinitesimal rotations, the linearized stationary Leray operator has a genuine non-symmetry kernel element.

Proof. Differentiate $\mathcal{F}(W_a) = 0$ at $a = 0$. Since $\mathcal{F}$ is C^1 on $\mathfrak{B}$, this gives (6.8). The final statement is exactly the nontriviality condition in Theorem 6.2. $\square$

Lemma 6.8 (Trajectories constrained to a stationary family are stationary). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family with $\dot{W}_a \neq 0$ on an interval. Suppose that a smooth solution of (6.2) has the form*

$$(6.9) \quad V(\tau) = W_{a(\tau)}$$

for a C^1 function $a(\tau)$. Then a is constant, and hence V is stationary.

Proof. Since $\mathcal{F}(W_a) = 0$ for every a , equation (6.2) gives

$$a'(\tau)\dot{W}_{a(\tau)} = \partial_\tau V(\tau) = \mathcal{F}(W_{a(\tau)}) = 0.$$

Because $\dot{W}_{a(\tau)} \neq 0$, we obtain $a'(\tau) = 0$. Thus a is constant on the interval. $\square$

Corollary 6.9 (Exact stationary-family subcase). *If an element of the residual class is itself an exact trajectory on an admissible stationary family, then it is a stationary centered profile. Hence it is excluded whenever it satisfies the stationary L^3 Liouville criterion, or more generally whenever the stationary profile belongs to one of the previously closed stationary or structured classes.*

Proof. Let $V(\tau) = W_{a(\tau)}$ be such an exact stationary-family trajectory. By Theorem 6.8, $a'(\tau) = 0$ on every interval on which the stationary-family parametrization is nondegenerate. Hence a is constant and V is a single stationary centered profile. If that stationary profile satisfies the stationary L^3 Liouville criterion or lies in one of the previously closed structured or ordered lower residual classes, the corresponding prior exclusion removes it from the residual class. $\square$

6.4. Stationary-hull profiles which occur as Seregin limits. The preceding lemma removes the possible internal dynamics along a stationary family: exact stationary-family motion is stationary by Theorem 6.8. The remaining question is attainability by blow-up: whether a degenerate stationary-family profile can occur as a Seregin limit of a finite-energy suitable weak solution while retaining local concentration. The following statement gives the precise Seregin-limit occurrence exclusion required to close the degenerate stationary-hull class with retained local concentration. Its proof is deferred until the terminal stratification theorem has been established.

Theorem 6.10 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. Let (W, Π) be a stationary centered profile satisfying:*

- (i) $W \in \mathfrak{B}$, $\mathcal{F}(W) = 0$, and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) is ε_* -concentrating;
- (iv) (W, Π) is not in the small, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, or rotational relative-equilibrium classes already removed earlier in the stratification.

Then no such (W, Π) exists.

Proof deferred. The proof is given after the terminal stratification theorem in [Theorem 10.27](#). The present statement records the exact R3 occurrence input that must be proved; no R3 exclusion is used before the terminal concentration theorem is available. $\square$

Remark 6.11 (Interpretation of the Seregin limit occurrence theorem). [Theorem 6.10](#) excludes occurrence as a Type I Seregin limit. It does not assert that degenerate stationary profiles cannot exist as elliptic solutions of the stationary centered equation. It asserts only that such profiles cannot occur as Type I blow-up limits with retained local concentration after all previously closed classes have been removed. This is the formulation required for the residual problem: the elliptic family may exist, but it cannot be reached by the local blow-up procedure coming from a finite-energy suitable weak solution.

6.5. Reduction of the degenerate stationary-hull concentration class.

Proposition 6.12 (R3 stationary-hull reduction). *Every retained stationary-hull residual profile is either a setup-closed stationary, tight, or structured profile, or it produces a stationary terminal concentration profile realized from the original blow-up sequence and satisfying the hypotheses of the terminal stratification theorem.*

Proof. Let $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By [Theorem 6.4](#), there is a stationary profile (W, Π) in the centered time-translation hull of (V, P) , and (W, Π) is ε_* -concentrating. The profile W belongs to $\mathfrak{B}$, satisfies $\mathcal{F}(W) = 0$, and lies on an admissible nontrivial stationary family. Moreover, by the ordered definition of $\mathcal{C}_{\text{stat}}^{\text{ret}}$, it is not in any of the previously closed lower classes.

By [Theorem 6.6](#), the hull profile (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point, after reselecting scales. Since local concentration is retained, (W, Π) satisfies every condition in [Theorem 6.10](#), except that the actual contradiction is deferred to the terminal stratification theorem. Thus the R3 branch has been reduced to a stationary terminal concentration profile realized from the original blow-up sequence; no R3 closure is used in this reduction step. $\square$

Proposition 6.13 (Already-closed integrable stationary-family subcase). *Let (W, Π) be a stationary centered profile lying on an admissible stationary family. If*

$$(6.10) \quad W \in L^3(\mathbb{R}^3),$$

then $W \equiv 0$. Hence no retained-concentration hull profile satisfying (6.10) can occur in $\mathcal{C}_{\text{stat}}^{\text{ret}}$.

Proof. The stationary physical representative associated with W is a backward self-similar Navier–Stokes solution. The assumption $W \in L^3(\mathbb{R}^3)$ places it in the stationary L^3 class already excluded by the classical Nečas–Růžička–Šverák Liouville theorem [7], equivalently by the endpoint ancient Liouville theorem already recorded in the setup paper and cited below as [Theorem 10.8](#). Hence $W \equiv 0$. A zero profile has zero pressure-normalized local CKN mass after choosing the admissible pressure gauge, so it cannot be ε_* -concentrating. $\square$

Remark 6.14 (Closure after terminal local concentration analysis). The preceding argument reduces the stationary-hull class with retained local concentration from a time-translation hull question to the question whether a stationary centered profile in a nontrivial stationary family can occur as a Seregin limit. [Theorem 6.10](#) is supplied only after the terminal stratification theorem has been proved.

Remark 6.15 (Residual decomposition statement). For the residual decomposition, the third line is reduced as follows and closed only after the terminal stratification theorem:

$$\begin{aligned} & \text{stationary-hull residual with retained concentration} \\ & \xrightarrow{\text{stationary-family Seregin limit reduction}} \text{terminal stationary concentration profile} \\ & \xrightarrow{\text{terminal closure}} \emptyset. \end{aligned}$$

The precise class is the *degenerate stationary-hull residual class with retained local concentration*.

7. REDUCTION OF DEGENERATE STATIONARY-HULL PROFILES WITH RETAINED CONCENTRATION

We record the terminal concentration input needed for the degenerate stationary-hull class. The actual closure is deferred until after the terminal stratification theorem. The argument is not specific to the existence of a non-isolated stationary family. Once a stationary-family element occurs as a Type I Seregin limit carrying retained local CKN concentration, it is a stationary terminal concentration profile. Such profiles are later excluded by the same local terminal concentration analysis and separated-profile exclusion used for the final residual class. No global coercive estimate for the original solution is used.

Throughout this section, a centered Type I profile satisfies

$$(7.1) \quad \partial_\tau U - \Delta U + \frac{1}{2} y \cdot \nabla U + \frac{1}{2} U + (U \cdot \nabla) U + \nabla \Pi = 0, \quad \nabla \cdot U = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$, with the usual local suitability and pressure-gauge conventions.

7.1. Deferred terminal concentration theorem for R3. We record the following specialization of the terminal local concentration theorem. It is stated here exactly in the form needed for the stationary-hull closure with retained local concentration, with proof deferred to the terminal section.

Theorem 7.1 (Deferred terminal concentration theorem for R3). *Let (U, Π) be a bounded centered Type I Seregin profile satisfying local suitability and carrying a retained local CKN concentration, i.e. for some compact parabolic cylinder $Q_r(z_0)$ and some $\varepsilon_* > 0$,*

$$(7.2) \quad \mathcal{E}_r(U, \Pi; z_0) \geq \varepsilon_*.$$

Assume that (U, Π) is outside the previously closed small-amplitude, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, and rotational relative-equilibrium classes. Then the following hold.

- (i) *Every terminal extraction of (U, Π) has the paired decomposition of [Theorem 9.11](#): a retained concentration set together with a residual remainder which is locally vanishing, diffuse, or noncompact after neighborhoods of the concentration set are removed.*
- (ii) *Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the retained compact CKN concentration [\(7.2\)](#).*
- (iii) *Mixed compact-diffuse and compact-noncompact residual profiles, and infinite chains of retained concentrating descendants, are impossible in the refined residual class.*
- (iv) *Finite separated families of retained concentration profiles are impossible.*
- (v) *Any remaining single retained concentration profile is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no loss of local compactness along escaping recenterings.*

(vi) Every retained terminally indecomposable concentration profile has a backward sequence of finite critical norm: there exist $\tau_k \rightarrow -\infty$ such that

$$(7.3) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

(vii) Retained terminal concentration profiles are either affine/parasitic lower-stratum profiles or have parasitic-free finite-shift mild physical pullbacks. Therefore the Albritton–Barker endpoint Liouville theorem applies to any normalized residual profile satisfying (7.3).

Proof deferred. This is the R3 specialization of [Theorem 10.24](#). It is stated here only as a reduction target for the stationary-hull section; its proof is the terminal stratification proof in [Sections 9](#) and [10](#). $\square$

Remark 7.2 (Local nature of the theorem). The theorem above is a local concentration statement. It uses the local suitable-weak-solution compactness and pressure normalization developed from the Caffarelli–Kohn–Nirenberg theory [\[2\]](#) together with the setup paper `proof_setup.tex`: local positive concentration and Type I Seregin compactness. It then uses paired terminal profile decomposition, sole-source exclusions, critical-tail compactification with realized critical-tail rigidity, recurrence for concentration chains, and the finite separated-profile theorem. It does not assert or require a global a priori bound for the original physical solution.

For reference, we record the stationary specialization of the atomic sequence- L^3 lemma used after the terminal theorem.

Lemma 7.3 (Indecomposable local profiles have backward sequence- L^3 control). *Let (U, Π) be a bounded smooth ancient solution of (7.1). Suppose that (U, Π) has retained local concentration and is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration in the parabolic CKN sense, and no escaping recentering sequence along which local compactness, suitability, or pressure control fails. Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$(7.4) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. This is the stationary-profile specialization of [Theorem 10.5](#). The hypotheses listed here are exactly the terminal indecomposability hypotheses used there: retained unit concentration, no retained concentrating tail limit, no diffuse exterior concentration, and no escaping recentering with local compactness or pressure-gauge failure. Applying [Theorem 10.5](#) gives (7.4). $\square$

7.2. Deferred stationary concentration statement.

Theorem 7.4 (No stationary residual concentration profile occurs as a Seregin limit). *Let (W, Π) be a stationary solution of the centered equation,*

$$(7.5) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Assume that:

- (i) (W, Π) is obtained as a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (ii) (W, Π) has retained local concentration in the sense of (7.2);
- (iii) (W, Π) is outside all previously closed lower classes listed in [Theorem 7.1](#);

Then no such (W, Π) exists.

Proof deferred. The proof is the stationary case of the R3 closure after [Theorem 10.24](#); see [Theorem 10.27](#). This theorem is kept here to name the exact stationary concentration statement used by the stationary-hull reduction. $\square$

7.3. Deferred degenerate stationary-hull closure statement.

Corollary 7.5 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. There is no stationary centered profile (W, Π) satisfying all of the following conditions:*

- (i) $W \in \mathfrak{B}$, W solves the stationary centered equation (7.5), and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) has retained local concentration;
- (iv) (W, Π) is outside the previously closed lower classes;

Proof deferred. The proof is included in the post-terminal R3 closure, Theorem 10.27. The stationary-family condition in (i) is additional structure; once the terminal theorem is available, the stationary concentration argument applies to these profiles in particular. $\square$

Proposition 7.6 (Stationary-hull reduction to the terminal concentration theorem). *Let $\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$ be the degenerate stationary-hull residual class with retained local concentration. Every element of this class produces a stationary terminal concentration profile realized from the original blow-up sequence and satisfying the hypotheses of Theorem 10.24.*

Proof. Let $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By the definition of the degenerate stationary-hull residual class with retained local concentration, the centered translation hull of (V, P) contains a stationary profile (W, Π) with retained local concentration which lies in an admissible nontrivial stationary family and is outside the previously closed lower classes. The scale-reselection lemma for centered hull limits shows that (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point.

Thus (W, Π) satisfies the hypotheses of the terminal stratification theorem in the stationary case. The actual emptiness conclusion is recorded after that terminal theorem as Theorem 10.27. $\square$

Remark 7.7 (How this enters the residual decomposition). The third residual line is reduced as follows and closed only after the terminal stratification theorem:

$$\begin{aligned} &\text{degenerate stationary-hull residual with retained concentration} \\ &\implies \text{stationary profile with retained concentration} \implies \emptyset. \end{aligned}$$

The second implication is proved by terminal local concentration analysis: decomposable profiles produce either an excluded separated concentration family, diffuse exterior concentration, or loss of local compactness; indecomposable profiles have backward sequence- L^3 control and are ruled out by the endpoint ancient Liouville theorem.

8. TAIL GEOMETRY OF THE GENERIC TERMINAL RESIDUAL

We now pass to the generic terminal residual class. The preliminary point is that the centered equation is not invariant under ordinary fixed spatial translations. The correct far-field recenterings are self-similar covariant translations.

Definition 8.1 (Covariant translations). For $a \in \mathbb{R}^3$, define

$$(T_a V)(y, \tau) := V(y + e^{\tau/2} a, \tau), \quad (T_a P)(y, \tau) := P(y + e^{\tau/2} a, \tau).$$

Lemma 8.2 (Covariant translation invariance). *If (V, P) solves (1.2), then $(T_a V, T_a P)$ also solves (1.2).*

Proof. Set $z = y + e^{\tau/2}a$. Then

$$\partial_\tau(T_a V)(y, \tau) = (\partial_\tau V)(z, \tau) + \frac{1}{2}e^{\tau/2}a \cdot \nabla V(z, \tau).$$

Moreover,

$$\frac{1}{2}y \cdot \nabla(T_a V)(y, \tau) = \frac{1}{2}z \cdot \nabla V(z, \tau) - \frac{1}{2}e^{\tau/2}a \cdot \nabla V(z, \tau).$$

The two additional transport terms cancel in $\partial_\tau T_a V + \frac{1}{2}y \cdot \nabla T_a V$. The Laplacian, the zeroth-order term, the convection term, the pressure gradient, and the divergence constraint are translation covariant. Substitution into (1.2) proves the claim. $\square$

Remark 8.3 (Fixed translates create a spurious drift). If one sets $V_n(y, \tau) = V(y + x_n, \tau)$ with fixed x_n , then the centered equation acquires the additional transport term

$$-\frac{1}{2}x_n \cdot \nabla V_n.$$

Thus fixed translates do not define a compact invariant far-field hull for the renormalized dynamics.

Definition 8.4 (Spacetime covariant action). For $s \in \mathbb{R}$, define the centered time translation

$$(\Theta_s V)(y, \tau) := V(y, \tau + s), \quad (\Theta_s P)(y, \tau) := P(y, \tau + s).$$

For a tail center $x \in \mathbb{R}^3$ observed at centered time σ , define the covariant tail recentering

$$\mathcal{T}_{(x, \sigma)} V(y, s) := V(y + e^{s/2}x, \sigma + s), \quad \mathcal{T}_{(x, \sigma)} P(y, s) := P(y + e^{s/2}x, \sigma + s).$$

Equivalently,

$$\mathcal{T}_{(x, \sigma)} = \Theta_\sigma T_{e^{-\sigma/2}x}.$$

Thus $\mathcal{T}_{(x, \sigma)}$ is a composition of exact symmetries of the centered equation. Ordinary parabolic recenterings are used below only for local terminal extraction before one has passed to a centered ancient profile; all escaping tail limits of centered profiles use $\mathcal{T}_{(x, \sigma)}$.

Lemma 8.5 (Semidirect covariance of the centered action). *The centered time translations and spatial covariant translations satisfy*

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}, \quad s \in \mathbb{R}, \quad a \in \mathbb{R}^3.$$

Equivalently, if $G = \mathbb{R}^3 \rtimes \mathbb{R}$ acts by

$$\mathcal{G}_{(a, s)} := \Theta_s T_a,$$

then the group law in this convention is

$$(8.1) \quad (a, s) \cdot (b, t) = (e^{-t/2}a + b, s + t), \quad \mathcal{G}_{(a, s)} \mathcal{G}_{(b, t)} = \mathcal{G}_{(a, s) \cdot (b, t)}.$$

In particular the subgroup $\{T_a : a \in \mathbb{R}^3\}$ is normal in the full spacetime covariant action, and the subspace of observables invariant under all spatial covariant translations is invariant under every Θ_s .

Proof. For a profile V ,

$$(\Theta_s T_a \Theta_{-s} V)(y, \tau) = V(y + e^{(\tau+s)/2}a, \tau) = (T_{e^{s/2}a} V)(y, \tau).$$

The pressure transforms identically. The asserted invariance of the spatially invariant observable subspace is the same identity written on the induced L^2 -representation. The group law (8.1) follows from

$$\Theta_s T_a \Theta_t T_b = \Theta_{s+t} T_{e^{-t/2}a + b}.$$

$\square$

Theorem 8.6 (Covariant observer calculus). *Let $Z := \mathbb{R}^3 \times \mathbb{R}$ denote the centered observer space, with points $z = (\xi, \sigma)$. For $a \in \mathbb{R}^3$ and $\sigma \in \mathbb{R}$,*

$$(T_a U)(y, \tau) = U(y + e^{\tau/2} a, \tau), \quad (\Theta_\sigma U)(y, \tau) = U(y, \tau + \sigma).$$

Then

$$\Theta_\sigma T_a = T_{e^{\sigma/2} a} \Theta_\sigma, \quad \Theta_\sigma T_a \Theta_{-\sigma} = T_{e^{\sigma/2} a}.$$

For $z = (\xi, \sigma)$, define the covariant observer

$$\mathcal{R}_z U := \Theta_\sigma T_{e^{-\sigma/2} \xi} U.$$

Equivalently,

$$(\mathcal{R}_{(\xi, \sigma)} U)(y, s) = U(y + e^{s/2} \xi, \sigma + s).$$

The covariant tube pulled back from $Q_r(0, 0)$ is

$$\mathcal{Q}_r^{\text{cov}}(\xi, \sigma) := \left\{ (Y, \tau) : |\tau - \sigma| < r^2, \ |Y - e^{(\tau - \sigma)/2} \xi| < r \right\}.$$

The map

$$(y, s) \mapsto (Y, \tau) = (y + e^{s/2} \xi, \sigma + s)$$

is a bijection from $Q_r(0, 0)$ onto $\mathcal{Q}_r^{\text{cov}}(\xi, \sigma)$ with spacetime Jacobian one. The same formulas apply to the pressure, modulo the admissible time-dependent gauges.

Proof. The semidirect identities are exactly [Theorem 8.5](#); written without conjugation,

$$(\Theta_\sigma T_a U)(y, \tau) = U(y + e^{(\tau + \sigma)/2} a, \tau + \sigma) = (T_{e^{\sigma/2} a} \Theta_\sigma U)(y, \tau).$$

Taking $a = e^{-\sigma/2} \xi$ gives

$$(\Theta_\sigma T_{e^{-\sigma/2} \xi} U)(y, s) = U(y + e^{(s + \sigma)/2} e^{-\sigma/2} \xi, \sigma + s) = U(y + e^{s/2} \xi, \sigma + s).$$

If $Y = y + e^{s/2} \xi$ and $\tau = \sigma + s$, then

$$y = Y - e^{(\tau - \sigma)/2} \xi, \quad s = \tau - \sigma,$$

so $|s| < r^2$ and $|y| < r$ are equivalent to the defining inequalities for $\mathcal{Q}_r^{\text{cov}}(\xi, \sigma)$. The transformation is a time-dependent spatial translation, hence has Jacobian one. $\square$

Definition 8.7 (Terminal profile class and local topology). For $M < \infty$, let $\mathfrak{X}_M$ be the class of pairs (U, Π) , modulo addition of arbitrary functions of time to Π , such that:

- (i) U is a smooth bounded ancient solution of (1.2) on $\mathbb{R}^3 \times \mathbb{R}$, with $\|U\|_{L^\infty} \leq M$;
- (ii) (U, Π) is locally suitable on every compact terminal cylinder;
- (iii) on every compact cylinder the pressure has an admissible $L^{3/2}$ -representative after subtracting a time-dependent gauge.

The topology on $\mathfrak{X}_M$ is

$$U_n \rightarrow U \quad \text{in } C_{\text{loc}}^\infty, \quad \Pi_n - c_n(\tau) \rightarrow \Pi \quad \text{in } L_{\text{loc}}^{3/2}$$

for suitable time-dependent gauges c_n . All profile limits, tail descendants, concentrating descendant hulls, and recurrent-core limits below are taken in this topology.

Definition 8.8 (Observer oscillation and spatial oscillation). Let $Q = B \times I$ be a bounded covariant terminal cylinder. For a velocity field U , define the constant-in-time oscillation functional

$$\text{osc}_{3, \text{ct}}(U; Q) := \inf_{c \in \mathbb{R}^3} \iint_Q |U - c|^3 dy d\tau.$$

Define also the spatial oscillation modulo time-dependent homogeneous gauges

$$\text{osc}_{3,\text{sp}}(U; Q) := \inf_{c \in L^3(I; \mathbb{R}^3)} \iint_Q |U(y, \tau) - c(\tau)|^3 dy d\tau.$$

Thus $\text{osc}_{3,\text{ct}}$ detects nonconstant time-dependent homogeneous modes, while $\text{osc}_{3,\text{sp}}$ is the quotient seminorm used to remove spatially homogeneous parasitic modes. A covariant observer $z \in Z$ is η -oscillatory for U if

$$\text{osc}_{3,\text{ct}}(U; \mathcal{Q}_1^{\text{cov}}(z)) \geq \eta.$$

On each compact observer window we fix a uniformly locally finite family of observer centers $\{z_k\} \subset Z$, subordinate to a bounded-overlap covariant unit-cylinder cover. The corresponding local oscillation measure on observer-center sets $E \subset Z$ is

$$\mu_U^{\text{osc}}(E) := \sum_{z_k \in E} \text{osc}_{3,\text{ct}}(U; \mathcal{Q}_1^{\text{cov}}(z_k)).$$

For a sequence U_n we write μ_n^{osc} for this measure. All uses of μ_n^{osc} are on compact observer-center windows or on specified escaping windows after restriction, so the bounded overlap gives uniform local finiteness. Replacing the cover by another bounded-overlap covariant unit cover changes only harmless constants and never changes the alternatives “positive threshold,” “vanishing threshold,” or “diffuse defect.” The concentration-set extraction uses $\text{osc}_{3,\text{ct}}$, because after parasitic-free normalization it detects genuine non-affine oscillation. Mesoscopic cylinders use $\text{osc}_{3,\text{sp}}$, equivalently the functional $\mathcal{A}_n(z, r)$ of [Theorem 9.33](#), because the minimal-scale quotient must ignore time-dependent homogeneous gauges. Thus the concentration-set construction and the mesoscopic affine quotient are kept distinct.

Lemma 8.9 (Local compactness and continuity of the profile transformations). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a sequence with the local suitability and pressure-gauge bounds supplied by the setup paper on every fixed compact cylinder. Then every subsequence has a further subsequence converging in the topology of $\mathfrak{X}_M$. Moreover the actions T_a , Θ_s , the covariant tail recenterings $\mathcal{T}_{(x,\sigma)}$ are continuous on $\mathfrak{X}_M$. The affine maps satisfy the bound-tracking continuity*

$$S_c : \mathfrak{X}_M \longrightarrow \mathfrak{X}_{M+|c|},$$

again with the pressure interpreted modulo the corresponding transformed time gauges.

Proof. The compactness assertion is the standard local Seregin compactness theorem applied on an exhausting sequence of compact cylinders, followed by a diagonal argument. The velocity convergence is upgraded to C_{loc}^∞ for bounded smooth centered profiles by interior parabolic estimates for (1.2); the pressure representatives converge weakly in $L_{\text{loc}}^{3/2}$ after the local gauges. The formulas defining T_a , Θ_s , $\mathcal{T}_{(x,\sigma)}$, and S_c are smooth changes of variables and affine pressure changes on each fixed compact cylinder. The first three preserve the L^∞ bound M . The affine map S_c subtracts the constant vector c , so its velocity bound is at most $M + |c|$. All four maps preserve the local topology and transform admissible gauges into admissible gauges. No global spatial norm or global pressure representative is used. $\square$

Definition 8.10 (Compact normalized descendant profile class). Let $\mathfrak{X}_M^\# \subset \mathfrak{X}_M$ denote the sequential closure, in the local topology of $\mathfrak{X}_M$, of all normalized Seregin profiles from the setup paper and all profiles obtained from them by the limiting operations used in this paper: centered time translations, covariant translations, covariant tail recenterings, terminal profile limits, concentrating descendant limits, recurrent-core limits, and the canonical affine normalization whenever the affine parameter is bounded. The closure is always taken with the local pressure gauges supplied by the normalized Seregin compactness theorem.

Lemma 8.11 (Compactness and continuity on the normalized profile class). *The space $\mathfrak{X}_M^\#$ is compact in the local topology inherited from $\mathfrak{X}_M$. The maps T_a , Θ_s , and the covariant recenterings used to form descendants act continuously on $\mathfrak{X}_M^\#$. The affine map satisfies the bound-tracking continuity*

$$S_c : \mathfrak{X}_M^\# \longrightarrow \mathfrak{X}_{M+|c|}^\#.$$

Proof. By definition, every sequence in $\mathfrak{X}_M^\#$ is a diagonal limit of normalized Seregin descendants with the same local L^∞ , local suitability, pressure-gauge, and interior smooth bounds on each fixed compact cylinder. Applying [Theorem 8.9](#) on an exhaustion of compact cylinders gives a further subsequential limit in $\mathfrak{X}_M$. The closure defining $\mathfrak{X}_M^\#$ contains that limit, so $\mathfrak{X}_M^\#$ is sequentially compact; the local topology is metrizable on bounded sets by the standard countable exhaustion, hence compact.

Continuity of the limiting operations is the continuity already proved in [Theorem 8.9](#), restricted to the closed normalized profile class. The affine map changes only the velocity bound and the explicit affine pressure gauge, so it maps $\mathfrak{X}_M^\#$ continuously into $\mathfrak{X}_{M+|c|}^\#$. $\square$

Definition 8.12 (Affine homogeneous symmetry). For $c \in \mathbb{R}^3$, define

$$(S_c V)(y, \tau) := V(y - 2c, \tau) - c, \quad (S_c P)(y, \tau) := P(y - 2c, \tau) + \frac{1}{2}c \cdot y.$$

Lemma 8.13 (Affine renormalized Galilean symmetry). *If (V, P) solves (1.2), then $(S_c V, S_c P)$ also solves (1.2).*

Proof. Set $U(y, \tau) = V(y - 2c, \tau) - c$, $\Pi(y, \tau) = P(y - 2c, \tau) + \frac{1}{2}c \cdot y$, and $z = y - 2c$. Then

$$\partial_\tau U = (\partial_\tau V)(z, \tau), \quad \Delta U = (\Delta V)(z, \tau), \quad \nabla U = (\nabla V)(z, \tau).$$

The nonlinear term is

$$(U \cdot \nabla)U = (V - c) \cdot \nabla V = (V \cdot \nabla)V - c \cdot \nabla V,$$

where all quantities on the right are evaluated at (z, τ) . The drift term satisfies

$$\frac{1}{2}y \cdot \nabla U = \frac{1}{2}(z + 2c) \cdot \nabla V = \frac{1}{2}z \cdot \nabla V + c \cdot \nabla V.$$

The two $c \cdot \nabla V$ terms cancel. Finally,

$$\frac{1}{2}U + \nabla \Pi = \frac{1}{2}(V - c) + \nabla P + \frac{1}{2}c = \frac{1}{2}V + \nabla P.$$

Substitution into (1.2) at the point (z, τ) gives the equation for (U, Π) . The divergence condition is unchanged by translation and subtracting a constant vector. $\square$

Definition 8.14 (Pressure gauges and affine quotient). On every ordinary local cylinder Q , pressure representatives are identified modulo time functions:

$$P_Q \sim P_Q + c_Q(\tau).$$

At the affine/parasitic quotient level, and only there, one also identifies pressure representatives modulo the affine spatial functions paired with spatially homogeneous parasitic velocities:

$$P_Q \sim P_Q + b_Q(\tau) \cdot y + c_Q(\tau).$$

The affine term is allowed exactly when it corresponds to a removable homogeneous mode in the velocity equation, as in [Theorem 8.16](#). Generic terminal residual branches are always considered after this quotient has either been fixed by the canonical parasitic-free representative or assigned to $\mathcal{C}_{\text{aff}}$.

Lemma 8.15 (Pressure atlas compatibility). *Let Q, Q' be overlapping local cylinders in a normalized Seregin extraction. Before affine quotienting, admissible pressure representatives satisfy*

$$P_Q - P_{Q'} = c_{Q,Q'}(\tau)$$

on $Q \cap Q'$. At the affine/parasitic quotient level they satisfy

$$P_Q - P_{Q'} = b_{Q,Q'}(\tau) \cdot y + c_{Q,Q'}(\tau)$$

with the affine term paired with the corresponding removable homogeneous velocity mode. The velocity equation is unchanged after the time-gauge identification, and after the affine identification it is interpreted in the quotient equation with the parasitic correction removed or assigned to $\mathcal{C}_{\text{aff}}$.

Proof. The time-gauge statement is the setup pressure atlas compatibility imported in [Theorem 1.3](#). The affine statement follows by substituting a spatially homogeneous velocity $b(\tau)$ into the centered equation: the only pressure gradient it can create is $-(\partial_\tau b + \frac{1}{2}b)$, hence an affine pressure in y , modulo a time function. This is exactly the quotient convention in [Theorem 8.14](#). All terminal limiting operations below choose their pressure gauges before taking compactness limits, and if the affine term is nontrivial the profile is assigned to $\mathcal{C}_{\text{aff}}$. $\square$

Definition 8.16 (Affine/parasitic lower stratum and residual normalization). The affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$ consists of the spatially homogeneous modes which are invisible to the spatial quotient seminorm $\text{osc}_{3,\text{sp}}$. It has two equivalent realizations.

- (i) A genuine centered profile belongs to $\mathcal{C}_{\text{aff}}$ if, after one affine homogeneous symmetry S_c , its velocity is spatially constant. In the parasitic-free representative this is the zero profile.
- (ii) A quotient limit belongs to $\mathcal{C}_{\text{aff}}$ if it is represented by a locally bounded spatially homogeneous velocity $b(\tau)$. In the weak centered equation this mode is paired with the affine pressure functional

$$(-\partial_\tau b(\tau) - \tfrac{1}{2}b(\tau)) \cdot y$$

in $\mathcal{D}'_\tau \mathcal{D}'_y$. If the mode is realized as a smooth suitable profile, this affine pressure is an ordinary local pressure; if it arises only as a measurable homogeneous gauge $c_n(\tau)$, it is used only in the quotient weak formulation and never as an independent $L_{\text{loc}}^{3/2}$ pressure.

In the ordered residual decomposition this stratum is part of the previously closed lower classes, not part of $\mathcal{C}_{\text{gen}}$. Equivalently, every generic residual profile is considered after the canonical affine normalization which removes its spatially homogeneous mild mode. Retained non-affine oscillation in the residual class is therefore always measured after this normalization.

Remark 8.17 (How affine constants are used). When a compact recurrent tail core is shown below to consist only of spatially homogeneous parasitic modes, the contradiction is not that an unnormalized nonzero constant has zero local L^3 -mass. The contradiction is that these modes have already been assigned to the closed lower stratum in [Theorem 8.16](#); after the canonical affine normalization they are the zero profile and cannot represent a generic residual profile carrying retained concentration or non-affine oscillation.

Lemma 8.18 (Homogeneous gauges generate only the affine/parasitic stratum). *Let $c_n : I \rightarrow \mathbb{R}^3$ be bounded in $L^\infty(I)$, and let*

$$C_n(y, \tau) := c_n(\tau).$$

Every local weak limit, time translate, or normalized noncompactness limit of (C_n) has vanishing spatial quotient oscillation on every compact cylinder and belongs to $\mathcal{C}_{\text{aff}}$. Conversely, after the canonical parasitic-free normalization, no nonzero spatially homogeneous mode remains in the residual profile class.

Proof. For every compact cylinder Q ,

$$\inf_{c(\tau)} \iint_Q |C_n(y, \tau) - c(\tau)|^3 dy d\tau = 0,$$

with the choice $c(\tau) = c_n(\tau)$. The same identity is stable under local weak limits and time recenterings. Hence every such limit has zero spatial quotient oscillation on every compact cylinder and is, by [Theorem 8.16](#), an affine/parasitic quotient mode. If a homogeneous mode is realized as a smooth centered solution, substituting $U = b(\tau)$ into (1.2) gives

$$\partial_\tau b + \frac{1}{2}b + \nabla P = 0,$$

so the pressure is affine in y up to a time gauge, exactly as in the definition of $\mathcal{C}_{\text{aff}}$. If b is only a bounded measurable gauge, the same identity is understood distributionally and only in the quotient formulation. The residual representative is defined after removal of this quotient direction; therefore a nonzero retained effect of b is assigned to the closed lower stratum, while the parasitic-free residual profile has no such mode. $\square$

Theorem 8.19 (Closed non-affine oscillation functional). *Let $Q \Subset \mathbb{R}^3 \times \mathbb{R}$ be a bounded connected parabolic cylinder. For $U \in \mathfrak{X}_M$, define*

$$d_Q(U) := \text{osc}_{3,\text{ct}}(U; Q)^{1/3}.$$

Then

$$|d_Q(U) - d_Q(V)| \leq \|U - V\|_{L^3(Q)}.$$

In particular d_Q is continuous under the local topology of $\mathfrak{X}_M$. Moreover,

$$d_Q(U) = 0 \iff U \text{ is spatially and temporally constant on } Q.$$

Fix once and for all a nested exhaustion

$$Q_1 \Subset Q_2 \Subset \dots, \quad \bigcup_{j=1}^{\infty} Q_j = \mathbb{R}^3 \times \mathbb{R},$$

by bounded connected parabolic cylinders. Define

$$\Phi(U) := \sum_{j=1}^{\infty} 2^{-j} \min\{1, d_{Q_j}(U)\}.$$

Then Φ is continuous on $\mathfrak{X}_M$, and

$$\Phi(U) = 0 \iff U \text{ is globally constant.}$$

Consequently, for every $\eta > 0$, the non-affine oscillation set

$$\mathfrak{A}_\eta := \{(U, \Pi) \in \mathfrak{X}_M : \Phi(U) \geq \eta\}$$

is closed in the local profile topology.

Proof. For every $c \in \mathbb{R}^3$, the triangle inequality in $L^3(Q)$ gives

$$\|U - c\|_{L^3(Q)} \leq \|U - V\|_{L^3(Q)} + \|V - c\|_{L^3(Q)}.$$

Taking the infimum over c gives

$$d_Q(U) \leq \|U - V\|_{L^3(Q)} + d_Q(V).$$

Interchanging U and V proves the Lipschitz estimate. Local smooth convergence implies local L^3 -convergence, so d_Q is continuous in the topology of $\mathfrak{X}_M$.

If $d_Q(U) = 0$, choose $c_k \in \mathbb{R}^3$ such that $\|U - c_k\|_{L^3(Q)} \rightarrow 0$. The sequence c_k is bounded; otherwise

$$\|U - c_k\|_{L^3(Q)} \geq |c_k| |Q|^{1/3} - \|U\|_{L^3(Q)} \rightarrow \infty,$$

which is impossible. Passing to a subsequence, $c_k \rightarrow c$, and hence $U = c$ almost everywhere on Q . Since U is smooth, $U \equiv c$ pointwise on Q . The converse is immediate.

Each summand in the definition of Φ is continuous and bounded by 2^{-j} . The Weierstrass test gives uniform convergence of the defining series, hence continuity of Φ . If $\Phi(U) = 0$, then $d_{Q_j}(U) = 0$ for every j , so U is constant on every Q_j . The cylinders are nested and overlap, so the constants agree and U is globally constant. The converse is again immediate. Closedness of $\mathfrak{A}_\eta$ follows from continuity of Φ . $\square$

Corollary 8.20 (Compact normalized non-affine oscillation sets). *For every $\eta > 0$,*

$$\mathfrak{A}_\eta^\# := \{(U, \Pi) \in \mathfrak{X}_M^\# : \Phi(U) \geq \eta\}$$

is compact in the normalized profile topology.

Proof. By Theorem 8.11, $\mathfrak{X}_M^\#$ is compact. By Theorem 8.19, Φ is continuous in the local topology. Therefore $\mathfrak{A}_\eta^\#$ is a closed subset of a compact space. $\square$

Theorem 8.21 (Affine normalization dichotomy). *Let S_c be the affine homogeneous symmetry of Theorem 8.12, and suppose the ordered lower strata include all constant centered profiles as in Theorem 8.16. For every $(U, \Pi) \in \mathfrak{X}_M$, exactly one of the following alternatives holds:*

- (i) $\text{osc}_{3,\text{sp}}(U; Q) = 0$ on every compact cylinder Q . Then U is a spatially homogeneous quotient mode, and (U, Π) belongs to the affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$.
- (ii) For the parasitic-free representative, after an admissible affine symmetry S_c , one has $\Phi(S_c U) > 0$. Equivalently, for some compact cylinder Q , $\text{osc}_{3,\text{ct}}(S_c U; Q) > 0$.

In the retained normalized residual space, alternative (i) has already been removed by the canonical affine/parasitic normalization, so only alternative (ii) can remain. Moreover, if $K \subset \mathfrak{X}_M$ is compact and

$$K \cap \{(U, \Pi) : \Phi(U) = 0\} = \emptyset,$$

then there exists $\eta_K > 0$ such that

$$\Phi(U) \geq \eta_K \quad \text{for every } (U, \Pi) \in K.$$

Proof. If $\text{osc}_{3,\text{sp}}(U; Q) = 0$ on every compact cylinder, then on each time slice U is spatially constant in the distributional sense. The local constants are compatible on overlapping cylinders, so $U(y, \tau) = b(\tau)$ locally in time for a measurable bounded function b . By Theorem 8.16, this is precisely the affine/parasitic quotient stratum; if it is realized by a smooth profile, the corresponding affine pressure is the one described there, and otherwise it is used only in the quotient weak formulation.

If the spatial oscillation does not vanish identically, U is not a spatially homogeneous quotient mode. After choosing the canonical parasitic-free representative and applying any admissible affine symmetry S_c , the resulting velocity is not globally constant. Hence, by Theorem 8.19, $\Phi(S_c U) > 0$. Conversely, if $\Phi(S_c U) = 0$, then $S_c U$ is globally constant, and the original profile is in $\mathcal{C}_{\text{aff}}$. This proves the dichotomy and the retained normalized statement. For the compact statement, Φ is continuous and strictly positive on K , so it attains a positive minimum

$$\eta_K := \min_{(U, \Pi) \in K} \Phi(U) > 0.$$

$\square$

Definition 8.22 (Asymptotic hull at infinity). Let V be a bounded smooth ancient solution of (1.2). Define

$$\mathcal{H}_\infty(V) := \{W : \exists (a_n, s_n) \in G, |(a_n, s_n)|_{\text{cov}} \rightarrow \infty, \mathcal{G}_{(a_n, s_n)} V \rightarrow W \text{ in } C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})\}.$$

Here $G = \mathbb{R}^3 \rtimes \mathbb{R}$ is the covariant spacetime group in Theorem 8.5, and any proper left-invariant metric may be used for $|(a, s)|_{\text{cov}}$. We write $\mathcal{A}_\infty(V) := \mathcal{H}_\infty(V)$ below for compatibility with the tail dichotomy notation. Thus the recurrent tail hull is invariant under the full covariant action, not merely under ordinary spatial translations.

Theorem 8.23 (Compactness of the asymptotic hull). *Let V be a bounded smooth ancient solution of (1.2). Then $\mathcal{A}_\infty(V) = \mathcal{H}_\infty(V)$ is nonempty and compact in $C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})$. Every $W \in \mathcal{A}_\infty(V)$ is a bounded smooth ancient solution of (1.2) with $\|W\|_{L^\infty} \leq \|V\|_{L^\infty}$.*

Proof. For any escaping covariant sequence (a_n, s_n) , the transformed solutions $\mathcal{G}_{(a_n, s_n)} V$ solve the same equation and have the same L^∞ bound by Theorems 8.2 and 8.5. On every compact cylinder, the coefficients of (1.2) are fixed and smooth. Interior parabolic estimates for the Stokes system with bounded smooth nonlinear data, combined with the pressure equation $-\Delta P = \partial_i \partial_j (V_i V_j)$ after local gauges, give uniform C^k -bounds on smaller compact cylinders for every k . A diagonal Arzela–Ascoli argument gives a subsequential limit in C_{loc}^∞ . Passing to the limit in the equation gives another bounded ancient solution. The same compactness argument applied to an arbitrary sequence of elements of $\mathcal{A}_\infty(V)$ proves compactness. $\square$

Definition 8.24 (Tail alternatives). Let $\mathcal{C}$ denote the set of constant vector fields. Define

$$\mathcal{C}_{\text{gen tail-const}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \subset \mathcal{C}\},$$

and

$$\mathcal{C}_{\text{gen tail-prof}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \text{ contains a nonconstant profile}\}.$$

Proposition 8.25 (Tail dichotomy and tail-zero normalization). *Every $V \in \mathcal{C}_{\text{gen}}$ belongs to $\mathcal{C}_{\text{gen tail-const}} \cup \mathcal{C}_{\text{gen tail-prof}}$. If $\mathcal{A}_\infty(V) = \{C_\infty\}$ is a singleton constant hull, then*

$$U := S_{C_\infty} V$$

solves (1.2) and satisfies

$$\mathcal{A}_\infty(U) = \{0\}.$$

Proof. The dichotomy is the definition. Since T_a and S_c commute,

$$T_a(S_{C_\infty} V) = S_{C_\infty}(T_a V).$$

Thus every far-field limit of $S_{C_\infty} V$ is

$$S_{C_\infty}(C_\infty) = 0.$$

The equation is preserved by Theorem 8.13. $\square$

Definition 8.26 (Tail-minimal recurrent core). A nonempty compact set $\mathcal{M} \subset \mathcal{A}_\infty(V)$ is a tail-minimal recurrent core if it is invariant under all covariant translations T_a , invariant under all time translations

$$(\Theta_s W)(y, \tau) := W(y, \tau + s),$$

and contains no proper nonempty compact subset with the same two invariance properties.

Lemma 8.27 (Pressure gradients in invariant local averages). *Let $\mathcal{M}$ be a compact set of smooth bounded solutions of (1.2), compact in the locally smooth topology and invariant under the covariant translations T_a and the time translations Θ_s . Let μ be a Borel probability measure*

on $\mathcal{M}$ invariant under both actions. For each $W \in \mathcal{M}$, let P_W be any smooth local pressure near $(0, 0)$ satisfying (1.2). Put

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0), \quad b_{ij}(W) := u_i(W)u_j(W).$$

Let Π_0 denote the orthogonal projection in $L^2(\mathcal{M}, \mu)$ onto the closed subspace of functions invariant under all spatial covariant translations T_a , and set

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Then the pressure contribution in the recurrent-core energy average is gauge-independent and satisfies

$$(8.2) \quad \int_{\mathcal{M}} u_i(W) p_i(W) d\mu(W) = \int_{\mathcal{M}} u_i^0(W) p_i^0(W) d\mu(W).$$

Equivalently, the nonzero spatial-frequency part of the pressure gradient does no work in invariant average:

$$(8.3) \quad \int_{\mathcal{M}} u_i(W) (p_i(W) - p_i^0(W)) d\mu(W) = 0.$$

Proof. The additive pressure gauge does not enter, because only ∇P_W appears. The pressure gradient is fixed by the velocity through the equation

$$\nabla P_W = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - \partial_\tau W - (W \cdot \nabla)W.$$

Thus $W \mapsto p_i(W)$ is a continuous bounded local observable on $\mathcal{M}$. The local pressure reconstruction step in `proof_setup.tex` supplies the local pressure representatives; the present argument uses only their gradients, so the additive gauges never enter.

Let U_a be the unitary representation of spatial covariant translations on $L^2(\mathcal{M}, \mu)$,

$$(U_a F)(W) := F(T_a W).$$

Let D_1, D_2, D_3 be its skew-adjoint infinitesimal generators. For a smooth local observable F , $D_k F$ is obtained by differentiating $F(T_{he_k} W)$ at $h = 0$. In particular, at $\tau = 0$,

$$D_k u_i(W) = \partial_k W_i(0, 0).$$

Hence

$$(8.4) \quad D_i u_i = 0 \quad \text{in } L^2(\mathcal{M}, \mu),$$

because each W is divergence-free.

Taking the spatial divergence of (1.2) gives the local pressure Poisson equation

$$(8.5) \quad -\Delta_y P_W = \partial_i \partial_j (W_i W_j).$$

Indeed, the divergence of $\partial_\tau W$, ΔW , and W is zero; the divergence of $y \cdot \nabla W$ is $\partial_i (y_j \partial_j W_i) = \partial_j W_j + y_j \partial_j \partial_i W_i = 0$; and $(W \cdot \nabla) W_i = \partial_j (W_j W_i)$. Applying the infinitesimal translation generators at $(0, 0)$ gives

$$(8.6) \quad D_i p_i = -D_i D_j b_{ij}.$$

The pressure gradient is curl-free, hence

$$(8.7) \quad D_k p_i = D_i p_k.$$

Let

$$A := -\sum_{k=1}^3 D_k^2.$$

By the spectral theorem for the commuting unitary group U_a , A is self-adjoint and nonnegative, and Π_0 is exactly the spectral projection of A at the eigenvalue 0. Define the nonzero spatial part

$$q_i := p_i - p_i^0 = (I - \Pi_0)p_i.$$

Equations (8.6) and (8.7) imply, on the nonzero spectral subspace,

$$q_k = D_k A^{-1} (I - \Pi_0) D_i D_j b_{ij}$$

in the standard regularized sense. Explicitly, with

$$q_{k,\varepsilon} := D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij}, \quad \varepsilon > 0,$$

the joint spectral resolution of D_1, D_2, D_3 gives

$$q_{k,\varepsilon} \longrightarrow q_k \quad \text{in } L^2(\mathcal{M}, \mu) \quad \text{as } \varepsilon \downarrow 0.$$

The convergence is precisely the multiplier convergence

$$\frac{\xi_k \xi_i \xi_j}{\varepsilon + |\xi|^2} \longrightarrow \frac{\xi_k \xi_i \xi_j}{|\xi|^2} \quad (\xi \neq 0),$$

together with $(I - \Pi_0)$, which removes the point $\xi = 0$. The curl-free relation (8.7) says that the nonzero spatial spectral part of p is parallel to ξ , and (8.6) fixes its longitudinal coefficient; there is therefore no additional harmonic component.

For each $\varepsilon > 0$, skew-adjointness of D_k and (8.4) give

$$\begin{aligned} \int_{\mathcal{M}} u_k q_{k,\varepsilon} d\mu &= \int_{\mathcal{M}} u_k D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu \\ &= - \int_{\mathcal{M}} (D_k u_k) (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu = 0. \end{aligned}$$

Passing to the limit in L^2 yields

$$\int_{\mathcal{M}} u_k q_k d\mu = 0,$$

which is (8.3). Since p_i^0 lies in the range of Π_0 and $u_i - u_i^0$ is orthogonal to that range,

$$\int_{\mathcal{M}} (u_i - u_i^0) p_i^0 d\mu = 0.$$

Combining this with (8.3) proves (8.2). $\square$

Theorem 8.28 (Rigidity of recurrent profile tail cores). *Let $\mathcal{M}$ be a compact tail-minimal recurrent core for (1.2) arising from the normalized Seregin setup in `proof_setup.tex`, with the affine zero mode interpreted by the affine symmetry Theorem 8.13. Equivalently, after applying one fixed affine normalization S_c , the spatially homogeneous part of the core satisfies the bounded mild centered formulation recorded in `proof_setup.tex`. Then every $W \in \mathcal{M}$ is spatially and temporally constant. More precisely, there exists $m \in \mathbb{R}^3$ such that*

$$W(y, \tau) \equiv m \quad \text{for every } W \in \mathcal{M}.$$

After applying the affine symmetry S_m , the core is reduced to the zero solution.

Proof. The group generated by covariant translations and time translations is amenable on the compact invariant set $\mathcal{M}$. Choose an invariant Borel probability measure μ with full support on a minimal component, and write

$$\langle F \rangle := \int_{\mathcal{M}} F(W) d\mu(W)$$

for local observables. Let P_W be a local pressure associated to W . As in [Theorem 8.27](#), set

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0),$$

and let Π_0 be the orthogonal projection onto observables invariant under all spatial covariant translations. Write

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Invariance gives

$$\langle \partial_\tau F(W)(0, 0) \rangle = 0, \quad \langle \partial_j F(W)(0, 0) \rangle = 0$$

for every smooth local observable F . These identities follow first for difference quotients generated by the group action and then by smoothness.

Evaluate [\(1.2\)](#) at $(0, 0)$. The drift term vanishes at $y = 0$, which gives

$$\partial_\tau W(0, 0) - \Delta W(0, 0) + \frac{1}{2}W(0, 0) + (W(0, 0) \cdot \nabla)W(0, 0) + \nabla P_W(0, 0) = 0.$$

Project this identity onto the spatially invariant subspace. The spatial derivative terms vanish under Π_0 : the Laplacian is $D_j D_j u_i$, and the nonlinear term is $D_j(u_j u_i)$ because $D_j u_j = 0$. Time translations do not literally commute with the covariant spatial translations; by [Theorem 8.5](#) they normalize them:

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}.$$

Hence the subspace of observables invariant under all T_a is invariant under Θ_s , and the orthogonal projection Π_0 commutes with the time-translation representation. Therefore $\Pi_0 \partial_\tau u_i = \partial_\tau u_i^0$. Thus

$$(8.8) \quad \partial_\tau u_i^0 + \frac{1}{2}u_i^0 + p_i^0 = 0 \quad \text{in } L^2(\mathcal{M}, \mu).$$

Taking the inner product of [\(8.8\)](#) with u_i^0 , summing in i , and using invariance under time translations,

$$\langle u_i^0 \partial_\tau u_i^0 \rangle = \frac{1}{2} \langle \partial_\tau |u^0|^2 \rangle = 0,$$

gives

$$(8.9) \quad \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Dot the equation with W , evaluate at $(0, 0)$, and average. The time term is

$$\langle W \cdot \partial_\tau W \rangle = \frac{1}{2} \langle \partial_\tau |W|^2 \rangle = 0.$$

The diffusion term gives

$$-W \cdot \Delta W = |\nabla W|^2 - \operatorname{div}(W \cdot \nabla W),$$

and therefore

$$-\langle W \cdot \Delta W \rangle = \langle |\nabla W|^2 \rangle.$$

The nonlinear term is a divergence:

$$W \cdot (W \cdot \nabla)W = \frac{1}{2}W \cdot \nabla |W|^2 = \frac{1}{2} \operatorname{div}(|W|^2 W),$$

so its average is zero. Decompose

$$p_i = p_i^0 + (p_i - p_i^0).$$

The nonzero spatial part of the pressure gradient does no work in invariant average by [Theorem 8.27](#), and the remaining spatial zero mode is evaluated by [\(8.9\)](#). Hence

$$\langle u_i p_i \rangle = \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Thus

$$(8.10) \quad \langle |\nabla W|^2 \rangle + \frac{1}{2} \langle |W|^2 \rangle - \frac{1}{2} \langle |u^0|^2 \rangle = 0.$$

Both terms are nonnegative. The second nonnegative quantity is

$$\langle |u|^2 \rangle - \langle |u^0|^2 \rangle = \langle |u - u^0|^2 \rangle,$$

because Π_0 is an orthogonal projection in $L^2(\mathcal{M}, \mu)$. Therefore

$$\langle |\nabla W|^2 \rangle = 0, \quad \langle |u - u^0|^2 \rangle = 0.$$

Since μ has full support and the observables are continuous in the local smooth topology, $\nabla W(0, 0) = 0$ for every $W \in \mathcal{M}$. Invariance moves $(0, 0)$ to any (y, τ) , so $\nabla W(y, \tau) = 0$ everywhere. Thus every element of $\mathcal{M}$ is spatially constant:

$$W(y, \tau) = a_W(\tau).$$

It remains only to identify the allowed spatially homogeneous zero mode. This is not a pressure-gauge assertion. By the normalization in the setup paper quoted in the statement, there is one fixed affine normalization S_c for which the spatially homogeneous representatives satisfy the bounded mild centered formulation recorded in `proof_setup.tex`. For a spatially constant mild centered solution $B(\tau)$, the nonlinear Duhamel term in the mild formulation vanishes, and the Ornstein–Uhlenbeck–Stokes semigroup acts on constants by

$$S(\tau - \sigma)B(\sigma) = e^{-(\tau - \sigma)/2}B(\sigma).$$

Thus

$$B(\tau) = e^{-(\tau - \sigma)/2}B(\sigma) \quad (\sigma < \tau).$$

Letting $\sigma \rightarrow -\infty$ and using boundedness of the ancient trajectory gives $B(\tau) = 0$ for every τ . Undoing the fixed affine normalization gives a single vector $m = c$ and

$$W(y, \tau) \equiv m \quad (W \in \mathcal{M}).$$

The affine symmetry S_m then maps the core to the zero solution. $\square$

9. TERMINAL CONCENTRATION PROFILES AND SEPARATED CENTERS

The terminal part of the proof is local in parabolic cylinders. It does not require a global L^3 bound, a global CKN budget, a finite critical-mass budget, or a global nonlinear profile decomposition for the original residual profile. Everything is formulated on fixed compact terminal cylinders after applying the recentering sequence under consideration.

Definition 9.1 (Parabolic recentering sequence). Let (V_n, P_n) be a terminal extraction sequence on expanding domains $D_n \uparrow \mathbb{R}^3 \times \mathbb{R}$, with local suitability and pressure-gauge control. A parabolic recentering sequence is a sequence

$$\mathbf{c} = (z_n)_{n \geq 1}, \quad z_n = (y_n, \tau_n) \in D_n.$$

Two recentering sequences are equivalent if

$$\sup_n \left(|y_n - y'_n| + |\tau_n - \tau'_n|^{1/2} \right) < \infty,$$

and asymptotically separated if the same parabolic distance tends to infinity. The recentered fields associated to $\mathbf{c}$ are denoted

$$V_n^{\mathbf{c}}(y, s) := V_n(y + y_n, s + \tau_n), \quad P_n^{\mathbf{c}}(y, s) := P_n(y + y_n, s + \tau_n) - a_n^{\mathbf{c}}(s),$$

where $a_n^{\mathbf{c}}$ is an admissible time-dependent local pressure gauge. All convergence assertions for a recentering sequence are made on fixed compact sets in the variables (y, s) .

Definition 9.2 (Combined parabolic-covariant descendant recentering). Let $g_n = (y_n, \tau_n)$ be an ordinary parabolic recentering sequence for a terminal extraction (V_n, P_n) , and let $\zeta_n = (\xi_n, \sigma_n)$ be a covariant tail recentering sequence in the centered limit generated by g_n . The combined parabolic-covariant descendant recentering is denoted

$$d_n = (g_n; \xi_n, \sigma_n)$$

and acts on the original terminal sequence by

$$V_n^d(y, s) := V_n(y + y_n + e^{s/2}\xi_n, s + \tau_n + \sigma_n),$$

with the pressure pulled back by the same formula and then adjusted by an admissible time-dependent gauge. Its relative distance from g_n is

$$|\xi_n| + |\sigma_n|^{1/2}.$$

Thus d_n is separated from g_n exactly when this relative distance tends to infinity. This definition is used only to lift descendants of a centered profile back to the original terminal extraction; ordinary first-generation terminal recenterings remain the ordinary parabolic recenterings of [Theorem 9.1](#).

Definition 9.3 (Atlas-dependent defect measure and concentrating recentering sequences). After choosing local pressure gauges $a_n(\tau)$, define

$$d\mu_n := \left(|V_n|^3 + |P_n - a_n(\tau)|^{3/2} \right) dy d\tau, \quad d\nu_n := |\nabla V_n|^2 dy d\tau.$$

The measure μ_n is the atlas-dependent Radon measure from [Theorem 3.1](#). It is used for local compactness and concentration-set extraction. Scale-invariant threshold and regularity statements are written with $\mathcal{E}_r$. The local mass of a recentering sequence $\mathfrak{c} = (z_n)$ is

$$m(\mathfrak{c}) := \limsup_{n \rightarrow \infty} \mu_n(Q_1(z_n)).$$

For a combined parabolic-covariant descendant recentering $d_n = (g_n; \xi_n, \sigma_n)$, the same notation means the mass of the pulled back defect measure in the combined coordinates:

$$m(\mathfrak{d}) := \limsup_{n \rightarrow \infty} \mu_n^{\mathfrak{d}}(Q_1(0, 0)),$$

where $\mu_n^{\mathfrak{d}}$ is obtained from μ_n by the coordinate change in [Theorem 9.2](#) and by the admissible pressure gauges used in that recentered frame. Thus first-generation recentering sequences and lifted covariant descendants use the same local unit-cylinder criterion, expressed in their own local coordinates. For a fixed small-data threshold $\varepsilon_{\text{sd}} > 0$, the recentering sequence is ε_{sd} -concentrating if $m(\mathfrak{c}) \geq \varepsilon_{\text{sd}}$, and locally vanishing at unit scale otherwise. It has loss of local compactness if

$$\limsup_{n \rightarrow \infty} \nu_n(Q_2(z_n)) = \infty$$

or if the local pressure gauges fail to be uniformly controlled in $L^{3/2}(Q_2(z_n))$.

Lemma 9.4 (Gauge compatibility and retained-concentration semicontinuity). *Let $(U_n, \Pi_n) \rightarrow (U, \Pi)$ in the topology of $\mathfrak{X}_M$ on a fixed compact cylinder containing $Q_1(0, 0)$. After choosing the pressure gauges used in that convergence, the following hold.*

- (i) *The pressure gauges are compatible with centered time shifts, covariant translations, covariant tail recenterings, affine normalization, and diagonal profile-tree extraction: applying any of these operations only changes the gauge by another admissible function of time.*

- (ii) *The local CKN quantity is lower semicontinuous in the direction needed for lifting concentrating limits:*

$$\mathcal{E}_1(U, \Pi; 0) \leq \liminf_{n \rightarrow \infty} \mathcal{E}_1(U_n, \Pi_n; 0).$$

Consequently, if the limit profile has $\mathcal{E}_1(U, \Pi; 0) \geq \eta$, then all sufficiently far approximating profiles have CKN mass at least any prescribed $\eta' < \eta$.

- (iii) *If retained concentration is carried by the velocity part, then it is closed under the same convergence:*

$$\liminf_{n \rightarrow \infty} \iint_{Q_1} |U_n|^3 \geq \eta \implies \iint_{Q_1} |U|^3 \geq \eta.$$

In applications where the lower bound is first obtained for velocity alone and then converted into a CKN lower bound, the descendant threshold is chosen below the resulting positive constant as in [Theorem 3.4](#).

Proof. Time shifts and covariant translations compose the pressure with a smooth local change of variables; affine normalization adds the explicit linear pressure $\frac{1}{2}c \cdot y$; and every local pressure is defined only up to a time-dependent function. Thus admissible gauges remain admissible under all operations listed in (i). For (ii), strong local convergence of U_n gives

$$\iint_{Q_1} |U_n|^3 \rightarrow \iint_{Q_1} |U|^3.$$

The pressure terms are convex in the weak $L^{3/2}$ topology after the fixed gauges, hence

$$\iint_{Q_1} |\Pi - c(\tau)|^{3/2} \leq \liminf_{n \rightarrow \infty} \iint_{Q_1} |\Pi_n - c_n(\tau)|^{3/2}.$$

Adding the two inequalities gives the claimed lower semicontinuity direction for $\mathcal{E}_1$. The velocity-only assertion follows directly from strong $L^3(Q_1)$ convergence. The statement is entirely local on Q_1 . $\square$

Lemma 9.5 (Setup compactness excludes retained local compactness failure). *Let (V_n, P_n) be a terminal sequence obtained from the Seregin setup in `proof_setup.tex`, or from a later terminal extraction of such a sequence. If a recentering sequence $\mathbf{c} = (z_n)$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness and its recentered fields converge locally in the Seregin topology to a suitable bounded terminal profile.*

Proof. Fix $R < \infty$. The recentered cylinders $Q_{2R}(z_n)$ are compact terminal cylinders in the variables of the chosen extraction. The setup paper gives local suitability, local pressure-gauge compactness, and the inherited Type I bound on every such compact cylinder. Therefore, after choosing the pressure gauges a_n^c , the sequence (V_n^c, P_n^c) has the standard local bounds needed for the Seregin compactness theorem on $Q_R(0, 0)$: strong local L^3 compactness for the velocity, weak local $L^{3/2}$ compactness for the pressure, and the local energy inequality in the limit. A diagonal argument over $R = 1, 2, \dots$ gives a suitable bounded terminal profile. The two failures used in the definition of loss of local compactness are precisely the negations of these local compactness and pressure-gauge conclusions, so a retained recentering sequence arising from the Seregin setup in `proof_setup.tex` cannot remain in the local-compactness-failure alternative. $\square$

Theorem 9.6 (Local concentration-set extraction). *Let $Z = \mathbb{R}^3 \times \mathbb{R}$ be the centered covariant observer space, and let $K_m \Subset Z$, $m \geq 1$, be a compact exhaustion. Let μ_n be the local defect measures of a terminal sequence, with local finite mass on compact observer windows:*

$$\sup_n \mu_n(K_m^+) < \infty \quad (m \geq 1),$$

where K_m^+ is a compact enlargement such that every covariant unit cylinder $\mathcal{Q}_1^{\text{cov}}(z)$, $z \in K_m$, lies in K_m^+ . The compact-window concentration sets are

$$\tilde{A}_{n,m}^\eta := \{z \in K_m : \mu_n(\mathcal{Q}_1^{\text{cov}}(z)) \geq \eta\}.$$

For the Hausdorff–Fell extraction we use the equivalent smoothed version below; thresholds are adjusted only by fixed cutoff constants. Fix $\chi \in C_c^\infty(Q_1(0,0))$, $0 \leq \chi \leq 1$, with $\chi \equiv 1$ on $Q_{1/2}(0,0)$. For $z \in Z$, put

$$\chi_z(Y, \tau) := \chi(\mathcal{R}_z^{-1}(Y, \tau)), \quad M_n(z) := \int \chi_z d\mu_n.$$

For $\eta > 0$, define the compact-window concentration set

$$A_{n,m}^\eta := \{z \in K_m : M_n(z) \geq \eta\}.$$

After passing to a subsequence, for every m and every rational $\eta > 0$, the compact sets $A_{n,m}^\eta$ converge in the Hausdorff–Fell topology to compact sets $A_m^\eta \subset K_m$. Moreover, for every open neighborhood $O \subset K_m$ of A_m^η ,

$$\limsup_{n \rightarrow \infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta.$$

Thus outside every neighborhood of the concentration set A_m^η , no η -concentrating covariant unit tube remains in the compact observer window K_m . The concentration set is the closed locus A_m^η , not a finite or countable enumeration of tubes.

Proof. For fixed n , the map $z \mapsto \chi_z$ is continuous in C_c on K_m , and all supports are contained in the compact enlargement K_m^+ . Since μ_n is a Radon measure finite on K_m^+ , M_n is continuous on K_m . Therefore $A_{n,m}^\eta$ is compact.

The hyperspace of compact subsets of the compact metric space K_m is compact in the Hausdorff–Fell topology. Hence, for each fixed pair (m, η) , a subsequence exists along which $A_{n,m}^\eta$ converges. A diagonal argument over $m \in \mathbb{N}$ and $\eta \in \mathbb{Q}_+$ gives simultaneous convergence.

Let $O \subset K_m$ be an open neighborhood of A_m^η . If the stated residual bound failed, there would exist $\delta > 0$, a subsequence n_k , and points $z_{n_k} \in K_m \setminus O$ such that

$$M_{n_k}(z_{n_k}) \geq \eta + \delta.$$

After replacing O by a slightly smaller open neighborhood of A_m^η , we may assume $K_m \setminus O$ is compact. Passing to a further subsequence,

$$z_{n_k} \rightarrow z_* \in K_m \setminus O.$$

For all large k ,

$$z_{n_k} \in A_{n_k,m}^{\eta+\delta/2}.$$

Hausdorff–Fell convergence gives $z_* \in A_m^{\eta+\delta/2}$. Since $A_m^{\eta+\delta/2} \subset A_m^\eta \subset O$, this contradicts $z_* \in K_m \setminus O$. The residual estimate follows. All bounds used here are local on K_m^+ ; no global CKN budget or countability of concentrating profiles is assumed. $\square$

Definition 9.7 (Residual measure after concentration-set removal). Let $U_\rho(A_{n,m}^\eta)$ be the ρ -neighborhood of the concentration set in the compact observer window K_m . The residual measure after deleting positive-threshold concentration neighborhoods is

$$\mu_{n,m,\eta,\rho}^{\text{rem}} := \mu_n \lfloor (K_m \setminus U_\rho(A_{n,m}^\eta)).$$

The terminal trichotomy uses this restricted measure: vanishing means $\mu_{n,m,\eta,\rho}^{\text{rem}}(K_m) \rightarrow 0$ along the selected extraction; diffuse defect means positive residual mass remains while no unit cylinder exceeds the concentration threshold after deletion; noncompact defect means that an escaping residual recentering loses local compactness or pressure-gauge control under its own normalization.

Lemma 9.8 (Oscillatory observers are CKN-concentrating). *Let μ_n be the local defect measure used in Theorem 9.6, so that μ_n dominates the velocity measure $|U_n|^3 dy d\tau$. If a covariant unit observer z satisfies*

$$\text{osc}_{3,\text{ct}}(U_n; \mathcal{Q}_1^{\text{cov}}(z)) \geq \eta,$$

then the same observer is CKN-concentrating at threshold η :

$$\mu_n(\mathcal{Q}_1^{\text{cov}}(z)) \geq \eta.$$

Consequently, after the CKN concentration sets of Theorem 9.6 are removed at every positive threshold, no positive constant-in-time oscillatory unit observer remains outside those sets.

Proof. By definition of the constant-in-time oscillation,

$$\text{osc}_{3,\text{ct}}(U_n; \mathcal{Q}_1^{\text{cov}}(z)) = \inf_{c \in \mathbb{R}^3} \iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n - c|^3 \leq \iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n|^3.$$

The local defect measure contains the velocity contribution, hence

$$\iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n|^3 \leq \mu_n(\mathcal{Q}_1^{\text{cov}}(z)).$$

Thus oscillation implies CKN concentration with the same threshold. The last statement follows by applying the local concentration-set extraction theorem to all rational positive thresholds and then lowering thresholds by harmless fixed cover and cutoff constants when the smoothed bump convention is used. $\square$

Lemma 9.9 (Spatial quotient defect measures are locally CKN-controlled). *On the same bounded-overlap covariant unit-cylinder cover, define the spatial quotient oscillation measure*

$$\nu_U^{\text{sp}}(E) := \sum_{z_k \in E} \text{osc}_{3,\text{sp}}(U; \mathcal{Q}_1^{\text{cov}}(z_k)).$$

Then ν_U^{sp} is locally finite on compact observer windows and is dominated, up to the fixed cover multiplicity, by the local velocity part of the defect measure. In particular, every compactness argument in Theorem 9.16 applies without change to ν_U^{sp} after concentration-set removal.

Proof. For each unit covariant cylinder,

$$\text{osc}_{3,\text{sp}}(U; \mathcal{Q}_1^{\text{cov}}(z_k)) \leq \iint_{\mathcal{Q}_1^{\text{cov}}(z_k)} |U|^3.$$

Summing over a bounded-overlap cover gives local finiteness and domination by the local velocity contribution, hence by the local defect measure. The proof of Theorem 9.16 uses only positivity, local finiteness, bounded-overlap domination, and weak compactness of normalized restrictions on the observer compactification. These properties hold for ν_U^{sp} exactly as they hold for the constant-in-time oscillation measure μ_U^{osc} . $\square$

Lemma 9.10 (Separation of recentered cylinders). *Let $\mathbf{c}_i = (z_{i,n})$ and $\mathbf{c}_j = (z_{j,n})$ be asymptotically separated recentering sequences. For every fixed $R < \infty$, the cylinder $Q_R(z_{j,n})$, written in the coordinates of $\mathbf{c}_i$, leaves every compact subset of the recentered variables. Consequently no positive compact velocity L^3 -mass assigned to $\mathbf{c}_j$ is visible on any fixed compact cylinder of the $\mathbf{c}_i$ -frame.*

Proof. In the $\mathbf{c}_i$ -coordinates the center of $Q_R(z_{j,n})$ is

$$(y_{j,n} - y_{i,n}, \tau_{j,n} - \tau_{i,n}).$$

The parabolic distance of this point from the origin tends to infinity by asymptotic separation. Hence, for every fixed compact terminal cylinder K , one has

$$K \cap (Q_R(z_{j,n}) - z_{i,n}) = \emptyset$$

for all sufficiently large n . This is a purely local statement about the relative positions of the two recentering sequences. It does not estimate the solution on all of space and does not use a global norm. $\square$

Theorem 9.11 (Paired terminal concentration-profile decomposition). *Every terminal concentration sequence admits a paired decomposition*

$$\text{retained concentration set} + \text{residual remainder}.$$

More precisely, after passing to a subsequence, the compact observer-window concentration sets A_m^η of Theorem 9.6 exist simultaneously for every m and every rational $\eta > 0$. Every observer sequence that remains in these sets at a fixed positive threshold and has no loss of local compactness generates a bounded retained terminal profile after local pressure gauges. Outside every neighborhood of the concentration set in each fixed compact observer window, the residual local CKN mass is below the chosen threshold in the precise concentration-deleted sense of $\mu_{n,m,\eta,\rho}^{\text{rem}}$ from Theorem 9.7. The residual remainder is then of exactly one of the following types:

$$\mathcal{R}_{\text{van}}, \quad \mathcal{R}_{\text{diff}}, \quad \mathcal{R}_{\text{noncomp}}.$$

Here $\mathcal{R}_{\text{van}}$ means the concentration-deleted residual measures vanish on compact observer windows and then on the selected expanding regions; $\mathcal{R}_{\text{diff}}$ means positive concentration-deleted residual mass remains while every fixed unit cylinder is below threshold; and $\mathcal{R}_{\text{noncomp}}$ means that some escaping residual recentering loses the local Seregin compactness or pressure-gauge control. Thus mixed configurations such as one compact concentration profile plus diffuse exterior remainder are part of the decomposition; they are not silently identified with the pure diffuse stratum.

Proof. Apply the local concentration-set extraction theorem, Theorem 9.6, on the compact observer exhaustion $\{K_m\}$. It produces closed concentration sets A_m^η and the residual estimate

$$\limsup_{n \rightarrow \infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta$$

outside every neighborhood O of A_m^η . This is the precise replacement for a maximal finite or countable tube family: concentrating centers are recorded by closed loci, and the residual is defined by restricting away from their neighborhoods.

If an observer sequence stays in a positive-threshold concentration set and has no loss of the local compactness specified in Theorem 9.3, then Theorem 9.5 and the standard diagonal Seregin compactness argument give a bounded retained terminal profile. If such an observer sequence loses local dissipation or pressure-gauge control, the corresponding remainder is $\mathcal{R}_{\text{noncomp}}$.

It remains to classify what is outside the concentration sets. If $\mu_{n,m,\eta,\rho}^{\text{rem}}(K_m) \rightarrow 0$ for every compact observer window and then no residual mass survives on the selected expanding terminal regions, the remainder is $\mathcal{R}_{\text{van}}$. If a positive amount of residual mass persists on expanding regions while every fixed compact observer window is below every concentration threshold after removing neighborhoods of the concentration sets, the remainder is $\mathcal{R}_{\text{diff}}$. These three residual states are mutually exclusive once the concentration sets are fixed, and they exhaust the residual possibilities. No step asserts that a diffuse or exterior remainder cannot coexist with compact concentration profiles. $\square$

Corollary 9.12 (Persistent local velocity mass forces retained concentration). *If a terminal sequence satisfies*

$$\liminf_{n \rightarrow \infty} \iint_{Q_1(0,0)} |V_n|^3 dy d\tau \geq \eta_0 > 0$$

with $\eta_0 \geq \varepsilon_{\text{sd}}$, then the terminal decomposition contains a local concentration profile unless the distinguished observer sequence has loss of local compactness. In the generic residual after

local vanishing and local compactness failure are excluded, the persistent compact velocity mass is therefore carried by the retained concentration set of [Theorem 9.11](#); the locus may contain one, finitely many, infinitely many, or continuum many concentrating observer classes, and the residual remainder may still be diffuse until the mixed compact–diffuse stratum is excluded.

Proof. The local defect measure μ_n dominates the velocity contribution. Hence the origin recentering sequence has μ -mass at least η_0 , and is ε_{sd} -concentrating unless it has loss of local compactness. If it has no loss of local compactness, concentration-set extraction places its observer class in a positive-threshold concentration set A_m^η of [Theorem 9.11](#). $\square$

Lemma 9.13 (Local vanishing and exterior alternatives do not carry compact concentration). *Let a terminal extraction inherit the positive compact velocity L^3 -mass supplied by the setup paper `proof_setup.tex`. Then the retained compact velocity mass cannot be carried solely by local vanishing, exterior escape, or diffuse exterior concentration.*

Proof. The assertion is local. By the local no-escape and concentration input in `proof_setup.tex`, failure of local vanishing at a singular terminal point gives a positive local velocity concentration sequence. The setup paper carries this through the Type I Seregin extraction as the compact velocity lower bound (1.5). Cover the fixed compact cylinder K in (1.5) by finitely many unit terminal cylinders. At least one cylinder in the cover carries a positive velocity lower bound, say $\eta_K > 0$. Since μ_n dominates $|V_n|^3 dy d\tau$, choosing $\varepsilon_{\text{sd}} \leq \eta_K$ makes that unit cylinder ε_{sd} -concentrating. Thus the distinguished compact concentration cannot be locally vanishing.

Local vanishing after removal of concentrating recentering sequences means that the residual profile has local CKN mass below ε_{sd} on each fixed compact terminal cylinder; it therefore cannot contain the fixed compact velocity lower bound. An exterior escaping concentration is visible only after its own recentering; relative to the distinguished compact terminal cylinder it escapes every bounded window. By [Theorem 9.10](#), it cannot be the sole location of a lower bound that remains in the distinguished compact window. Finally, diffuse exterior concentration is defined by persistence through expanding regions while all fixed unit cylinders are locally vanishing. If such a sequence contained the compact lower-bound threshold from `proof_setup.tex`, the positive concentration input in that framework would select a unit terminal cylinder with mass at least the retained threshold, contradicting local vanishing. Thus none of these alternatives can be the unique location of the retained compact velocity mass. $\square$

Theorem 9.14 (Local exclusion of noncompact and nonconcentrating terminal alternatives). *For terminal sequences generated by the Seregin setup theorem in `proof_setup.tex`, loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are not possible sole locations of the retained compact velocity mass.*

Proof. Retained recentering sequences with loss of local compactness are removed by [Theorem 9.5](#). Local vanishing, exterior escape, and diffuse exterior concentration are removed by [Theorem 9.13](#). The proof uses only the positive concentration input and the local compactness framework in `proof_setup.tex`, and fixed compact terminal cylinders. $\square$

Definition 9.15 (Observer compactification). We fix a metrizable compactification $\overline{Z}$ of the observer space $Z = \mathbb{R}^3 \times \mathbb{R}$ with the following properties.

- (i) Compact observer balls of finite covariant radius embed continuously.
- (ii) The covariant observer maps generated by T_a and Θ_s extend continuously to $\overline{Z}$.
- (iii) Unit-cylinder concentration evaluation maps are upper semicontinuous after the fixed smoothing convention of [Theorem 9.6](#).
- (iv) The normalized restrictions of locally finite defect measures are tight as probability measures on $\overline{Z}$.

Such a compactification is obtained by embedding Z into the product of compact intervals generated by a countable separating family of bounded continuous observer functions, including the smoothed unit-cylinder evaluation functions and the coordinate cutoffs for a compact exhaustion, and then taking the closure.

Theorem 9.16 (Diffuse-defect compactness). *Let μ_n^{osc} be the residual constant-in-time oscillation measures of Theorem 8.8, after removing neighborhoods of all positive-threshold concentration sets from Theorem 9.6. By Theorem 9.8, this removal also removes all positive-threshold oscillatory unit observers on compact windows. Assume there are observer regions $E_n \subset Z$, escaping every compact subset of Z , and a number $m_0 > 0$ such that*

$$\mu_n^{\text{osc}}(E_n) \geq m_0,$$

while no unit observer in E_n carries a positive oscillation threshold:

$$\sup_{z \in E_n} \text{osc}_{3,\text{ct}}(U_n; \mathcal{Q}_1^{\text{cov}}(z)) \longrightarrow 0.$$

Let $\bar{Z}$ be the compact metrizable observer compactification of Theorem 9.15, and define probability measures

$$\lambda_n := \frac{\mu_n^{\text{osc}} \lfloor E_n}{\mu_n^{\text{osc}}(E_n)} \quad \text{on } \bar{Z}.$$

After passing to a subsequence,

$$\lambda_n \rightharpoonup \lambda$$

weakly as probability measures on $\bar{Z}$. The limit satisfies

$$\lambda(\bar{Z}) = 1, \quad \lambda(B) = 0 \quad \text{for every } B \Subset Z.$$

Thus the limit is a nonzero diffuse-defect profile supported at the boundary of the observer compactification, not a local concentration profile.

Proof. The measures λ_n are probability measures on the compact metric space $\bar{Z}$, so weak compactness gives a convergent subsequence. Testing against the constant function 1 preserves total mass and gives $\lambda(\bar{Z}) = 1$.

Let $B \Subset Z$. Since E_n escapes every compact subset of Z , one has $B \cap E_n = \emptyset$ for all sufficiently large n . Hence $\lambda_n(B) = 0$ eventually, and the portmanteau theorem gives $\lambda(B) = 0$. The additional unit-scale diffuseness assumption gives the stronger moving-window estimate

$$\sup_{z \in E_n} \lambda_n(B_Z(z, 1)) \leq \frac{C}{m_0} \sup_{\zeta \in E_n} \text{osc}_{3,\text{ct}}(U_n; \mathcal{Q}_1^{\text{cov}}(\zeta)) \longrightarrow 0.$$

Here $B_Z(z, 1)$ is any fixed unit ball for a proper metric on Z , and C is the bounded-overlap constant of the observer cover. The restriction $\lambda_n = \mu_n^{\text{osc}} \lfloor E_n / \mu_n^{\text{osc}}(E_n)$ is why only cover centers lying in E_n enter the estimate. Therefore no positive unit-scale local oscillation profile is hidden inside the escaping windows. The proof is local on the chosen observer windows and uses no global critical-mass or energy budget. $\square$

Definition 9.17 (Diffuse-defect profile class). Let $\mathfrak{D}_M$ be the compact closure, in the weak topology of probability measures on $\bar{Z}$ together with the local $\mathfrak{X}_M$ -topology for the associated velocity hulls, of all normalized diffuse-defect profiles produced by Theorem 9.16. The diffuse-defect mass functional is normalized to be identically one on $\mathfrak{D}_M$. This compactness is the product of weak compactness of probability measures on the compact observer space $\bar{Z}$ and the local Seregin compactness of bounded velocity hulls in $\mathfrak{X}_M$, after the admissible pressure gauges. The covariant observer group

$$G := \mathbb{R}^3 \rtimes \mathbb{R}$$

acts on $\mathfrak{D}_M$ by the observer maps generated by T_a and Θ_s .

Theorem 9.18 (Recurrent diffuse-defect measure). *The covariant observer action of G is continuous on $\mathfrak{D}_M$. Every nonzero diffuse-defect profile $\lambda \in \mathfrak{D}_M$ generates a G -invariant Borel probability measure $\mathbb{P}$ on $\mathfrak{D}_M$. The normalized diffuse-defect mass remains one under this averaging.*

Proof. Continuity is part of the compactified observer topology in [Theorem 9.17](#): the action is continuous on the compact observer compactification, and the associated velocity hulls transform continuously in the local smooth topology with the admissible pressure gauges. The group $G = \mathbb{R}^3 \rtimes \mathbb{R}$, with the semidirect law from [Theorem 8.5](#), is solvable and hence amenable. Let (F_α) be a left Følner net for G , with Haar measure $|F_\alpha|$, and define probability measures on $\mathfrak{D}_M$ by

$$\mathbb{P}_\alpha := \frac{1}{|F_\alpha|} \int_{F_\alpha} \delta_{g\lambda} dg.$$

Compactness of $\mathfrak{D}_M$ implies weak-* compactness of its probability measures, so a subnet converges weakly to a probability measure $\mathbb{P}$. For every continuous observable F on $\mathfrak{D}_M$ and every fixed $h \in G$,

$$\int F(h\omega) d\mathbb{P}_\alpha(\omega) - \int F(\omega) d\mathbb{P}_\alpha(\omega) = \frac{1}{|F_\alpha|} \int_{F_\alpha} (F(hg\lambda) - F(g\lambda)) dg.$$

The Følner property makes the right-hand side tend to zero. Passing to the weak limit gives G -invariance of $\mathbb{P}$. Since the diffuse-defect mass is identically one on $\mathfrak{D}_M$, its average against $\mathbb{P}$ is also one. $\square$

Theorem 9.19 (Diffuse-defect trichotomy). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a normalized terminal residual sequence with $\|U_n\|_{L^\infty} \leq M$. Suppose that, after removing the concentration sets at every positive affine-normalized threshold, there are escaping observer windows E_n such that*

$$\mu_n^{\text{osc}}(E_n) \geq m_0 > 0.$$

Then, after passing to a subsequence, the residual diffuse defect falls into one of the following ordered alternatives:

- D1. Regenerated concentration. *There exist covariant observers z_n and a threshold $\eta > 0$ such that*

$$\text{osc}_{3,\text{ct}}(U_n; \mathcal{Q}_1^{\text{cov}}(z_n)) \geq \eta.$$

This produces a retained concentrating descendant.

- D2. Affine/parasitic collapse. *Every velocity tangent associated with the diffuse-defect hull is locally constant after the admissible affine normalization; the defect belongs to the affine/parasitic lower stratum of [Theorem 8.16](#).*

- D3. Critical diffuse tail. *There is a nonzero recurrent diffuse-defect profile supported on $\bar{Z} \setminus Z$, carrying normalized diffuse oscillation mass one, annihilating every fixed unit concentration observable, and not contained in the affine/parasitic lower stratum.*

Equivalently, residual diffuse defects are contained in the union of the regenerated concentration alternative, the affine/parasitic stratum, and the critical diffuse-tail class $\mathcal{C}_{\text{critdiff}}$.

Proof. If D1 occurs, the asserted trichotomy holds. Assume D1 does not occur. Then, for every fixed $\eta > 0$, all sufficiently large n satisfy

$$\sup_{z \in E_n} \text{osc}_{3,\text{ct}}(U_n; \mathcal{Q}_1^{\text{cov}}(z)) < \eta.$$

The hypotheses of [Theorem 9.16](#) apply to $\mu_n^{\text{osc}}|_{E_n}$, and hence produce a diffuse-defect profile λ with total mass one and zero mass on each compact subset of Z . By [Theorem 9.18](#), the orbit closure of λ carries a G -invariant probability measure $\mathbb{P}$.

If every local velocity tangent in the support of $\mathbb{P}$, after the canonical affine normalization, is locally constant, then [Theorem 8.16](#) places the defect in the affine lower stratum. This is D2. If not, the invariant diffuse boundary profile has positive normalized spatial quotient oscillation mass, no unit-scale concentration observable, and non-affine velocity content. This is precisely the critical diffuse-tail alternative D3. The alternatives are ordered by first extracting all regenerated concentration and then all affine/parasitic collapse; after those removals, the remaining nonzero diffuse profile is $\mathcal{C}_{\text{critdiff}}$. $\square$

Definition 9.20 (Critical-tail compactification). The class $\mathcal{C}_{\text{critdiff}}$ consists of recurrent diffuse-defect profiles produced by [Theorem 9.19](#) which are supported on the boundary of the covariant observer compactification, carry normalized spatial quotient oscillation mass one, annihilate every fixed unit concentration observable, and are not affine/parasitic.

The critical-tail compactification $\mathfrak{T}_M$ is the compact profile class of triples

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda)$$

generated by such critical diffuse profiles after macroscopic blow-down. Here $\bar{W}$ is the barycentric macroscopic velocity, $\mathcal{Y}_{x,s}$ is the Young measure recording unresolved bounded macroscopic oscillation, and Λ is the log-scale diffuse measure recording mass escaping every fixed multiplicative annulus. The coherent subspace $\mathfrak{T}_{\text{coh}}$ is the special case

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}, \quad \Lambda = 0.$$

The purely log-diffuse subspace is denoted $\mathfrak{T}_{\text{logdiff}}$, and the oscillatory Young-tail subspace is denoted $\mathfrak{T}_{\text{Young}}$. Convergence in $\mathfrak{T}_M$ means local weak-* convergence of the bounded velocity representatives, pressure convergence modulo time and affine gauges in the distribution topology needed for the limiting equation, vague convergence of the defect measures, weak convergence of the Young measures against bounded continuous test functions, convergence of the logarithmic annular coordinates, and convergence in the affine pressure quotient fixed in [Theorem 8.14](#).

For coherent elements, the corresponding macroscopic tangent pair (W, Q) belongs to $\mathcal{C}_{\text{cohcrit}}$ and satisfies on annular cylinders $\{0 < r_0 < |x| < r_1 < \infty\} \times I$

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \quad \nabla \cdot W = 0$$

in distributions, with nonzero spatial quotient oscillation and recurrence under the induced logarithmic dilation flow.

9.1. Log-annular virial identities for recurrent critical defects. Throughout this subsection let $\chi \in C_c^\infty((-2, 2))$ satisfy $0 \leq \chi \leq 1$ and $\chi \equiv 1$ on $(-1, 1)$. For $\rho \in \mathbb{R}$ and $A \geq 1$, set

$$\phi_{\rho,A}(y) := \chi\left(\frac{\log |y| - \rho}{A}\right), \quad R := e^\rho.$$

The cutoff is used only on annuli avoiding the origin. When differentiating with respect to ρ , A is kept fixed.

Lemma 9.21 (Log-annular oscillation virial identity). *Let $(U, \Pi) \in \mathfrak{X}_M$ be a smooth bounded solution of (1.2). Fix $A \geq 1$, $\rho \in \mathbb{R}$, and put $\phi = \phi_{\rho,A}$. Define the weighted affine gauge*

$$c_{\rho,A}(\tau) := \frac{\int \phi(y) U(y, \tau) dy}{\int \phi(y) dy}, \quad V_{\rho,A}(y, \tau) := U(y, \tau) - c_{\rho,A}(\tau),$$

so that

$$\int \phi V_{\rho,A} dy = 0.$$

Set

$$\begin{aligned} E_{\rho,A}(\tau) &:= \int \phi |V_{\rho,A}|^2 dy, & \mathcal{V}_{\rho,A}(\tau) &:= R^{-1} E_{\rho,A}(\tau), \\ D_{\rho,A}(\tau) &:= \int \phi |\nabla U|^2 dy, & \mathcal{D}_{\rho,A}(\tau) &:= R^{-1} D_{\rho,A}(\tau). \end{aligned}$$

Then $\mathcal{V}_{\rho,A}$ is locally absolutely continuous in τ , and for a.e. τ , after subtracting an arbitrary pressure gauge $\pi(\tau)$, one has

$$\begin{aligned} (9.1) \quad \partial_\tau \mathcal{V}_{\rho,A} + 2\mathcal{D}_{\rho,A} + \frac{1}{2} \partial_\rho \mathcal{V}_{\rho,A} &= R^{-1} \int (\Delta \phi) |V_{\rho,A}|^2 dy \\ &+ R^{-1} \int |V_{\rho,A}|^2 U \cdot \nabla \phi dy \\ &+ 2R^{-1} \int (\Pi - \pi(\tau)) V_{\rho,A} \cdot \nabla \phi dy. \end{aligned}$$

Moreover

$$(9.2) \quad |\nabla \phi_{\rho,A}(y)| \leq \frac{C_\chi}{A|y|}, \quad |\Delta \phi_{\rho,A}(y)| \leq \frac{C_\chi}{A|y|^2} + \frac{C_\chi}{A^2|y|^2},$$

and

$$y \cdot \nabla \phi_{\rho,A}(y) = \frac{1}{A} \chi' \left(\frac{\log |y| - \rho}{A} \right), \quad \partial_\rho \phi_{\rho,A} = -y \cdot \nabla \phi_{\rho,A}.$$

Proof. Since $c_{\rho,A}(\tau)$ is the weighted least-squares affine gauge,

$$\int \phi V_{\rho,A} dy = 0.$$

Hence the derivative of the gauge drops out:

$$\frac{d}{d\tau} \int \phi |U - c_{\rho,A}|^2 = 2 \int \phi V_{\rho,A} \cdot \partial_\tau U.$$

Substitute the centered equation

$$\partial_\tau U = \Delta U - (U \cdot \nabla) U - \frac{1}{2} y \cdot \nabla U - \frac{1}{2} U - \nabla \Pi.$$

The diffusion term gives

$$2 \int \phi V \cdot \Delta U = -2 \int \phi |\nabla U|^2 + \int (\Delta \phi) |V|^2.$$

The convective term gives, using $U \cdot \nabla U = U \cdot \nabla V$ and $\nabla \cdot U = 0$,

$$-2 \int \phi V \cdot (U \cdot \nabla U) = - \int \phi U \cdot \nabla |V|^2 = \int |V|^2 U \cdot \nabla \phi.$$

The centered drift and linear terms give

$$\begin{aligned} - \int \phi V \cdot (y \cdot \nabla U) - \int \phi V \cdot U &= -\frac{1}{2} \int \phi y \cdot \nabla |V|^2 - \int \phi |V|^2 \\ &= \frac{1}{2} \int y \cdot \nabla \phi |V|^2 + \frac{1}{2} \int \phi |V|^2. \end{aligned}$$

The terms involving $c_{\rho,A}$ vanish because $\int \phi V = 0$. Finally, for any scalar gauge $\pi(\tau)$,

$$-2 \int \phi V \cdot \nabla \Pi = 2 \int (\Pi - \pi(\tau)) V \cdot \nabla \phi,$$

because $\nabla \cdot V = 0$ and

$$\int \pi(\tau) V \cdot \nabla \phi \, dy = \pi(\tau) \int \nabla \cdot (\phi V) \, dy = 0.$$

Combining these identities yields

$$\begin{aligned} \partial_\tau E_{\rho,A} + 2D_{\rho,A} &= \int (\Delta \phi) |V|^2 + \int |V|^2 U \cdot \nabla \phi + 2 \int (\Pi - \pi) V \cdot \nabla \phi \\ &\quad + \frac{1}{2} \int \phi |V|^2 + \frac{1}{2} \int y \cdot \nabla \phi |V|^2. \end{aligned}$$

Since

$$\partial_\rho \phi_{\rho,A} = -y \cdot \nabla \phi_{\rho,A}$$

and the weighted least-squares condition also removes the $\partial_\rho c_{\rho,A}$ term,

$$\partial_\rho E_{\rho,A} = \int \partial_\rho \phi_{\rho,A} |V|^2 = - \int y \cdot \nabla \phi_{\rho,A} |V|^2.$$

Therefore

$$\frac{1}{2} R^{-1} \left(\int \phi |V|^2 + \int y \cdot \nabla \phi |V|^2 \right) = -\frac{1}{2} \partial_\rho (R^{-1} E_{\rho,A}) = -\frac{1}{2} \partial_\rho \mathcal{V}_{\rho,A}.$$

Thus, after multiplying the energy identity by R^{-1} , the centered drift and zeroth-order contribution is $-\frac{1}{2} \partial_\rho \mathcal{V}_{\rho,A}$. Moving it to the left gives (9.1). The cutoff bounds follow by direct differentiation in polar coordinates. $\square$

Remark 9.22 (Virial normalization). The normalization $R^{-1} \int \phi |U - c|^2$ is the correct scale for an actual spatial tail of size $|y|^{-1}$. It is not automatically a compact observable for an arbitrary bounded residual profile. In the critical-tail reduction below, one uses it either on selected scale-critical defect windows or as an observable on the compact recurrent defect hull after the required scale-critical virial tightness has been verified. Without that compactness, the virial identity remains valid for smooth annular cutoffs but cannot by itself close the critical-tail alternatives.

Lemma 9.23 (Realized log-window virial compactness). *Let $\mathcal{D} \in \mathfrak{T}_M$ be a recurrent critical-tail defect generated by a normalized terminal sequence realized from the original blow-up sequence after concentration-set removal and affine normalization. Then, along the selected defect subsequence and after the admissible local pressure gauges of Theorem 1.18, the annular virial observables $\mathcal{V}_{\rho,A}$, the local dissipation $\mathcal{D}_{\rho,A}$, and the right-hand side flux contributions*

$$R^{-1} \int (\Delta \phi) |V|^2, \quad R^{-1} \int |V|^2 U \cdot \nabla \phi, \quad 2R^{-1} \int (\Pi - \pi) V \cdot \nabla \phi$$

appearing in (9.1) are compact in the local distributional topology and define continuous observables on the closed hull of $\mathcal{D}$ under the joint time-translation and log-translation flow. Equivalently, the hypothesis on legitimate compact observability invoked in Theorem 9.24 is automatic for realized recurrent defects, after either ordinary minimal mesoscopic-scale reduction or assignment to the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$.

Proof. Each ingredient is bounded by the uniform L^∞ bound on the realizing sequence and by the local pressure-gauge compactness inherited through Theorem 1.18. Specifically, on every fixed log-annular cylinder $\Sigma_{\rho,A} = \{e^{\rho-A} \leq |y| \leq e^{\rho+A}\} \times [s_0, s_1]$, the cutoff ϕ and its derivatives are supported away from the spatial origin and satisfy the scaled bounds $|\nabla \phi| \lesssim |y|^{-1}$, $|\Delta \phi| \lesssim |y|^{-2}$ from (9.2). Combined with $\|U_n\|_{L^\infty(\Sigma_{\rho,A})} \leq M$, the integrals $\int \phi |V_{\rho,A}|^2$, $\int (\Delta \phi) |V|^2$, and $\int |V|^2 U \cdot \nabla \phi$ are uniformly bounded. The pressure flux term $\int (\Pi - \pi) V \cdot \nabla \phi$ involves only the cutoff gradient, supported away from the origin; using the local pressure gauge subtraction

$\Pi - \pi$ recorded in [Theorems 1.19](#) and [9.45](#), the corresponding integrals are uniformly bounded in L^1 . Local dissipation $\int \phi |\nabla U|^2$ is uniformly bounded along the realizing sequence by the local energy inequality of the suitable weak formulation.

Continuity in the local distributional topology is the standard fact that each integral is a continuous bilinear pairing of the L^∞ field against a smooth compactly supported test function, with the pressure gauge chosen to make $\Pi - \pi$ locally $L^{3/2}$ -compact, see [Theorem 9.46](#). The joint time-translation and log-translation flow extends these observables to the recurrent hull by density. If at any stage the homogeneous gauge sequence loses compactness, [Theorem 9.49](#) assigns the descendant to $\mathcal{C}_{\text{aff}}$ and the assertion is recorded as the affine alternative. In the complementary case the observability hypothesis of [Theorem 9.24](#) is verified, and the reduction provided by the minimal mesoscopic-scale theorem [Theorem 9.54](#) closes the diagram. $\square$

Lemma 9.24 (Recurrent virial equality case and reduction). *Let $\mathcal{D} \in \mathfrak{T}_M$ be a recurrent critical-tail defect generated by a bounded normalized terminal sequence after concentration-set removal and affine normalization. By [Theorem 9.23](#) the log-annular virials in [Theorem 9.21](#) are legitimate compact observables on the recurrent hull of $\mathcal{D}$ in the sense of [Theorem 9.22](#), after either minimal mesoscopic-scale reduction or assignment to $\mathcal{C}_{\text{aff}}$. Let $\mathbb{P}$ be an invariant probability measure on that hull for the joint time-translation and log-translation flow. Assume also that the averaged pressure gauges are chosen by a local annular pressure decomposition, so that the three right-hand side terms in [\(9.1\)](#) have limits as defect observables. Then the following conclusions hold.*

(i) Zero-flux equality case. *If*

$$(9.3) \quad \lim_{A \rightarrow \infty} \int \left| R^{-1} \int (\Delta \phi) |V|^2 + R^{-1} \int |V|^2 U \cdot \nabla \phi + 2R^{-1} \int (\Pi - \pi) V \cdot \nabla \phi \right| d\mathbb{P} = 0,$$

then every local tangent in the support of $\mathbb{P}$ has zero spatial gradient. Consequently

$$\mathcal{D} \in \mathcal{C}_{\text{aff}}.$$

(ii) Tight Dirac flux reduction. *Suppose [\(9.3\)](#) fails, and suppose the nonzero limiting flux is neither concentrated on a positive-threshold unit concentrating observer nor lost to logarithmic infinity. Suppose moreover that the annular blow-downs selected by this nonzero flux generate a Dirac Young measure. Then the selected annular blow-downs converge strongly in L^3_{loc} on macroscopic annular cylinders to a nonzero coherent critical tail. Hence*

$$\mathcal{D} \in \mathcal{C}_{\text{cohcrit}}.$$

The minimal mesoscopic-scale reduction theorem proved below supplies the tight-Dirac alternative for realized recurrent defects. Combined with that theorem, the preceding alternatives give the inclusion

$$\mathcal{D} \in \mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Proof. Integrate [\(9.1\)](#) against the invariant measure $\mathbb{P}$. The averaged $\partial_\tau \mathcal{V}_{\rho,A}$ term vanishes by time-translation invariance. The averaged $\partial_\rho \mathcal{V}_{\rho,A}$ term vanishes by log-translation invariance. Therefore

$$2 \int \mathcal{D}_{\rho,A} d\mathbb{P} = \int \mathcal{F}_{\rho,A} d\mathbb{P},$$

where $\mathcal{F}_{\rho,A}$ denotes the sum of the three right-hand side flux terms in [\(9.1\)](#). Under [\(9.3\)](#) and by nonnegativity of the dissipation,

$$\lim_{A \rightarrow \infty} \int \mathcal{D}_{\rho,A} d\mathbb{P} = 0.$$

Choose a countable exhausting family of ordinary unit covariant cylinders. For each cylinder, choose A large enough that the corresponding cutoff is identically one on a fixed enlargement.

The preceding limit then implies that, for $\mathbb{P}$ -a.e. profile, the local dissipation $\int_Q |\nabla U|^2$ vanishes on every cylinder in the exhausting family. Hence $\nabla U = 0$ distributionally on every compact cylinder. Thus every local tangent is spatially constant. By the affine normalization dichotomy, this is precisely the affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$.

We now prove the tight Dirac flux reduction. Since the averaged flux does not vanish and is not concentrated on a unit concentrating observer, there are logarithmic annuli with nonzero normalized flux but with no positive-threshold unit constant-in-time oscillatory subcylinder. Since the flux is not lost to logarithmic infinity, these annuli may be selected with a tight macroscopic scale $R_n \rightarrow \infty$. Passing to the corresponding macroscopic variables gives bounded blown-down fields

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s)$$

on every fixed annular cylinder. Terminal compactness gives a weak-* L_{loc}^∞ limit and an associated Young measure. By the Dirac hypothesis on the selected flux, the Young measure is δ_W almost everywhere. Uniform boundedness then upgrades convergence to strong L_{loc}^3 convergence on annular cylinders. Passing to the limit in the blown-down equation, the viscous and nonlinear terms vanish by the factors R_n^{-2} and R_n^{-1} , and the pressure gauges converge in annular distributions. The limit therefore satisfies

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \quad \nabla \cdot W = 0,$$

on macroscopic annular cylinders. The selected flux is nonzero, so the limit has nonzero spatial quotient oscillation. Thus $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$. $\square$

9.2. Logarithmic escape as a hidden-scale obstruction. The logarithmic escape alternative is not ruled out below by a separate global tail estimate. The point of the next reduction is narrower and more useful: once tight annular descendants have no hidden-scale variance, a nonzero logarithmic escape component cannot survive. Thus the log-diffuse alternative and the Young alternative are both reduced to the same minimal mesoscopic-scale theorem.

Definition 9.25 (Log-diffuse recurrent critical defect). A realized recurrent critical-tail defect is log-diffuse if, after projecting the normalized diffuse oscillation measures to the logarithmic radial variable $\rho = \log |y|$, their log marginals λ_n satisfy

$$\sup_{\rho_0 \in \mathbb{R}} \lambda_n([\rho_0 - A, \rho_0 + A]) \rightarrow 0 \quad \text{for every fixed } A < \infty.$$

Equivalently, the critical-tail compactification has a nonzero logarithmic escape component Λ . This alternative is denoted by $\mathcal{C}_{\text{log diff}}$.

Definition 9.26 (Log-window descendant). Let λ_n be the normalized logarithmic defect marginals of a log-diffuse critical defect. A log-window descendant is obtained by choosing numbers $A \geq 1$, centers ρ_n , and intervals

$$I_n = [\rho_n - A, \rho_n + A]$$

with $\lambda_n(I_n) > 0$, restricting the defect measures to I_n , and renormalizing:

$$\tilde{\lambda}_n := \frac{\lambda_n \lfloor I_n}{\lambda_n(I_n)}.$$

After applying the logarithmic recentering $\rho \mapsto \rho - \rho_n$, the renormalized measures are tight in the fixed interval $[-A, A]$. Any critical-tail profile obtained as a subsequential limit of these normalized restrictions is called a log-window descendant of the original log-diffuse defect.

Lemma 9.27 (Nonzero logarithmic escape has tight log-window descendants). *Let $\mathcal{D} \in \mathcal{C}_{\log \text{ diff}}$ be a nonzero realized recurrent critical-tail defect after concentration-set removal and affine normalization. Then, for some fixed $A \geq 1$, $\mathcal{D}$ has a nonzero log-window descendant. The descendant is a tight macroscopic annular critical-tail realization and retains normalized spatial quotient oscillation mass.*

Proof. The logarithmic marginals λ_n are probability measures. Fix any $A \geq 1$. Cover $\mathbb{R}_\rho$ by the intervals

$$I_{k,A} := [2Ak - A, 2Ak + A], \quad k \in \mathbb{Z}.$$

For each n , at least one interval has positive λ_n -mass. Choose one such interval and denote it by $I_n = [\rho_n - A, \rho_n + A]$. Restricting to I_n , renormalizing, and translating ρ_n to the origin gives probability measures supported in the fixed compact interval $[-A, A]$. Compactness of the observer boundary profile class and the local Seregine compactness of $\mathfrak{X}_M$ give a subsequential limit. By construction, the limiting profile has total normalized defect mass one, hence is nonzero.

The logarithmic recentering fixes the selected multiplicative annulus, so the resulting critical-tail realization is tight in logarithmic scale. The original diffuse defect was measured after affine normalization by the oscillation defect measure; the normalized restriction preserves total oscillation mass equal to one. Hence the descendant carries nonzero spatial quotient oscillation. No positive lower bound for the unnormalized mass $\lambda_n(I_n)$ is required, because the descendant is a normalized boundary profile. $\square$

Lemma 9.28 (Renormalized log-window heredity). *Let $\mathcal{D} \in \mathcal{C}_{\log \text{ diff}}$ be a log-diffuse defect realized from the original blow-up sequence, and let $\mathcal{F} \subset \mathfrak{X}_M$ be a closed union of alternatives which is closed under retained descendants and normalized support profiles. Suppose that every nonzero normalized log-window descendant of $\mathcal{D}$ belongs to $\mathcal{F}$. Then the original log-diffuse defect itself has no surviving mass outside $\mathcal{F}$. In particular, if $\mathcal{F}$ has already been excluded by the lower-stratum closure theorems, then $\mathcal{D}$ is impossible.*

Proof. It remains to verify that the selected logarithmic windows remain tied to the original defect even if their unnormalized mass tends to zero. Let Λ be the compactified logarithmic defect measure associated with $\mathcal{D}$. If $\Lambda(\mathfrak{X}_M \setminus \mathcal{F}) > 0$, then by regularity of Borel probability measures on the compact defect space there is a compact set

$$K \Subset \mathfrak{X}_M \setminus \mathcal{F}$$

with $\Lambda(K) > 0$. Pulling K back to the approximating observer compactification and to the logarithmic coordinate gives, after passing to a subsequence, positive-measure windows I_n whose normalized restrictions converge to a probability measure supported in K . Equivalently, one obtains a nonzero normalized log-window descendant outside $\mathcal{F}$.

This contradicts the hypothesis that every nonzero log-window descendant lies in $\mathcal{F}$. Hence Λ is supported on $\mathcal{F}$. Since $\mathcal{F}$ is closed under retained descendants and normalized support profiles and is already excluded, the original defect has no admissible surviving normalized support profile. The fact that $\lambda_n(I_n)$ may tend to zero does not affect the argument: the normalized restriction records a support profile of the original probability measure, and a positive original defect measure cannot be supported entirely on excluded closed alternatives. $\square$

Theorem 9.29 (Logarithmic escape reduces to hidden-scale compactness). *Assume [Theorem 9.56](#). Then every nonzero log-diffuse critical-tail defect generated after concentration-set removal and affine normalization has a nonzero normalized support profile in*

$$\mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Consequently, after the concentration, affine/parasitic, and coherent critical-tail alternatives have been removed, no log-diffuse alternative remains.

Proof. Let $\mathcal{D} \in \mathcal{C}_{\log \text{ diff}}$ be nonzero. By [Theorem 9.27](#), nonzero tight log-window descendants exist. Let $\tilde{\mathcal{D}}$ be any such descendant. If the descendant regenerates a positive-threshold unit concentrating observer, then it belongs to $\mathcal{C}_{\text{conc}}$. If its spatial quotient oscillation vanishes, then [Theorem 8.21](#) places it in $\mathcal{C}_{\text{aff}}$. It remains to consider the case in which it is neither concentrating nor affine.

Because $\tilde{\mathcal{D}}$ was obtained by restriction to a fixed logarithmic window, its logarithmic escape component is zero. The only remaining possible obstruction in the critical-tail compactification is a non-Dirac annular Young defect. By [Theorem 9.56](#), every tight annular blow-down generated by the descendant has no hidden-scale variance. Hence [Theorem 9.32](#) makes the Young measure Dirac. The coherent extraction [Theorem 9.58](#) then gives a nonzero coherent critical tail; thus $\tilde{\mathcal{D}} \in \mathcal{C}_{\text{cohcrit}}$.

In all cases the log-window descendant lies in the closed excluded set

$$\mathcal{F} := \mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

By [Theorem 9.28](#), if every normalized log-window descendant of the original log-diffuse defect lies in $\mathcal{F}$, then the original defect has no surviving mass outside $\mathcal{F}$. Once the concentration, affine/parasitic, and coherent critical-tail alternatives are removed, this makes the original log-diffuse defect impossible. $\square$

Corollary 9.30 (Log-diffuse alternative exclusion from minimal mesoscopic scale reduction). *Inside the retained residual decomposition,*

$$\mathcal{C}_{\log \text{ diff}} = \emptyset.$$

Equivalently, after affine/parasitic lower-stratum removal, logarithmic hidden escape is excluded by the same hidden-scale compactness theorem that removes the Young alternative.

Proof. By [Theorem 9.54](#), [Theorem 9.56](#) applies to all tight annular descendants. Hence [Theorem 9.29](#) reduces every nonzero log-diffuse defect to a descendant in $\mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}$. The concentration alternative has been removed by terminal concentration-set extraction, the affine stratum by [Theorem 9.61](#), and the coherent critical-tail alternative by [Theorem 9.67](#). Hence no log-diffuse alternative remains. $\square$

9.3. Hidden-scale Young variance.

Definition 9.31 (Hidden-scale variance). Let W_n be a macroscopic critical-tail blow-down on a compact annular cylinder

$$K \Subset \{0 < |x| < \infty\} \times I.$$

For $0 < \delta < 1$, set

$$\mathfrak{H}_K(W_n, \delta) := \sup_{\substack{|h| < \delta \\ |\theta| < \delta}} \|W_n(\cdot + h, \cdot + \theta) - W_n\|_{L^3(K)},$$

where the translated functions are evaluated on a fixed slightly larger annular cylinder containing all $(x + h, s + \theta)$ with $(x, s) \in K$ and $|h| + |\theta| < \delta$. The sequence has no hidden-scale variance on K if

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \mathfrak{H}_K(W_n, \delta) = 0.$$

It has no hidden-scale variance if this holds on every compact macroscopic annular cylinder.

Lemma 9.32 (Hidden-scale variance is the Young obstruction). *Let W_n be uniformly bounded in L_{loc}^∞ on macroscopic annular cylinders and suppose*

$$W_n \rightharpoonup^* \bar{W} \quad \text{in } L_{\text{loc}}^\infty.$$

On a compact annular cylinder K , the following are equivalent:

- (i) $W_n \rightarrow \bar{W}$ strongly in $L^3(K)$;
- (ii) the generated Young measure is Dirac on K :

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \quad \text{for a.e. } (x,s) \in K;$$

- (iii) W_n has no hidden-scale variance on K .

Consequently a realized Young critical tail is precisely a failure of the hidden-scale variance condition on some compact annular cylinder.

Proof. The uniform $L^\infty(K)$ bound gives a uniform $L^3(K)$ bound. The Fréchet–Kolmogorov compactness criterion on $L^3(K)$, applied after extending the functions to a fixed slightly larger cylinder, says that this bounded family is relatively compact in $L^3(K)$ if and only if the translation modulus in [Theorem 9.31](#) tends to zero. Thus (iii) is equivalent to relative compactness in $L^3(K)$. Any strong subsequential limit must agree with the weak-* limit $\bar{W}$, so relative compactness is equivalent to strong convergence to $\bar{W}$. This proves (i) $\Leftrightarrow$ (iii).

Strong $L^3(K)$ convergence implies convergence in measure, hence the associated Young measure is $\delta_{\bar{W}}$. Conversely, if the generated Young measure is Dirac and $\{W_n\}$ is uniformly bounded in $L^\infty(K)$, then $W_n \rightarrow \bar{W}$ in measure on K . Uniform boundedness gives uniform integrability of $|W_n - \bar{W}|^3$, so Vitali convergence yields strong $L^3(K)$ convergence. Hence (i) $\Leftrightarrow$ (ii). $\square$

Definition 9.33 (Multiscale affine oscillation). Let (U_n, Π_n) be a terminal sequence. For a covariant observer $z = (\xi, \sigma)$ and $r \geq 1$, define the critical-tail cylinder

$$\mathcal{Q}_{r,\text{ct}}^{\text{cov}}(z) := \mathcal{R}_z(B_r(0) \times (-1, 1)).$$

Define

$$\mathcal{A}_n(z, r) := r^{-3} \text{osc}_{3,\text{sp}}(\mathcal{R}_z U_n; B_r(0) \times (-1, 1)).$$

The exponent 3 is the spatial dimension; the centered time interval is kept at fixed size because critical-tail blow-downs are spatial/logarithmic blow-downs, not parabolic rescalings. Allowing the minimizing constant to depend on time removes the harmless spatially homogeneous affine mode. Unit concentration-set removal is formulated with $\text{osc}_{3,\text{ct}}$; after the affine/parasitic quotient has been applied, the mesoscopic residual analysis is formulated with $\text{osc}_{3,\text{sp}}$, and $\sup_z \mathcal{A}_n(z, 1)$ is below the retained spatial-oscillation threshold on the residual exterior window, after the fixed cover constants are absorbed into the threshold.

Definition 9.34 (Minimal mesoscopic oscillation scale). Let $R_n \rightarrow \infty$ be the macroscopic scale of a tight critical-tail blow-down. A minimal mesoscopic oscillation scale is a sequence of covariant centers z_n , radii r_n , and a number $\varepsilon_0 > 0$ such that

$$1 \ll r_n \ll R_n, \quad \mathcal{A}_n(z_n, r_n) \geq \varepsilon_0,$$

and, after replacing r_n by a dyadic comparable scale if necessary,

$$\sup_z \sup_{1 \leq r \leq r_n/2} \mathcal{A}_n(z, r) < \varepsilon_0/2$$

on the selected residual exterior window. Thus all smaller retained subscales are affine-flat, while the scale r_n is the first scale at which non-affine oscillation is visible.

Lemma 9.35 (Finite-overlap control of affine oscillation). Let $I \Subset \mathbb{R}$, let $f \in L_{\text{loc}}^\infty(\mathbb{R}^3 \times I)$ with $\|f\|_{L^\infty} \leq M$, and let

$$B_1(a_0), B_1(a_1), \dots, B_1(a_N)$$

be a finite chain of unit balls such that

$$|B_1(a_j) \cap B_1(a_{j+1})| \geq \kappa > 0 \quad (j = 0, \dots, N-1).$$

Assume that for each j there exists $c_j \in L^\infty(I; \mathbb{R}^3)$, $|c_j| \leq M$, such that

$$\int_I \int_{B_1(a_j)} |f(y, s) - c_j(s)|^3 dy ds \leq \delta.$$

Then, for $0 \leq j, k \leq N$,

$$(9.4) \quad \int_I |c_j(s) - c_k(s)|^3 ds \leq C(N, \kappa) \delta,$$

and

$$(9.5) \quad \int_I \int_{B_1(a_j)} |f(y, s) - c_k(s)|^3 dy ds \leq C(N, \kappa) \delta.$$

Consequently, if all unit cylinders in a bounded chain have affine-normalized oscillation $o(1)$, then all their local affine gauges agree up to $o(1)$ in $L^3(I)$.

Proof. For neighboring indices put $E_j = B_1(a_j) \cap B_1(a_{j+1})$. Since $|E_j| \geq \kappa$, the pointwise inequality

$$|c_j(s) - c_{j+1}(s)|^3 \leq 4\kappa^{-1} \int_{E_j} (|f(y, s) - c_j(s)|^3 + |f(y, s) - c_{j+1}(s)|^3) dy$$

holds for a.e. s . Integrating in s gives

$$\int_I |c_j - c_{j+1}|^3 ds \leq C_\kappa \delta.$$

Summing along the chain and using

$$\left| \sum_{\ell=j}^{k-1} (c_\ell - c_{\ell+1}) \right|^3 \leq |j - k|^2 \sum_{\ell=j}^{k-1} |c_\ell - c_{\ell+1}|^3$$

proves (9.4). Finally,

$$|f - c_k|^3 \leq 4|f - c_j|^3 + 4|c_j - c_k|^3$$

on $B_1(a_j) \times I$, and (9.5) follows from (9.4). $\square$

Lemma 9.36 (Spacetime finite-overlap translation control). *Let $K \Subset \mathbb{R}^3 \times I$, let $(h_n, \theta_n) \rightarrow 0$, and let f_n be uniformly bounded in L^∞ on a fixed neighborhood of*

$$K \cup (K + (h_n, \theta_n)).$$

Assume that every unit covariant cylinder in a finite spacetime chain covering the segment joining $(x, s) \in K$ to $(x + h_n, s + \theta_n)$ has spatial quotient oscillation $o(1)$. Then there are spatially homogeneous functions $a_n, b_n \in L^\infty(I; \mathbb{R}^3)$ such that

$$\|f_n - a_n\|_{L^3(K)} + \|f_n(\cdot + h_n, \cdot + \theta_n) - b_n\|_{L^3(K)} = o(1),$$

and

$$\|f_n(\cdot + h_n, \cdot + \theta_n) - f_n - (b_n - a_n)\|_{L^3(K)} = o(1).$$

Consequently, if the translated difference has a positive $L^3(K)$ lower bound while no unit concentrating observer is present, then that lower bound is carried by the spatially homogeneous time-dependent mode $b_n - a_n$. In the ordered residual decomposition this descendant belongs to the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$.

Proof. Cover K and $K + (h_n, \theta_n)$ by finitely many unit spacetime cylinders contained in the fixed neighborhood. Refine the cover so that any two successive cylinders in the chain have spatial overlap bounded below and time overlap bounded below by constants depending only on K . On each cylinder choose a near-minimizing spatially homogeneous gauge. Applying [Theorem 9.35](#) on each time-overlap interval and summing along the finite chain shows that all gauges on the chain agree in L^3 on their common time intervals up to $o(1)$. Gluing the finitely many local gauges by this overlap relation gives global-in-space functions $a_n(s)$ on the part of the chain covering K and $b_n(s)$ on the part covering $K + (h_n, \theta_n)$. The first displayed estimate follows from the local small oscillation and finite overlap. Subtracting the two approximations gives the second displayed estimate.

If the translated difference remains bounded below, then $\|b_n - a_n\|_{L^3(\text{proj}_s K)}$ stays bounded below. The corresponding descendant has zero $\text{osc}_{3,\text{sp}}$ on every compact cylinder but a nonzero homogeneous time-dependent component. That is the affine/parasitic lower stratum in the ordered profile decomposition. $\square$

Lemma 9.37 (Translation defect produces concentration, affine collapse, or a mesoscopic radius). *Let $K \Subset \{0 < |x| < \infty\} \times I$, let*

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s), \quad R_n \rightarrow \infty,$$

be a bounded macroscopic blow-down, and suppose that for some $(h_n, \theta_n) \rightarrow 0$ and $\eta_0 > 0$,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

Set

$$\ell_n := R_n(|h_n| + |\theta_n|^{1/2}).$$

The observer metric used here is the fixed parabolic metric on covariant observer space,

$$d_{\text{obs}}((h, \theta), (0, 0)) \simeq |h| + |\theta|^{1/2}.$$

Multiplication by R_n converts this macroscopic observer displacement into the corresponding centered spatial/logarithmic tail scale. This is an observer blow-down scale, not a Navier–Stokes parabolic rescaling of the centered time variable. Then, after passing to a subsequence, exactly one of the following alternatives occurs.

- (i) *A positive-threshold unit concentrating observer is regenerated.*
- (ii) *The defect is carried only by a spatially homogeneous time-dependent affine mode; equivalently the descendant is assigned to $\mathcal{C}_{\text{aff}}$.*
- (iii) *$1 \ll \ell_n \ll R_n$, and there are covariant centers z_n , radii $r_n \sim \ell_n$, and $\varepsilon_0 = \varepsilon_0(\eta_0, K) > 0$ such that*

$$\mathcal{A}_n(z_n, r_n) \geq \varepsilon_0.$$

Proof. Since $(h_n, \theta_n) \rightarrow 0$, $\ell_n/R_n \rightarrow 0$. First assume ℓ_n is bounded. The two points compared in the original centered variables are then connected, on the compact set K , by a uniformly finite chain of covariant unit cylinders. If a unit cylinder in that chain has a positive constant-in-time oscillation lower bound, alternative (i) holds. Otherwise the spatial quotient oscillation on every cylinder in the chain tends to zero. By [Theorem 9.36](#), the displayed translation defect can persist only through a spatially homogeneous time-dependent mode. This is alternative (ii). Thus, if neither (i) nor (ii) occurs, $\ell_n \rightarrow \infty$.

In that remaining case $1 \ll \ell_n \ll R_n$. Cover K and its translated copy by finitely many balls of radius comparable to $|h_n| + |\theta_n|^{1/2}$ in macroscopic variables. Pulling this cover back to the centered variables gives covariant cylinders of radius comparable to ℓ_n . If all such cylinders had spatial quotient oscillation below a sufficiently small $\varepsilon_0 = \varepsilon_0(\eta_0, K)$, then [Theorem 9.36](#), after rescaling the finite overlap constants, would again force the translated fields to differ by only an

affine/parasitic time mode. This is alternative (ii), contrary to the present case. Therefore one such cylinder has $\mathcal{A}_n(z_n, r_n) \geq \varepsilon_0$ with $r_n \sim \ell_n$, proving (iii). $\square$

Lemma 9.38 (Failure of hidden-scale compactness produces a minimal mesoscopic scale). *Let $W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s)$ be a tight macroscopic annular blow-down after concentration-set removal. If W_n fails the hidden-scale variance condition on some compact annular cylinder K , then either a positive-threshold unit concentrating observer is regenerated, the affine stratum $\mathcal{C}_{\text{aff}}$ is reached by a descendant, or there is a minimal mesoscopic oscillation scale in the sense of [Theorem 9.34](#).*

Proof. Failure of hidden-scale variance gives a compact annular cylinder K , a number $\eta_0 > 0$, and translations $(h_n, \theta_n) \rightarrow 0$ such that, after passing to a subsequence,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

By [Theorem 9.37](#), the translation defect either regenerates a unit concentrating observer, reaches $\mathcal{C}_{\text{aff}}$, or gives centers z_n , radii ρ_n with $1 \ll \rho_n \ll R_n$, and $\varepsilon_0 > 0$ such that

$$\mathcal{A}_n(z_n, \rho_n) \geq \varepsilon_0.$$

In the last case, let r_n be the smallest dyadic radius in $[1, C\rho_n]$ for which

$$\sup_z \mathcal{A}_n(z, r_n) \geq \varepsilon_0$$

on the selected residual exterior window. Unit concentration-set removal excludes $r_n = O(1)$, so $r_n \rightarrow \infty$. Since $r_n \lesssim \rho_n$ and $\rho_n/R_n \rightarrow 0$, one has $r_n/R_n \rightarrow 0$. By dyadic minimality, all smaller scales up to $r_n/2$ have oscillation below $\varepsilon_0/2$, after slightly lowering ε_0 . This is a minimal mesoscopic oscillation scale. $\square$

Definition 9.39 (Minimal-scale spatial blow-down). Let z_n and r_n be a minimal mesoscopic oscillation scale. Write

$$U_n^{z_n} := \mathcal{R}_{z_n} U_n, \quad \Pi_n^{z_n} := \mathcal{R}_{z_n} \Pi_n.$$

Choose measurable velocity gauges $c_n : (-T, T) \rightarrow \mathbb{R}^3$, with $|c_n| \leq M$, which are minimizing, or within $o(1)$ of minimizing, the affine oscillation on each compact time interval. The minimal-scale spatial blow-down is

$$V_n(x, s) := U_n^{z_n}(r_n x, s), \quad W_n(x, s) := V_n(x, s) - c_n(s).$$

Thus V_n is the ungauged blow-down and W_n is the affine-normalized velocity used to measure non-affine oscillation. The associated scaled pressure is

$$Q_n(x, s) := r_n^{-1} \Pi_n^{z_n}(r_n x, s),$$

modulo the admissible addition of affine functions of x and arbitrary functions of s . Affine pressure terms are always paired with the corresponding spatially homogeneous velocity gauge in the weak formulation; no pointwise time derivative of c_n is required. This is not the Navier–Stokes parabolic scaling. It is the spatial/logarithmic tail blow-down adapted to the centered residual equation. The pressure normalization is chosen so that $\nabla_x Q_n$ is the rescaled pressure-gradient term appearing in the centered blow-down equation.

Remark 9.40 (First-bad pressure and gauge convention). Throughout the minimal mesoscopic-scale argument, all distributional equations, pressure Poisson equations, and local energy inequalities are written for the ungauged field V_n and its genuine scaled pressure Q_n . The field $W_n = V_n - c_n(s)$ is used only for quotient quantities: spatial oscillation, Young measures, and variance after the homogeneous gauge sequence has either compactified in time or has been assigned to the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$. Thus an “affine pressure paired with c_n ” is quotient

notation. It is not asserted to be an $L_{\text{loc}}^{3/2}$ pressure, and no pointwise derivative $\partial_s c_n$ is used unless compactness of c_n has already been obtained.

Lemma 9.41 (Minimal-scale blow-down equation). *Let V_n, W_n, Q_n, c_n be the minimal-scale blow-downs of Theorem 9.39. On every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, V_n and W_n are divergence-free, $\|V_n\|_{L^\infty(K)} \leq M$, $\|W_n\|_{L^\infty(K)} \leq 2M$, and the ungauged field satisfies*

$$(9.6) \quad \partial_s V_n - r_n^{-2} \Delta V_n + \frac{1}{2} x \cdot \nabla V_n + \frac{1}{2} V_n + r_n^{-1} (V_n \cdot \nabla) V_n + \nabla Q_n = 0$$

in distributions. The affine-normalized field $W_n = V_n - c_n(s)$ is used for oscillation and Young-measure compactness. Its equation is understood only in the quotient sense made precise in Theorem 9.42; no pointwise derivative of c_n and no $L_{\text{loc}}^{3/2}$ affine pressure for W_n is assumed. Moreover

$$(9.7) \quad -\Delta Q_n = r_n^{-1} \partial_i \partial_j (V_{n,i} V_{n,j})$$

in distributions, after the same gauges.

Proof. The covariant recentering $\mathcal{R}_{z_n}$ is an exact symmetry of the centered equation, so $(U_n^{z_n}, \Pi_n^{z_n})$ solves (1.2). Since centered time is not rescaled,

$$\begin{aligned} \partial_s V_n &= (\partial_\tau U_n^{z_n})(r_n x, s), & \Delta_y U_n^{z_n} &= r_n^{-2} \Delta_x V_n, \\ y \cdot \nabla_y U_n^{z_n} &= x \cdot \nabla_x V_n, & (U_n^{z_n} \cdot \nabla_y) U_n^{z_n} &= r_n^{-1} (V_n \cdot \nabla_x) V_n. \end{aligned}$$

If $Q_n = r_n^{-1} \Pi_n^{z_n}(r_n x, s)$, then

$$\nabla_x Q_n = (\nabla_y \Pi_n^{z_n})(r_n x, s).$$

Substituting these identities gives the scaled equation for V_n . We do not treat $W_n = V_n - c_n(s)$ as carrying an ordinary pressure unless the gauges have additional time regularity. Instead, all PDE identities below are first written for the ungauged field V_n , and the passage to W_n is made only after quotienting out spatially homogeneous modes. Taking the divergence of the ungauged scaled equation gives (9.7); the linear drift has zero divergence contribution because V_n is divergence-free. The L^∞ bounds follow from $\|U_n\|_{L^\infty} \leq M$ and $|c_n| \leq M$. $\square$

Lemma 9.42 (Minimal-scale quotient formulation). *Let V_n, W_n, c_n, Q_n be as in Theorem 9.41. The minimal-scale analysis uses the following quotient formulation.*

- (i) *The distributional equation and local energy inequality are always applied to the ungauged field V_n and its pressure Q_n .*
- (ii) *The affine-normalized field $W_n = V_n - c_n(s)$ is used only for oscillation functionals and Young measures, which are invariant under spatially homogeneous shifts after the gauge sequence is either compact or assigned to $\mathcal{C}_{\text{aff}}$.*
- (iii) *If c_n is not compact in L_{loc}^3 in time, the homogeneous gauge sequence itself generates an affine/parasitic descendant. If $c_n \rightarrow c$ strongly in L_{loc}^3 , all Young-measure variances and strong compactness statements pass from V_n to W_n by deterministic translation of the generated Young measures.*

Thus the symbol “affine pressure paired with c_n ” means a quotient functional, not an asserted $L_{\text{loc}}^{3/2}$ pressure unless the time regularity of c_n has been separately proved.

Proof. Statement (i) is exactly the construction in Theorem 9.41. The local energy inequality is stable under the minimal-scale spatial blow-down for V_n , because V_n is obtained from an actual centered solution by an exact covariant recentering and a spatial scaling; no subtraction of c_n is involved.

For (ii) and (iii), observe that subtracting a spatially homogeneous function does not change any $\text{osc}_{3,\text{sp}}$ -type seminorm in which the same class of homogeneous gauges is minimized over.

If the sequence c_n is not compact in L^3_{loc} , [Theorem 9.49](#) places the resulting homogeneous descendant in $\mathcal{C}_{\text{aff}}$. If $c_n \rightarrow c$ strongly, then [Theorem 9.50](#) identifies the Young measure of W_n with the deterministic translate of the Young measure of V_n , and the variance densities agree. Hence all compactness and variance conclusions used for W_n follow from identities proved for V_n , without requiring $\partial_s c_n$ to be a function. $\square$

Lemma 9.43 (Covariance of the whole-space Riesz pressure). *At the finite-energy Leray level one may choose the standard whole-space pressure representative*

$$p(\cdot, t) = \mathcal{R}_i \mathcal{R}_j (u_i u_j)(\cdot, t)$$

modulo a function of time. This representative is preserved, modulo the corresponding time gauge, by physical Type I scaling, the centered self-similar change of variables, centered time translation, spatial translation, and the covariant translations used in this paper. Local pressure representatives are restrictions of this whole-space representative, followed only by the admissible addition of functions of time.

Proof. Let K_{ij} be the Calderón–Zygmund pressure kernel. It is homogeneous of degree -3 :

$$K_{ij}(\lambda x) = \lambda^{-3} K_{ij}(x), \quad \lambda > 0.$$

Therefore, if

$$u^\lambda(x, t) = \lambda u(x_0 + \lambda x, t_0 + \lambda^2 t), \quad p^\lambda(x, t) = \lambda^2 p(x_0 + \lambda x, t_0 + \lambda^2 t),$$

then

$$p^\lambda(\cdot, t) = \mathcal{R}_i \mathcal{R}_j (u_i^\lambda u_j^\lambda)(\cdot, t)$$

modulo the scaled time gauge. Spatial translations commute with convolution by K_{ij} , and time translations only relabel the time variable. The centered change of variables is exactly a time-dependent parabolic scaling with the pressure factor $P(y, \tau) = (-t)p(x, t)$; hence the same homogeneity gives the centered Riesz representative at each fixed centered time. The covariant translations used below are compositions of these exact spatial/time changes and the admissible pressure gauges recorded in the centered equation. Since all additional gauges are functions of time, their gradients vanish and their restrictions to compact cylinders are precisely the local pressure gauges used in the Seregin topology. $\square$

Lemma 9.44 (Harmonic remainders in normalized pressure limits). *Let (U_n, Π_n) be a sequence of normalized profiles obtained from the whole-space representative in [Theorem 9.43](#), before applying any homogeneous velocity quotient. Suppose that $U_n \rightarrow U$ locally smoothly and that, after time-dependent gauges, $\Pi_n \rightarrow \Pi$ locally in $L^{3/2}$ or weakly in $L^{3/2}$. Then*

$$\Pi(\cdot, s) - \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s)$$

is a time-dependent constant. After the affine/parasitic quotient is applied, the only additional admissible harmonic pressure direction is a linear function $\beta(s) \cdot y$; this linear part is exactly the pressure paired with the spatially homogeneous parasitic velocity mode and is absent in the parasitic-free representative.

Proof. Before the affine quotient, the pressure of each approximating profile is the whole-space Riesz representative, modulo a time gauge. Since the velocities are uniformly bounded in L^∞ , these representatives have uniformly bounded BMO seminorms on $\mathbb{R}^3$:

$$\|\mathcal{R}_i \mathcal{R}_j (U_{n,i} U_{n,j})(\cdot, s)\|_{\text{BMO}} \leq CM^2.$$

The BMO bound is invariant under the exact scalings and translations in [Theorem 9.43](#). Hence, after passing to a subsequence at each fixed time interval and using a diagonal extraction, the whole-space representatives converge weakly-* in local BMO-duality to a whole-space BMO

pressure P^* , modulo time gauges. The local $L^{3/2}$ -pressure limit Π agrees with P^* on each compact cylinder, up to the same time-dependent gauge, because both are local limits of the same normalized representatives. Passing to the limit in the pressure Poisson equation gives

$$-\Delta P^*(\cdot, s) = \partial_i \partial_j (U_i U_j)(\cdot, s)$$

in distributions. Thus the difference

$$h(\cdot, s) := P^*(\cdot, s) - \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s)$$

is harmonic in space and belongs to $\text{BMO}(\mathbb{R}^3)$ modulo an additive constant. An entire harmonic BMO function is constant: indeed, for each ball $B_R(x)$, the interior estimate for harmonic functions and the John–Nirenberg control of mean oscillation give

$$|\nabla h(x, s)| \leq CR^{-1} \|h(\cdot, s)\|_{\text{BMO}(\mathbb{R}^3)}$$

with C independent of R ; letting $R \rightarrow \infty$ yields $\nabla h(\cdot, s) = 0$. Since Π and P^* differ only by the local time gauge, before affine quotienting the difference between Π and $\mathcal{R}_i \mathcal{R}_j (U_i U_j)$ is spatially constant.

Equivalently, if a non-affine harmonic component were present on some compact ball and time interval, elliptic regularity would give a point where a nonzero spatial derivative of order at least one remains. Testing against compactly supported functions with zero spatial mean, and with zero first moments when isolating the affine quotient, would detect this component. Such a component cannot arise from the uniformly BMO whole-space Riesz representatives plus time-only gauges described in [Theorem 9.43](#).

The affine/parasitic normalization is the only subsequent operation that changes the velocity by a spatially homogeneous function. For a smooth homogeneous mode $b(s)$, substitution into the centered equation gives the affine pressure gradient

$$\nabla_y [(-\partial_s b - \tfrac{1}{2}b) \cdot y].$$

For merely bounded measurable homogeneous gauges this expression is used only as the quotient functional specified in [Theorems 8.16, 9.40 and 9.42](#). Thus the normalized profile class contains no non-affine harmonic pressure remainders: only time gauges and, inside $\mathcal{C}_{\text{aff}}$, the explicit affine parasitic pressure direction are allowed. $\square$

Proposition 9.45 (Normalized pressure representation modulo affine parasitic modes). *Every normalized Seregin, terminal, covariant descendant, minimal-scale blow-down, or concentrating descendant profile used in the residual profile class is represented by a pressure of the form*

$$(9.8) \quad \Pi(\cdot, s) = \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s) + \beta(s) \cdot y + \alpha(s),$$

where α is an arbitrary time gauge and $\beta(s) \cdot y$ is exactly the affine pressure paired with the spatially homogeneous parasitic velocity mode. Outside the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$, the representative is taken with $\beta = 0$.

Proof. The whole-space Riesz representative and its covariance under the exact operations are given by [Theorem 9.43](#). The exclusion of non-affine harmonic pressure remainders in normalized limits is [Theorem 9.44](#). The only additional pressure direction created by affine/parasitic normalization is the explicit linear pressure paired with the homogeneous velocity gauge. Hence (9.8) is the complete normalized pressure representation. If $\beta \neq 0$ has a retained effect, the associated profile is assigned to $\mathcal{C}_{\text{aff}}$; in the parasitic-free representative used outside $\mathcal{C}_{\text{aff}}$, one sets $\beta = 0$. $\square$

Lemma 9.46 (Scaled pressure compactness modulo affine gauges). *Let (U_n, Π_n) be bounded centered solutions with $\|U_n\|_{L^\infty} \leq M$, let z_n be covariant centers, let $r_n \rightarrow \infty$, and set*

$$\tilde{U}_n(y, s) := (\mathcal{R}_{z_n} U_n)(y, s), \quad \tilde{\Pi}_n(y, s) := (\mathcal{R}_{z_n} \Pi_n)(y, s).$$

For

$$\tilde{Q}_n(x, s) := r_n^{-1} \tilde{\Pi}_n(r_n x, s)$$

and every compact cylinder $K = B_R \times I \Subset \mathbb{R}^3 \times \mathbb{R}$, there are affine functions

$$L_n(x, s) = b_n(s) \cdot x + a_n(s)$$

such that

$$(9.9) \quad \tilde{Q}_n - L_n \rightarrow 0 \quad \text{strongly in } L^{3/2}(K).$$

The same statement holds for the minimal-scale pressures Q_n in the quotient sense of [Theorem 9.42](#). If the affine parts L_n carry a nonzero retained descendant, then that descendant lies in the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$. Otherwise the affine parts may be discarded in the affine-normalized quotient, and the non-affine pressure converges strongly to zero.

Proof. The covariant recentering is an exact symmetry, so it suffices to work with a single bounded centered solution (U, Π) . For a.e. s , the pressure Poisson equation is

$$-\Delta \Pi(\cdot, s) = \partial_i \partial_j (U_i U_j)(\cdot, s)$$

in $\mathbb{R}^3$. Let

$$\Pi^{\text{loc}}(\cdot, s) := \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s)$$

be the whole-space Calderón–Zygmund representative. Since $\|U\|_{L^\infty} \leq M$,

$$\|\Pi^{\text{loc}}(\cdot, s)\|_{\text{BMO}(\mathbb{R}^3)} \leq CM^2$$

uniformly in s . Hence, by the John–Nirenberg inequality, for every ball B_ρ ,

$$\inf_{a \in \mathbb{R}} \|\Pi^{\text{loc}}(\cdot, s) - a\|_{L^{3/2}(B_\rho)} \leq CM^2 |B_\rho|^{2/3}.$$

After the scaling $x \mapsto r_n x$, this gives

$$\inf_{a_n(s)} \|r_n^{-1} \Pi^{\text{loc}}(r_n \cdot, s) - a_n(s)\|_{L^{3/2}(B_R)} \leq C_{R,M} r_n^{-1} \rightarrow 0.$$

The bound is uniform for $s \in I$, hence after choosing the measurable minimizing averages $a_n(s)$ one also has

$$\|r_n^{-1} \Pi^{\text{loc}}(r_n \cdot, s) - a_n(s)\|_{L^{3/2}(B_R \times I)} \leq C_{R,M,I} r_n^{-1} \rightarrow 0.$$

By [Theorem 9.45](#), the only remaining pressure component in the normalized profile class is

$$\beta(s) \cdot y + \alpha(s).$$

After scaling,

$$r_n^{-1} (\beta(s) \cdot r_n x + \alpha(s)) = \beta(s) \cdot x + r_n^{-1} \alpha(s),$$

which is an affine function of x . Taking this function as L_n , the remaining pressure is exactly the scaled Leray representative already estimated above, and it tends to zero strongly in $L^{3/2}(B_R \times I)$. This proves (9.9). The minimal-scale pressure is used only up to the affine quotient described in [Theorem 9.42](#); in that quotient the same estimate applies. $\square$

Lemma 9.47 (Pressure–velocity product convergence). *Let $K \Subset \mathbb{R}^3 \times \mathbb{R}$. Suppose*

$$Q_n \rightarrow Q \quad \text{strongly in } L^{3/2}(K), \quad W_n \rightharpoonup^* W \quad \text{in } L^\infty(K),$$

with $\sup_n \|W_n\|_{L^\infty(K)} < \infty$. Then

$$Q_n W_n \rightarrow QW \quad \text{in } \mathcal{D}'(K).$$

Proof. Let $\varphi \in C_c^\infty(K; \mathbb{R}^3)$. Then

$$\iint_K (Q_n W_n - QW) \cdot \varphi = \iint_K (Q_n - Q) W_n \cdot \varphi + \iint_K Q(W_n - W) \cdot \varphi.$$

The first term tends to zero by the strong $L^{3/2}$ convergence of Q_n and the uniform L^∞ bound on W_n . Since $Q\varphi \in L^1(K)$, the second term tends to zero by weak-* convergence in $L^\infty(K)$. $\square$

Lemma 9.48 (Pressure compactness at a minimal mesoscopic scale). *Let Q_n be as in Theorem 9.41. For every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, exactly one of the following holds:*

- (i) *the affine pressure/velocity gauges produce a retained descendant in $\mathcal{C}_{\text{aff}}$;*
- (ii) *after discarding those affine gauges in the affine-normalized quotient, there exists a pressure limit Q such that*

$$Q_n \rightarrow Q \quad \text{strongly in } L^{3/2}(K).$$

In fact, in the non-affine quotient one may take $Q = 0$. Consequently, if $W_n \rightharpoonup^ \bar{W}$ in L_{loc}^∞ , then*

$$Q_n W_n \rightharpoonup Q \bar{W} \quad \text{in distributions on } K.$$

Proof. This is Theorem 9.46 applied to the minimal-scale scaling. If the affine gauges retain nonzero descendant concentration, we are in alternative (i). Otherwise the affine gauges are precisely the quotient directions removed by affine normalization, and the remaining pressure converges strongly to 0 in $L^{3/2}(K)$. Strong convergence of the pressure and weak-* convergence of the uniformly bounded velocity give the product convergence in distributions by Theorem 9.47. $\square$

Lemma 9.49 (Homogeneous gauge compactness or affine stratum). *Let $c_n : I \rightarrow \mathbb{R}^3$ be uniformly bounded in $L^\infty(I)$ on every compact interval $I \Subset \mathbb{R}$. Then either c_n is precompact in $L_{\text{loc}}^3(\mathbb{R}; \mathbb{R}^3)$ after passing to a subsequence, or a time recentering of the spatially homogeneous profiles*

$$C_n(y, s) := c_n(s)$$

generates a nonzero affine/parasitic descendant. In the ordered residual decomposition the second alternative belongs to $\mathcal{C}_{\text{aff}}$.

Proof. By the Fréchet–Kolmogorov criterion on compact time intervals, failure of precompactness in L_{loc}^3 is equivalent, after passing to a subsequence, to the existence of an interval I , translations $\theta_n \rightarrow 0$, and $\eta > 0$ such that

$$\|c_n(\cdot + \theta_n) - c_n\|_{L^3(I)} \geq \eta.$$

The associated spatially homogeneous profiles $C_n(y, s) = c_n(s)$ have zero spatial quotient oscillation on every compact cylinder, while the displayed lower bound gives a nonzero homogeneous time-dependent descendant. This is exactly the affine/parasitic lower stratum. If this alternative does not occur, the time translation modulus of c_n vanishes on every compact interval; boundedness then gives L_{loc}^3 precompactness. $\square$

Lemma 9.50 (Variance is invariant under homogeneous velocity gauges). *Let V_n be uniformly bounded in L_{loc}^∞ , let $c_n \rightarrow c$ strongly in $L_{\text{loc}}^3(\mathbb{R})$, and set*

$$W_n(x, s) := V_n(x, s) - c_n(s).$$

If V_n generates a Young measure $\nu_{x,s}$ with barycenter $\bar{V}$, then W_n generates the translated Young measure

$$\tilde{\nu}_{x,s}(A) = \nu_{x,s}(A + c(s))$$

with barycenter $\bar{W} = \bar{V} - c(s)$. The variance densities agree:

$$\langle |\xi - \bar{V}(x, s)|^2 \rangle_{\nu_{x,s}} = \langle |\xi - \bar{W}(x, s)|^2 \rangle_{\tilde{\nu}_{x,s}}.$$

Consequently any distributional variance inequality proved for V_n passes unchanged to the affine-normalized fields W_n .

Proof. After passing to a subsequence, $c_n(s) \rightarrow c(s)$ a.e. and in measure on every compact time interval. For every continuous compactly supported function $F(\xi)$ and every compactly supported test function $\varphi(x, s)$,

$$\iint \varphi F(W_n) dx ds = \iint \varphi F(V_n - c_n(s)) dx ds \rightarrow \iint \varphi \int F(\xi - c(s)) d\nu_{x,s}(\xi) dx ds.$$

This is precisely the translated Young measure. Translation by a deterministic vector preserves centered second moments, giving the variance identity. The distributional inequality involves only this variance density and therefore is unchanged. $\square$

Lemma 9.51 (Barycenter equation and variance inequality). *Let V_n, W_n, Q_n, c_n be a minimal-scale blow-down. Then either the sequence has a retained affine/parasitic descendant in $\mathcal{C}_{\text{aff}}$, or, after passing to a subsequence, $c_n \rightarrow c$ strongly in L^3_{loc} ,*

$$V_n \rightharpoonup^* \bar{V}, \quad W_n \rightharpoonup^* \bar{W} \quad \text{in } L^\infty_{\text{loc}},$$

and V_n, W_n generate Young measures $\nu_{x,s}$ and $\tilde{\nu}_{x,s}$, respectively. Set

$$E(x, s) := \langle |\xi|^2 \rangle_{\tilde{\nu}_{x,s}}, \quad \mathcal{V}(x, s) := E(x, s) - |\bar{W}(x, s)|^2.$$

Then $\bar{V}$ satisfies the coherent transport equation with pressure Q . The affine-normalized barycenter $\bar{W} = \bar{V} - c(s)$ satisfies the same equation only as a quotient modulo spatially homogeneous modes; after passing to the parasitic-free representative outside $\mathcal{C}_{\text{aff}}$, this quotient equation is represented as

$$(9.10) \quad \partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \quad \nabla \cdot \bar{W} = 0,$$

and the nonnegative variance density $\mathcal{V} \in L^\infty_{\text{loc}}$ satisfies

$$(9.11) \quad \partial_s \mathcal{V} + \frac{1}{2} x \cdot \nabla \mathcal{V} + \mathcal{V} \leq 0$$

in distributions.

Proof. By [Theorem 9.49](#), either the homogeneous gauges generate an affine/parasitic descendant, or $c_n \rightarrow c$ strongly in L^3_{loc} after passing to a subsequence. We work in the second case. The weak-* compactness of V_n, W_n and generation of Young measures are standard consequences of the uniform L^∞_{loc} bound. Testing [\(9.6\)](#) against a compactly supported divergence-free vector field, the viscous term is bounded by

$$Cr_n^{-2} \|W_n\|_{L^\infty},$$

and the nonlinear term by

$$Cr_n^{-1} \|V_n\|_{L^\infty}^2.$$

Both tend to zero. The pressure compactness in [Theorem 9.48](#) gives either a retained descendant in $\mathcal{C}_{\text{aff}}$, or the pressure gradient limit in the affine-normalized quotient. Thus $\bar{V}$ satisfies the coherent transport equation. Since $\bar{W} = \bar{V} - c(s)$, passing from $\bar{V}$ to $\bar{W}$ changes only the spatially homogeneous quotient component. We do not differentiate c as a function. Instead, [Theorem 9.42](#) says that this is an affine/parasitic quotient direction. If it has retained contribution, the profile is in $\mathcal{C}_{\text{aff}}$; otherwise the canonical parasitic-free representative removes it, and the representative satisfies [\(9.10\)](#).

We next pass to the local energy inequality for the ungauged fields V_n . Dot (9.6) with V_n . In distributional local-energy form, using $\nabla \cdot V_n = 0$, one obtains

$$\begin{aligned} \partial_s \frac{|V_n|^2}{2} - r_n^{-2} \Delta \frac{|V_n|^2}{2} + r_n^{-2} |\nabla V_n|^2 + \frac{1}{2} x \cdot \nabla \frac{|V_n|^2}{2} + \frac{1}{2} |V_n|^2 + \operatorname{div}(Q_n V_n) \\ = -r_n^{-1} \operatorname{div} \left(\frac{1}{2} |V_n|^2 V_n \right). \end{aligned}$$

There is no affine pressure work term in this identity; the homogeneous gauge has not been subtracted. When tested against nonnegative compactly supported scalar functions, the viscous diffusion and nonlinear-flux terms vanish in the limit because of the factors r_n^{-2} and r_n^{-1} , while the dissipation term is nonnegative and may be dropped. The strong pressure convergence gives

$$\operatorname{div}(Q_n V_n) \rightarrow \operatorname{div}(Q \bar{V}).$$

Therefore

$$\partial_s \frac{E_V}{2} + \frac{1}{2} x \cdot \nabla \frac{E_V}{2} + \frac{1}{2} E_V + \operatorname{div}(Q \bar{V}) \leq 0$$

in distributions, where

$$E_V(x, s) := \langle |\xi|^2 \rangle_{\nu_{x,s}}$$

is the second moment of the Young measure generated by V_n . On the other hand, dotting the barycenter equation for $\bar{V}$ with $\bar{V}$ gives

$$\partial_s \frac{|\bar{V}|^2}{2} + \frac{1}{2} x \cdot \nabla \frac{|\bar{V}|^2}{2} + \frac{1}{2} |\bar{V}|^2 + \operatorname{div}(Q \bar{V}) = 0.$$

Subtracting gives the variance inequality for the variance of V_n . By Theorem 9.50, the same variance density is $\mathcal{V}$, the variance of W_n . This proves (9.11). $\square$

Lemma 9.52 (Bounded ancient variance Liouville theorem). *Let $F \in L_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})$ satisfy*

$$0 \leq F \leq C < \infty$$

and

$$\partial_s F + \frac{1}{2} x \cdot \nabla F + F \leq 0$$

in distributions on $\mathbb{R}^3 \times \mathbb{R}$. Then $F \equiv 0$.

Proof. Step 1: renormalization. Set $G(x, s) := e^s F(x, s)$, so that $0 \leq G \leq C e^s$ and, for every smooth nonnegative compactly supported test function ψ ,

$$\langle G, (-\partial_s - \frac{1}{2} x \cdot \nabla - \frac{3}{2}) \psi \rangle = \langle F, (-\partial_s - \frac{1}{2} x \cdot \nabla - \frac{1}{2}) (e^s \psi) \rangle = \langle F, e^s (-\partial_s - \frac{1}{2} x \cdot \nabla - \frac{1}{2}) \psi - e^s \psi \rangle.$$

Adding the two contributions reduces to $\langle F, e^s (-\partial_s - \frac{1}{2} x \cdot \nabla - \frac{3}{2}) \psi \rangle$, and the distributional hypothesis $\partial_s F + \frac{1}{2} x \cdot \nabla F + F \leq 0$ applied to $e^s \psi \geq 0$ gives

$$(9.12) \quad \partial_s G + \frac{1}{2} x \cdot \nabla G \leq 0 \quad \text{in distributions on } \mathbb{R}^3 \times \mathbb{R}.$$

Step 2: backward adjoint semigroup. The forward characteristic flow of $\tilde{L} := \partial_s + \frac{1}{2} x \cdot \nabla$ starting from (x_0, s_0) is $x(\sigma) = e^{(\sigma-s_0)/2} x_0$. Define, for $T \geq 0$, the backward adjoint operator on test functions

$$(\mathcal{S}_T \psi)(x, s) := e^{-3T/2} \psi(e^{-T/2} x, s - T).$$

The prefactor $e^{-3T/2}$ is exactly the Jacobian of $x \mapsto e^{-T/2}x$ on $\mathbb{R}^3$, so $\int \mathcal{S}_T \psi \, dx \, ds = \int \psi \, dx \, ds$ and the family $\{\mathcal{S}_T\}_{T \geq 0}$ is the adjoint of the forward characteristic flow on $L^1(\mathbb{R}^3 \times \mathbb{R})$. In particular, by the substitution $x = e^{T/2}x'$, $s = s' + T$ (Jacobian $e^{3T/2}$),

$$(9.13) \quad \langle G, \mathcal{S}_T \psi \rangle = \iint G(e^{T/2}x', s' + T) \psi(x', s') \, dx' \, ds'.$$

Step 3: monotonicity in T . Fix $\psi \geq 0$ smooth and compactly supported. A direct computation using the chain rule shows that $\frac{d}{dT} \mathcal{S}_T \psi = -\tilde{L}^*(\mathcal{S}_T \psi)$ where $\tilde{L}^* = -\partial_s - \frac{1}{2}x \cdot \nabla - \frac{3}{2}$; equivalently,

$$\frac{d}{dT} \langle G, \mathcal{S}_T \psi \rangle = \langle G, -\tilde{L}^*(\mathcal{S}_T \psi) \rangle = \langle \tilde{L}G, \mathcal{S}_T \psi \rangle.$$

Since $\mathcal{S}_T \psi \geq 0$ and $\tilde{L}G \leq 0$ distributionally by (9.12),

$$(9.14) \quad \frac{d}{dT} \langle G, \mathcal{S}_T \psi \rangle \leq 0.$$

Step 4: Lebesgue limit and conclusion. Pick a standard nonnegative mollifier $\psi_0 \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$ of unit total mass, fix $(x_1, s_1) \in \mathbb{R}^3 \times \mathbb{R}$ and $T \geq 0$, and set $(x_0, s_0) := (e^{-T/2}x_1, s_1 - T)$. Apply (9.14) to the rescaled test function $\psi_\varepsilon(x, s) = \varepsilon^{-4} \psi_0((x - x_0)/\varepsilon, (s - s_0)/\varepsilon)$: since $T \mapsto \langle G, \mathcal{S}_T \psi_\varepsilon \rangle$ is nonincreasing on $[0, \infty)$,

$$\langle G, \mathcal{S}_T \psi_\varepsilon \rangle \leq \langle G, \psi_\varepsilon \rangle.$$

By (9.13),

$$\langle G, \mathcal{S}_T \psi_\varepsilon \rangle = \iint G(e^{T/2}x', s' + T) \psi_\varepsilon(x', s') \, dx' \, ds'$$

which approximates $G(e^{T/2}x_0, s_0 + T) = G(x_1, s_1)$ at the rate dictated by Lebesgue differentiation. Similarly $\langle G, \psi_\varepsilon \rangle \rightarrow G(x_0, s_0)$ as $\varepsilon \downarrow 0$ at common Lebesgue points. Therefore, at every common Lebesgue point of G ,

$$G(x_1, s_1) \leq G(e^{-T/2}x_1, s_1 - T).$$

Translating back through $G = e^s F$,

$$(9.15) \quad e^{s_1} F(x_1, s_1) \leq e^{s_1 - T} F(e^{-T/2}x_1, s_1 - T),$$

hence $F(x_1, s_1) \leq e^{-T} F(e^{-T/2}x_1, s_1 - T) \leq C e^{-T}$ using the hypothesis $F \leq C$ a.e. Letting $T \rightarrow \infty$ gives $F(x_1, s_1) = 0$ at every Lebesgue point, hence almost everywhere. The distributional inequality is satisfied by the zero a.e. representative, so $F \equiv 0$. $\square$

Lemma 9.53 (Minimal-scale blow-downs are strongly compact). *Every minimal-scale blow-down W_n either has a retained affine/parasitic descendant in $\mathcal{C}_{\text{aff}}$, or has, after passing to a subsequence, a limit $\bar{W}$ such that*

$$W_n \rightarrow \bar{W} \quad \text{strongly in } L_{\text{loc}}^3(\mathbb{R}^3 \times \mathbb{R}).$$

The limit satisfies (9.10).

Proof. By Theorem 9.51, either the affine/parasitic alternative has occurred, or the Young-measure variance $\mathcal{V}$ satisfies the hypotheses of Theorem 9.52. Hence $\mathcal{V} \equiv 0$. Thus

$$\tilde{\nu}_{x,s} = \delta_{\bar{W}(x,s)} \quad \text{for a.e. } (x, s).$$

Uniform L_{loc}^∞ boundedness then upgrades convergence in measure to strong convergence in L_{loc}^3 . The limit equation has already been proved in Theorem 9.51. $\square$

Theorem 9.54 (Minimal mesoscopic oscillation scale reduction). *Every minimal mesoscopic oscillation scale generated by a realized recurrent critical-tail extraction after concentration-set removal and affine normalization has a descendant in*

$$\mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Proof. Let z_n, r_n be a minimal mesoscopic oscillation scale. If, before passing to the minimal-scale blow-down, some unit covariant observer in the selected residual window carries a positive constant-in-time oscillation threshold, then the concentration alternative is regenerated and the descendant lies in $\mathcal{C}_{\text{conc}}$. It remains to consider the case in which unit concentration has been removed at all fixed positive thresholds.

Form the minimal-scale blow-down W_n of [Theorem 9.39](#). By [Theorem 9.53](#), either the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$ is reached, or, after passing to a subsequence,

$$W_n \rightarrow W \quad \text{strongly in } L_{\text{loc}}^3,$$

and (W, Q) satisfies the coherent critical-tail transport equation

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \quad \nabla \cdot W = 0.$$

The minimal-scale lower bound and strong convergence give

$$\text{osc}_{3,\text{sp}}(W; B_1(0) \times (-1, 1)) \geq \varepsilon_0.$$

Thus the limit is not spatially affine/parasitic on the unit critical-tail window. If this non-affine lower bound were lost in a further descendant, the descendant would be assigned to $\mathcal{C}_{\text{aff}}$ by [Theorem 8.21](#). Otherwise the strongly realized nonzero limit is a coherent critical-tail descendant, hence belongs to $\mathcal{C}_{\text{cohcrit}}$. Therefore every minimal mesoscopic oscillation scale is assigned to $\mathcal{C}_{\text{conc}}$, $\mathcal{C}_{\text{aff}}$, or $\mathcal{C}_{\text{cohcrit}}$, as claimed. $\square$

Remark 9.55 (Critical-tail reduction theorem). [Theorem 9.54](#) is the critical-tail alternative-reduction theorem proved above. It is a local multiscale rigidity statement: oscillation first appearing at an intermediate scale $1 \ll r_n \ll R_n$ must either regenerate a concentrating observer, collapse into the affine/parasitic lower stratum, or produce a coherent critical tail. The log-diffuse alternative has been reduced above to the same hidden-scale compactness statement through tight log-window descendants.

Theorem 9.56 (No hidden-scale variance from minimal mesoscopic scale reduction). *Every tight macroscopic annular blow-down generated by a realized recurrent critical-tail extraction after concentration-set removal and affine normalization has no hidden-scale variance on compact macroscopic annular cylinders.*

Proof. Suppose that hidden-scale variance fails. By [Theorem 9.38](#), either a unit concentrating observer is regenerated, the affine stratum is reached, or a minimal mesoscopic oscillation scale exists. The first alternative contradicts concentration-set removal. The affine alternative is removed by [Theorem 9.61](#). In the third alternative, [Theorem 9.54](#) gives a descendant in $\mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}$. The concentration case again contradicts concentration-set removal; the affine case is removed by [Theorem 9.61](#); and the coherent critical-tail case is closed by [Theorem 9.67](#). Thus all alternatives are impossible, and hidden-scale variance cannot fail. $\square$

Theorem 9.57 (Critical-tail compactification extraction). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a bounded normalized terminal residual sequence, $\|U_n\|_{L^\infty} \leq M$. Assume that, after removal of all affine-normalized concentrating covariant observers, there remains a nonzero diffuse exterior oscillation defect. Then, after passing to a subsequence, the sequence generates a nonzero element*

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M.$$

Moreover, the barycentric velocity satisfies the linear critical-tail transport law

$$\partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \quad \nabla \cdot \bar{W} = 0$$

in distributions on macroscopic annular cylinders, and the full defect $(\mathcal{Y}, \Lambda)$ is transported by the induced log-dilation flow.

Proof. Let ν_n^{sp} be the spatial quotient oscillation defect measure after concentration-set removal, as in [Theorem 9.9](#). By the mixed-diffuse hypothesis there are escaping observer windows E_n and $m_0 > 0$ such that

$$\nu_n^{\text{sp}}(E_n) \geq m_0.$$

Normalize

$$\lambda_n := \frac{\nu_n^{\text{sp}} \lfloor E_n}{\nu_n^{\text{sp}}(E_n)}.$$

By [Theorem 9.9](#) and the proof of [Theorem 9.16](#), after passing to a subsequence these probability measures converge to a boundary diffuse-defect profile. Because all positive-threshold concentrating observers have been removed in the stronger $\text{osc}_{3,\text{ct}}$ -sense, the limit gives no mass to ordinary unit concentrating observers in Z .

We now compactify the critical boundary without imposing annular tightness. Apply Prokhorov compactness first in the observer compactification and then in the logarithmic radial compactification. If the normalized defect is tight in a fixed multiplicative annulus, select macroscopic scales $R_n \rightarrow \infty$, observers z_n , affine gauges $c_n(s)$, and pressure gauges. On fixed annular cylinders set, schematically,

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s).$$

The uniform bound $\|U_n\|_{L^\infty} \leq M$ gives weak-* compactness of W_n in L_{loc}^∞ , and the fundamental theorem of Young measures gives, after passing to a further subsequence, a barycenter $\bar{W}$ and Young measure $\mathcal{Y}_{x,s}$. If the sequence is strongly compact in L_{loc}^3 , then $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$; otherwise $\mathcal{Y}$ records the unresolved oscillation defect. If the defect is not tight in any fixed logarithmic annulus, the escaping logarithmic mass is recorded by Λ . These alternatives are exhaustive and define an element of $\mathfrak{T}_M$. Since the normalized defect mass is one, $\mathcal{T} \neq 0$.

It remains to identify the equation satisfied by the barycenter. After the change $y = y_n + R_n x$ and the admissible affine/pressure normalizations, the centered equation becomes, in distributions on fixed macroscopic annuli,

$$\partial_s W_n - R_n^{-2} \Delta_x W_n + R_n^{-1} W_n \cdot \nabla_x W_n + \frac{1}{2} x \cdot \nabla_x W_n + \frac{1}{2} W_n + \nabla_x Q_n = o(1).$$

For every compactly supported test function, the viscous and nonlinear contributions are bounded respectively by

$$C R_n^{-2} \|W_n\|_{L^\infty}, \quad C R_n^{-1} \|W_n\|_{L^\infty}^2,$$

and hence both tend to zero. The pressure gauges give distributional compactness of the remaining gradients on annuli. Passing to the weak-* barycentric limit yields the displayed linear critical-tail transport equation. Applying the same weak formulation to transported test functions shows that $(\mathcal{Y}, \Lambda)$ is transported by the induced log-dilation flow. This construction is local on macroscopic annuli and uses no global critical norm, CKN budget, or energy estimate. $\square$

Corollary 9.58 (Coherent extraction under annular tightness and strong compactness). *If the critical-tail defect $\mathcal{T} \in \mathfrak{T}_M$ produced by [Theorem 9.57](#) satisfies $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{W(x,s)}$ for a.e. (x, s) , then there exist macroscopic scales, covariant observers, affine gauges, and pressure gauges for which the blown-down fields converge strongly in L_{loc}^3 and weak-* in L_{loc}^∞ to a nonzero coherent critical tail $(W, Q) \in \mathcal{C}_{\text{coherit}}$.*

Proof. This is the coherent subcase of [Theorem 9.57](#). The condition $\Lambda = 0$ rules out loss of mass across logarithmic scale, and the Dirac-valued Young measure gives strong L_{loc}^3 convergence by uniform L^∞ boundedness. Nontriviality follows from the normalized affine oscillation mass. $\square$

Theorem 9.59 (Coherent critical-tail classification). *Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be a coherent critical tail. In logarithmic radial variables*

$$x = e^\rho \omega, \quad Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s),$$

its compact log-translation/time-translation hull belongs to exactly one of

$$\mathcal{C}_{\text{affine}}, \quad \mathcal{C}_{\text{homcrit}}, \quad \mathcal{C}_{\text{logper}}, \quad \mathcal{C}_{\text{ap}}.$$

Here $\mathcal{C}_{\text{homcrit}}$ is the homogeneous (-1) -tail class, $\mathcal{C}_{\text{logper}}$ is the nonzero log-periodic tail class, and $\mathcal{C}_{\text{ap}}$ is the remaining aperiodic minimal log-translation hull class.

Proof. Let

$$\mathcal{H}(Z) := \overline{\{Z(\rho + a, \omega, s + b) : a, b \in \mathbb{R}\}}$$

be the compact logarithmic translation hull. If the affine-normalized oscillation vanishes on every compact log-cylinder, then the tail is affine/parasitic. Otherwise, if every element of $\mathcal{H}(Z)$ is independent of ρ , then W is (-1) -homogeneous and belongs to $\mathcal{C}_{\text{homcrit}}$. If the log-translation orbit has a nonzero period, then W is discretely self-similar at infinity and belongs to $\mathcal{C}_{\text{logper}}$. If neither alternative holds, compact-flow decomposition gives a minimal compact invariant subsystem with no nonzero log period; this is the aperiodic log-translation class $\mathcal{C}_{\text{ap}}$. These alternatives exhaust compact log-translation hulls. $\square$

Theorem 9.60 (Realized critical-tail rigidity by stratification). *Let $\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M$ be a nonzero realized critical-tail profile generated by a bounded normalized residual Navier–Stokes terminal sequence after terminal concentration-set removal. Then exactly one of the following ordered profile alternatives occurs:*

- (a) $\mathcal{T}$ regenerates affine-normalized local concentration;
- (b) $\mathcal{T}$ is affine/parasitic;
- (c) $\mathcal{T}$ is log-diffuse;
- (d) $\mathcal{T}$ is a Young critical tail;
- (e) $\mathcal{T}$ is a coherent homogeneous critical tail;
- (f) $\mathcal{T}$ is a coherent log-periodic critical tail;
- (g) $\mathcal{T}$ is a coherent aperiodic critical tail.

In the ordered refined decomposition, the log-diffuse, Young, homogeneous, log-periodic, and aperiodic critical-tail alternatives are lower structured strata. Thus every nonzero realized critical-tail profile which remains after these lower strata have been removed either regenerates local concentration or collapses to the affine/parasitic lower stratum.

Proof. If the covariant observer orbit closure of $\mathcal{T}$ intersects a constant-in-time concentration set $\{U : \text{osc}_{3,\text{ct}}(U; Q_1^{\text{cov}}) \geq \eta\}$ for some $\eta > 0$, then alternative (a) holds. This case is removed first.

If every spatial quotient oscillation observable vanishes on the profile, then [Theorems 8.16](#) and [8.21](#) place the profile in the affine/parasitic lower stratum. This is alternative (b). It remains to consider the case in which the profile has nonzero non-affine diffuse content and does not regenerate unit-scale concentration.

By [Theorem 9.57](#), the residual normalized diffuse mass is represented by the critical-tail compactification. Thus the boundary mass has only two mechanisms: mass escaping every bounded logarithmic annulus, recorded by Λ , and annular blow-down mass, recorded by $(\bar{W}, \mathcal{Y})$. If $\Lambda \neq 0$, the profile is log-diffuse, giving (c). If $\Lambda = 0$ but $\mathcal{Y}_{x,s}$ is not a Dirac mass on a set of positive macroscopic measure, the profile is a Young critical tail, giving (d).

The remaining case is coherent: $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$ a.e. Then [Theorem 9.58](#) gives a coherent critical tail $(\bar{W}, Q) \in \mathcal{C}_{\text{cohcrit}}$. Applying [Theorem 9.59](#) gives exactly one of the homogeneous, log-periodic, or aperiodic coherent alternatives, namely (e), (f), or (g), unless the

affine/parasitic case already occurred. The alternatives are mutually exclusive after the priority order above: regenerated concentration is removed first, then affine collapse, then log-diffuse mass, then Young oscillation, and finally the coherent log-translation classification. The final assertion follows from this ordered classification. $\square$

9.4. Complete closure of lower alternatives. The critical-tail classification above is a classification theorem rather than an emptiness theorem. The complete residual decomposition is closed by local lower-stratum exclusions together with the minimal mesoscopic-scale theorem [Theorem 9.54](#). That theorem first gives hidden-scale compactness through [Theorem 9.56](#). The log-diffuse alternative is reduced to hidden-scale compactness by the log-window descendant selection of [Theorem 9.29](#); the Young alternative is reduced to it by the Fréchet–Kolmogorov/Young-measure equivalence in [Theorem 9.32](#). The critical-tail reduction theorem used in both reductions is [Theorem 9.54](#). These are local statements about realized residual descendants; none is a global $L_t^\infty L_x^3$, finite-energy-tail, or global CKN-budget estimate.

Theorem 9.61 (Oscillatory entry normalization). *Let (U, Π) be a retained residual profile after the earlier lower classes recorded in the setup paper `proof_setup.tex` have been removed. Then exactly one of the following holds:*

- (i) *after canonical affine/parasitic normalization and an admissible affine symmetry S_c , the profile retains positive constant-in-time non-affine oscillation:*

$$\Phi(S_c U) > 0;$$

- (ii) *all spatial oscillations $\text{osc}_{3,\text{sp}}(U; Q)$ vanish on compact cylinders, and the profile belongs to the affine/parasitic quotient stratum $\mathcal{C}_{\text{aff}}$.*

Consequently the retained normalized residual decomposition has

$$\mathcal{C}_{\text{aff}} \cap \mathcal{R}^\#(\mathcal{S}; I, J) = \emptyset.$$

Proof. The dichotomy is exactly [Theorem 8.21](#). If all spatial oscillations $\text{osc}_{3,\text{sp}}(U; Q)$ vanish on compact cylinders, then the profile is a spatially homogeneous quotient mode and belongs to $\mathcal{C}_{\text{aff}}$. If this alternative is not available after lower-stratum removal, the parasitic-free representative has $\Phi(S_c U) > 0$ for an admissible affine symmetry S_c . A profile in $\mathcal{C}_{\text{aff}}$ therefore cannot carry retained normalized residual concentration, so $\mathcal{C}_{\text{aff}} \cap \mathcal{R}^\#(\mathcal{S}; I, J) = \emptyset$. $\square$

Corollary 9.62 (Young alternative exclusion from minimal mesoscopic scale reduction).

$$\mathcal{C}_{\text{Young}} = \emptyset.$$

Proof. Let a retained residual profile generate a nonzero Young critical-tail alternative. By [Theorem 9.30](#), the log-diffuse alternative is already empty, so the defect has a tight macroscopic annular realization. Let W_n be the corresponding annular blow-down and $\mathcal{Y}_{x,s}$ its Young measure. By [Theorems 9.54](#) and [9.56](#), W_n has no hidden-scale variance on compact annular cylinders. Hence [Theorem 9.32](#) gives

$$\mathcal{Y}_{x,s} = \delta_{\tilde{W}(x,s)} \quad \text{for a.e. } (x, s),$$

contradicting the definition of $\mathcal{C}_{\text{Young}}$. Therefore $\mathcal{C}_{\text{Young}} = \emptyset$. $\square$

Theorem 9.63 (Critical-tail rigidity from minimal mesoscopic scale reduction). *Every nonzero recurrent critical-tail defect generated by a realized residual terminal sequence belongs to*

$$\mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Equivalently, after regenerated concentration, affine/parasitic collapse, and coherent critical tails have been removed, no recurrent critical-tail defect remains.

Proof. Let $\mathcal{D} \in \mathfrak{T}_M$ be a nonzero recurrent critical-tail defect. By [Theorem 9.60](#), exactly one ordered alternative occurs: regenerated concentration, affine/parasitic collapse, log-diffuse defect, Young critical tail, or coherent critical tail. The first two alternatives are precisely $\mathcal{C}_{\text{conc}}$ and $\mathcal{C}_{\text{aff}}$. The log-diffuse alternative is empty by [Theorem 9.30](#). The Young alternative is empty by [Theorem 9.62](#). The only remaining nonempty critical-tail alternative is therefore the coherent alternative $\mathcal{C}_{\text{cohcrit}}$. This proves the inclusion. Once $\mathcal{C}_{\text{conc}}$, $\mathcal{C}_{\text{aff}}$, and $\mathcal{C}_{\text{cohcrit}}$ are removed, the inclusion leaves no recurrent critical-tail defect. $\square$

Lemma 9.64 (Bounded velocity gauges for realized blow-downs). *Let $f : E \rightarrow \mathbb{R}^3$ be measurable on a finite positive-measure set E , and assume $|f| \leq M$ a.e. Then the minimization problem*

$$\inf_{c \in \mathbb{R}^3} \int_E |f - c|^3$$

has a minimizer in $\overline{B_M(0)}$. Consequently every affine-normalized blow-down of a sequence satisfying $\|U_n\|_{L^\infty} \leq M$ may be represented, without increasing any local oscillation functional, by velocity gauges c_n with $|c_n| \leq M$. If the normalization is performed time-slice by time-slice, the same conclusion holds for a measurable choice $c_n(s)$, with $|c_n(s)| \leq M$ for a.e. s .

Proof. The functional is continuous and coercive in c , so a minimizer exists. Let $\pi : \mathbb{R}^3 \rightarrow \overline{B_M(0)}$ be the nearest-point projection. Since $\overline{B_M(0)}$ is closed and convex, for every $v \in \overline{B_M(0)}$ and every $c \in \mathbb{R}^3$,

$$|v - \pi(c)| \leq |v - c|.$$

Taking $v = f(y)$ and integrating gives

$$\int_E |f - \pi(c)|^3 \leq \int_E |f - c|^3.$$

Thus any minimizing sequence may be projected into $\overline{B_M(0)}$, and a minimizer can be chosen there. Applying this to the local velocity fields used to define the affine-normalized observer oscillations gives the asserted bounded representatives. The time-slice version is the same argument on each slice, with measurability obtained by the usual measurable-selection theorem for a continuous coercive finite-dimensional minimization problem. The projection step is pointwise in s , so the bound $|c_n(s)| \leq M$ is preserved. $\square$

Theorem 9.65 (Bounded-origin realization of coherent critical tails). *Let a coherent critical tail be generated by a bounded normalized residual Navier–Stokes terminal sequence with:*

- (i) *locally bounded velocity representatives;*
- (ii) *pressure gauges bounded modulo time and affine functions on each fixed time window;*
- (iii) *convergence on every fixed annulus/log-window;*
- (iv) *the normalized support-profile property realized from the original blow-up sequence.*

Then it is realized, after subsequence extraction and admissible gauges, as a weak- L_{loc}^∞ limit on full macroscopic compact cylinders, including cylinders meeting the macroscopic origin. In particular, every coherent critical tail has a representative which is locally bounded near $x = 0$ on compact macroscopic time intervals, and whose restrictions to all tail windows reproduce the coherent tail.*

Proof. Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be a coherent critical tail generated by a bounded terminal sequence $(U_n, \Pi_n) \subset \mathfrak{X}_M$, with $\|U_n\|_{L^\infty} \leq M$. By coherent extraction, after passing to a subsequence there are macroscopic scales $R_n \rightarrow \infty$, covariant observer centers (y_n, σ_n) , affine velocity gauges, and pressure gauges such that on every compact annular cylinder

$$A_{r_0, r_1, I} := \{r_0 < |x| < r_1\} \times I, \quad 0 < r_0 < r_1 < \infty,$$

the fields

$$W_n(x, s) := U_n(y_n + R_n x, \sigma_n + s) - c_n(s)$$

converge to W strongly in $L^3(A_{r_0, r_1, I})$ and weak-* in $L^\infty(A_{r_0, r_1, I})$.

We first check that the same gauges which realize the annular coherent tail are bounded on compact time intervals. Fix a compact interval I and use the annulus $1 < |x| < 2$. The weak-* L^∞ convergence on $\{1 < |x| < 2\} \times I$ implies, after discarding finitely many n , that

$$\|W_n\|_{L^\infty(\{1 < |x| < 2\} \times I)} \leq C_I.$$

For a.e. $s \in I$, if $|c_n(s)| > M + C_I$, then for every x with $1 < |x| < 2$,

$$|U_n(y_n + R_n x, \sigma_n + s) - c_n(s)| \geq |c_n(s)| - M > C_I,$$

contradicting the preceding L^∞ bound. Hence

$$|c_n(s)| \leq M + C_I \quad \text{for a.e. } s \in I.$$

This argument keeps the annular representative fixed. The bounded-gauge lemma above shows, in addition, that the affine-normalized construction may always be started with bounded canonical gauges.

Therefore, on every full macroscopic cylinder

$$K_{R, I} := B_R(0) \times I,$$

one has

$$|W_n(x, s)| \leq |U_n(y_n + R_n x, \sigma_n + s)| + |c_n(s)| \leq 2M + C_I$$

for a.e. $(x, s) \in K_{R, I}$. The terminal sequence is defined on the whole ancient space, or on expanding domains containing $y_n + R_n B_R$ for all large n ; hence W_n is defined on $K_{R, I}$ for all sufficiently large n .

Choose an exhaustion of $\mathbb{R}^3 \times \mathbb{R}$ by full compact macroscopic cylinders $K_j = B_j(0) \times [-j, j]$. Banach–Alaoglu gives, by a diagonal extraction that does not change notation, a field

$$W^{\text{full}} \in L_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R}),$$

such that

$$W_n \rightharpoonup^* W^{\text{full}} \quad \text{in } L^\infty(K_j)$$

for every j . The subsequence still has the annular strong L_{loc}^3 convergence to the coherent tail W . Therefore, for every annular cylinder $A_{r_0, r_1, I}$,

$$W^{\text{full}} = W \quad \text{a.e. on } A_{r_0, r_1, I}.$$

Since the annuli exhaust $(\mathbb{R}^3 \setminus \{0\}) \times \mathbb{R}$, the annular coherent tail is the restriction of the full weak-* limit W^{full} away from $x = 0$.

We take W^{full} as the full-cylinder representative of the coherent tail. It is locally bounded on every compact macroscopic cylinder, including cylinders meeting $x = 0$, and it realizes the same annular tail because it agrees with W on all punctured compact cylinders. Pressure gauges are not needed for the boundedness conclusion; if one records the pressure as well, the local pressure-gauge compactness in the terminal profile class gives distributional pressure-gradient limits on the same full cylinders after the usual time-dependent gauges. The construction uses only the uniform local boundedness inherited from the Type I terminal profile class and finite-dimensional affine normalization; it uses no global critical norm or global tail budget. $\square$

Lemma 9.66 (Minimal coherent hulls inherit bounded-origin vanishing). *Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be realized by the bounded-origin theorem, and write*

$$Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s).$$

Let $\mathcal{M}$ be a compact minimal invariant subset of the logarithmic translation hull generated by Z . Then every $Z_* \in \mathcal{M}$ is itself realized by a full locally bounded macroscopic tangent near the origin, and

$$Z_*(\rho, \cdot, \cdot) \rightarrow 0 \quad \text{as } \rho \rightarrow -\infty$$

in the local logarithmic topology. Consequently $0 \in \mathcal{H}(Z_*)$. If $\mathcal{M} = \mathcal{H}(Z_*)$ is minimal, then $\mathcal{M} = \{0\}$.

Proof. Elements of the logarithmic hull are limits of logarithmic translations of the same coherent tail realized from the original blow-up sequence. Logarithmic translation corresponds to choosing another macroscopic scale in the bounded-origin realization. The bounded-origin theorem gives L_{loc}^∞ bounds on every full compact cylinder, including neighborhoods of the macroscopic origin, with constants that are stable under passing to local hull limits. Hence each hull element Z_* corresponds to a velocity

$$W_*(r\omega, s) = r^{-1} Z_*(\log r, \omega, s)$$

which is locally bounded near $r = 0$.

For $r = e^\rho \downarrow 0$, local boundedness gives

$$|Z_*(\rho, \omega, s)| = r |W_*(r\omega, s)| \leq C_K r$$

on every compact angular-time set K . Thus $Z_*(\rho, \cdot, \cdot) \rightarrow 0$ locally as $\rho \rightarrow -\infty$, and the zero profile belongs to the orbit closure $\mathcal{H}(Z_*)$. If Z_* lies in a minimal compact invariant set $\mathcal{M}$, then $\mathcal{H}(Z_*) = \mathcal{M}$. Since $\{0\}$ is closed and invariant, minimality forces $\mathcal{M} = \{0\}$. $\square$

Corollary 9.67 (Coherent critical-tail alternative closure). *One has*

$$\mathcal{C}_{\text{homcrit}} = \emptyset, \quad \mathcal{C}_{\text{logper}} = \emptyset, \quad \mathcal{C}_{\text{ap}} = \emptyset.$$

Proof. Let (W, Q) be a coherent critical tail and use the bounded-origin realization theorem to view W as locally bounded near $x = 0$ on compact time intervals.

If $W \in \mathcal{C}_{\text{homcrit}}$, then

$$W(r\omega, s) = r^{-1} A(\omega, s).$$

Local boundedness near $r = 0$ gives

$$|A(\omega, s)| = r |W(r\omega, s)| \leq Mr$$

for small r . Letting $r \downarrow 0$ gives $A = 0$, hence $W \equiv 0$, contradicting the nonzero spatial quotient oscillation of a realized coherent tail. Thus $\mathcal{C}_{\text{homcrit}} = \emptyset$.

If $W \in \mathcal{C}_{\text{logper}}$, set

$$Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s).$$

Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \rightarrow 0$ as $\rho \rightarrow -\infty$ in the local logarithmic topology. If Z has nonzero period L , then

$$Z(\rho, \omega, s) = Z(\rho - kL, \omega, s)$$

for every integer k . If $L > 0$, send $k \rightarrow +\infty$; if $L < 0$, send $k \rightarrow -\infty$. In both cases the orbit approaches $r = 0$, where bounded-origin realization gives zero. Hence $Z(\rho, \omega, s) = 0$ for every ρ , so $W \equiv 0$, again contradicting nonzero critical tail mass. Thus $\mathcal{C}_{\text{logper}} = \emptyset$.

If $W \in \mathcal{C}_{\text{ap}}$, let $\mathcal{M}$ be the nonzero minimal aperiodic logarithmic subsystem used to define this alternative and choose $Z_* \in \mathcal{M}$. By [Theorem 9.66](#), $0 \in \mathcal{H}(Z_*)$. Since $\mathcal{M}$ is minimal, $\mathcal{H}(Z_*) = \mathcal{M}$; hence the closed invariant set $\{0\}$ contained in $\mathcal{M}$ forces $\mathcal{M} = \{0\}$. This contradicts the nonzero affine-normalized oscillation required in $\mathcal{C}_{\text{ap}}$. Therefore $\mathcal{C}_{\text{ap}} = \emptyset$. $\square$

Corollary 9.68 (Critical-tail exclusion, complete form). *No nonzero realized critical-tail profile remains after terminal concentration-set removal and affine/parasitic lower-stratum removal.*

Proof. Let $\mathcal{T} \in \mathfrak{T}_M$ be a remaining nonzero realized critical-tail profile. By [Theorem 9.60](#), exactly one ordered alternative occurs: regenerated concentration, affine/parasitic collapse, log-diffuse defect, Young critical tail, coherent homogeneous tail, coherent log-periodic tail, or coherent aperiodic tail.

Regenerated concentration contradicts terminal concentration-set removal. The affine/parasitic alternative is impossible by [Theorem 9.61](#). The log-diffuse alternative is impossible by [Theorem 9.30](#), and the Young alternative is impossible by [Theorem 9.62](#). The three coherent alternatives are impossible by [Theorem 9.67](#). All alternatives are therefore impossible, so no such $\mathcal{T}$ remains. $\square$

Theorem 9.69 (No mixed compact–diffuse residual profile). *Let (U, Π) be a retained terminal profile in the refined residual class, after the affine normalization of [Theorem 8.16](#). Then U cannot have a diffuse exterior oscillation remainder, nor an escaping retained noncompact recentering remainder, coexisting with its compact retained core. Equivalently, the paired terminal configurations*

$$\text{one compact concentration core} + \mathcal{R}_{\text{diff}}, \quad \text{one compact concentration core} + \mathcal{R}_{\text{noncomp}},$$

and their finite, infinite, or continuum concentration-set analogues do not occur in the refined residual class.

Proof. The noncompact escaping case is local. In this theorem the noncompact remainder is retained in the velocity or affine-oscillation sense. If an escaping recentering with retained mass loses the local dissipation or pressure-gauge bounds in [Theorem 9.3](#), then after its own local normalization it is a retained compact-mass recentering sequence. This contradicts [Theorem 9.5](#). Thus it remains to exclude a compact core plus diffuse exterior remainder.

Assume that such a diffuse exterior remainder exists. By the paired concentration-set decomposition [Theorem 9.11](#), remove neighborhoods of the concentration set at every positive constant-in-time oscillation threshold. The mixed assumption says that a nonzero diffuse oscillation residual remains. Apply [Theorem 9.19](#).

If regenerated concentration occurs, then a positive oscillatory observer survives after all positive-threshold concentration sets were removed, contradicting the definition of the diffuse residual. If affine/parasitic collapse occurs, then [Theorems 8.16](#) and [10.11](#) places the relevant descendant hull in an already closed lower stratum. This cannot occur inside $\mathcal{C}_{\text{gen}}$.

The only remaining alternative is a critical diffuse tail. By [Theorem 9.57](#), it generates a nonzero $\mathcal{T} \in \mathfrak{T}_M$. This contradicts [Theorem 9.68](#). Hence no mixed compact–diffuse residual profile exists. $\square$

Definition 9.70 (Finite separated family of retained concentration profiles). A finite separated family of retained concentration profiles is a finite family of recentering sequences. For the original terminal extraction these are the parabolic recenterings of [Theorem 9.1](#); for descendants of a centered profile they are the covariant tail recenterings of [Theorem 8.4](#). We write them uniformly as $g_{1,n}, \dots, g_{J,n}$, $J \geq 2$, such that:

- (i) the recentering sequences are mutually separated;
- (ii) each recentered sequence converges locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo gauges to a terminal profile (U_j, Π_j) ;
- (iii) each limiting profile has retained local concentration:

$$\mathcal{E}_1(U_j, \Pi_j; 0) \geq \varepsilon_* \quad (j = 1, \dots, J).$$

This is a finite local family, not a statement that all concentration profiles in the whole space have been counted. The separated-profile proof below treats the possible continuation of escaping recenterings by local concentration alternatives, not by a global counting estimate.

10. SEPARATED CONCENTRATION PROFILES, INDECOMPOSABILITY, AND SEQUENCE- L^3

We now close the terminal alternatives carrying retained local concentration. The proof uses no global critical-mass budget. Instead, finite separated concentration families are treated through local recentering and compactness. Iterating retained tail limits from such a finite family has only local profile outcomes: an indecomposable retained concentration profile, a mixed compact–diffuse or compact–noncompact successor, or a chain of retained concentrating descendants. The mixed successors are excluded by [Theorem 9.69](#); infinite chains of retained concentrating descendants and finite loops are excluded by the compact recurrence theorem.

Definition 10.1 (Tail limits and terminal indecomposability). Let (U, Π) be a terminal profile with retained local concentration. A retained concentrating tail limit is a limit of covariant tail recenterings

$$\mathcal{T}_{(x_n, \tau_n)}(U, \Pi), \quad |x_n| + |\tau_n|^{1/2} \rightarrow \infty,$$

in the local smooth velocity topology and the local $L^{3/2}$ pressure topology modulo gauges, with retained local concentration at the origin. An ordinary fixed spatial shift is not used in this definition; by [Theorems 8.3](#) and [8.4](#), it would introduce an unbounded spurious drift for escaping centers. Diffuse exterior concentration means that there are $\tau_n \rightarrow -\infty$, $R_n \rightarrow \infty$, and $\eta > 0$, such that

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq \eta,$$

while every unit-scale exterior covariant recentering sequence based in the parabolic exterior windows

$$\{|y| > R_n\} \times (\tau_n - 1, \tau_n + 1)$$

has pressure-normalized parabolic CKN mass below the retained-concentration threshold, unless that recentering sequence has loss of local compactness. Thus the word “diffuse” is always parabolic: localization is tested by unit CKN cylinders, not only by spatial balls on the distinguished time slices. An escaping non-compact recentering sequence is a covariant tail recentering sequence for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails. A retained concentration profile is terminally indecomposable if it has no retained concentrating tail limit, no diffuse exterior concentration, and no escaping non-compact recentering sequence.

Lemma 10.2 (Time-slice concentration gives parabolic CKN concentration). *Let (U, Π) be a bounded smooth solution of (1.2) with $\|U\|_{L^\infty} \leq M$. If*

$$\int_{B_1(0)} |U(y, 0)|^3 dy \geq \varepsilon_0 > 0,$$

then there are $\delta = \delta(M, \varepsilon_0) > 0$ and $c_0 = c_0(M, \varepsilon_0) > 0$ such that, after the standard local pressure gauge,

$$\iint_{B_1(0) \times (-\delta^2, 0)} |U|^3 dy d\tau \geq c_0, \quad \text{and hence} \quad \mathcal{E}_1(U, \Pi; 0) \geq c_0.$$

The same constants apply to every covariant tail recentering $\mathcal{T}_{(x, \sigma)}(U, \Pi)$.

Proof. Apply interior parabolic regularity on $B_2 \times [-2\delta_0, \delta_0]$, with $\delta_0 > 0$ universal and small. On this cylinder the drift $-\frac{1}{2}y \cdot \nabla U$ has bounded smooth coefficients, and the nonlinear forcing is bounded by M^2 . After choosing local pressure gauges, the standard interior estimates give

$$\|\partial_\tau U\|_{L^\infty(B_1 \times [-\delta_0, 0])} + \|\nabla U\|_{L^\infty(B_1 \times [-\delta_0, 0])} \leq C(M)$$

where $C(M)$ is independent of the particular profile. Hence the map $\tau \mapsto \int_{B_1} |U(y, \tau)|^3 dy$ is uniformly continuous with a modulus depending only on M and $|B_1|$. Choose $\delta \leq \delta_0$ so that

$$\int_{B_1} |U(y, \tau)|^3 dy \geq \frac{1}{2} \varepsilon_0 \quad -\delta^2 < \tau < 0.$$

Integrating in time gives a positive lower bound for the velocity part of $\mathcal{E}_1(U, \Pi; 0)$; the pressure contribution is nonnegative after any admissible gauge. The constants are unchanged under $\mathcal{T}_{(x, \sigma)}$, because that recentering is an exact composition of the centered symmetries in [Theorem 8.4](#). $\square$

Lemma 10.3 (One retained tail limit gives two separated concentration profiles). *Let U be a retained concentration terminal profile. If U has a retained concentrating tail limit, then the corresponding terminal extraction contains a finite separated family of retained concentration profiles.*

Proof. The origin recentering sequence carries retained concentration:

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Let W be a retained concentrating tail limit. By definition there are covariant relative tail recentering parameters $\zeta_n = (x_n, \tau_n)$, with $|x_n| + |\tau_n|^{1/2} \rightarrow \infty$, such that $\mathcal{T}_{\zeta_n}(U, \Pi) \rightarrow (W, \Pi_W)$ in the terminal topology and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

By local convergence of velocity and pressure after gauges,

$$\mathcal{E}_1(\mathcal{T}_{\zeta_n} U, \mathcal{T}_{\zeta_n} \Pi; 0) \geq \frac{1}{2} \varepsilon_*$$

for all sufficiently large n . Thus, for large n , the origin recentering sequence and the covariant tail sequence ζ_n are separated and have retained local concentration. They form a finite separated two-profile concentration family. $\square$

Proposition 10.4 (No finite separated family and local exclusions imply indecomposability). *Every single retained concentration terminal profile arising from a refined residual candidate is terminally indecomposable.*

Proof. Let U be a single retained concentration terminal profile. If U had a retained concentrating tail limit, [Theorem 10.3](#) would produce a finite two-profile separated family with retained concentration, contradicting [Theorem 10.23](#). If U had diffuse exterior concentration or an escaping non-compact recentering sequence, the corresponding terminal extraction would be a mixed profile configuration consisting of the retained compact core U plus a diffuse or noncompact tail remainder. This is excluded by [Theorem 9.69](#). Hence no retained concentrating tail limit, diffuse exterior concentration, or escaping non-compact recentering sequence exists, and U is terminally indecomposable. $\square$

Lemma 10.5 (Indecomposable sequence- L^3 completeness). *Let (U, Π) be a bounded smooth ancient solution of (1.2). Suppose U has retained local concentration and is terminally indecomposable in the sense of [Theorem 10.1](#). Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We prove the contrapositive. Define

$$A(\tau) := \int_{\mathbb{R}^3} |U(y, \tau)|^3 dy \in [0, \infty].$$

If the conclusion fails, then

$$\liminf_{\tau \rightarrow -\infty} A(\tau) = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $A(\tau_n) \rightarrow \infty$, allowing the value $+\infty$. Since U is bounded by some M , for each finite R ,

$$\int_{|y| \leq R} |U(y, \tau_n)|^3 dy \leq M^3 |B_R|.$$

Choose $R_n \rightarrow \infty$ so slowly that, after discarding finitely many n ,

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

If $A(\tau_n) = +\infty$, take $R_n = n$; otherwise choose R_n with $M^3 |B_{R_n}| \leq A(\tau_n)/2$.

Define the parabolic exterior concentration level

$$m_n^{\text{par}} := \sup_{\substack{|x| > R_n - 2 \\ |s - \tau_n| \leq 1}} \mathcal{E}_1(\mathcal{T}_{(x,s)}(U, \Pi); 0),$$

where $\mathcal{E}_1$ is computed after the admissible local pressure gauge in the recentered variables. This is the correct quantity for the terminal decomposition: it tests unit parabolic CKN cylinders, not merely spatial time-slices.

First suppose $\limsup_{n \rightarrow \infty} m_n^{\text{par}} \geq \varepsilon_*$. Passing to a subsequence, choose (x_n, s_n) , with

$$|x_n| > R_n - 2, \quad |s_n - \tau_n| \leq 1,$$

such that

$$\mathcal{E}_1(\mathcal{T}_{(x_n, s_n)}(U, \Pi); 0) \geq \frac{1}{2} \varepsilon_*.$$

Since $R_n \rightarrow \infty$ and $\tau_n \rightarrow -\infty$, these are escaping covariant tail recenterings. If they have loss of local compactness, terminal indecomposability is contradicted. Otherwise local compactness gives a subsequential terminal limit (W, Π_W) , and lower semicontinuity of the pressure-normalized CKN quantity gives

$$\mathcal{E}_1(W, \Pi_W; 0) \geq c\varepsilon_*$$

after lowering the retained threshold by a fixed harmless factor. Hence W is a retained concentrating covariant tail limit, contradicting terminal indecomposability.

It remains to consider the case in which $m_n^{\text{par}} < \varepsilon_*$ for all large n , after lowering the threshold if necessary. If some unit exterior covariant recentering sequence in the windows

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

has loss of local compactness or pressure-gauge compactness, terminal indecomposability is contradicted by the noncompact defect alternative. If the exterior mass is critical only after log-window normalization, the resulting normalized support profile is a critical-tail defect and is one of the already excluded terminal alternatives. If no such recentering loses compactness and no critical-tail normalization is selected, then the exterior time-slice lower bound

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1$$

together with the parabolic smallness $m_n^{\text{par}} < \varepsilon_*$ is exactly the parabolic diffuse exterior concentration alternative in [Theorem 10.1](#). This again contradicts terminal indecomposability. Both parabolic alternatives are impossible, so the desired sequence exists. $\square$

Proposition 10.6 (Stability of the bounded mild formulation). *Let (U_m, Π_m) be normalized parasitic-free centered profiles whose physical pullbacks are bounded mild Navier–Stokes solutions on every finite terminal interval. Assume $U_m \rightarrow U$ locally smoothly in centered variables and that the U_m are uniformly bounded on compact centered time intervals. Then the physical pullback of U is also bounded mild on every finite terminal interval. More precisely, for every $T < 0$, after setting*

$$u^T(x, s) := (-T - s)^{-1/2} U\left(\frac{x}{\sqrt{-T - s}}, -\log(-T - s)\right), \quad s < 0,$$

and translating the interval so that its terminal time is 0, one has, for all $-\infty < s < t < 0$,

$$(10.1) \quad u^T(t) = e^{(t-s)\Delta} u^T(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (u^T \otimes u^T)(\sigma) d\sigma$$

in the bounded mild duality sense used in the setup paper. The same conclusion is preserved by centered time translations, covariant translations, retained tail limits, concentrating descendants, recurrent-core limits, and affine normalization outside $\mathcal{C}_{\text{aff}}$.

Proof. For each m , the normalized setup representative satisfies the Duhamel formula on every finite physical interval. Fix $T < 0$ and $-\infty < s < t < 0$. Pull the centered convergence back to the compact physical time interval $[s, t]$ after the finite terminal shift. The velocities converge locally smoothly and are uniformly bounded; hence

$$u_m^T \otimes u_m^T \rightarrow u^T \otimes u^T$$

locally in every finite L^q , and in particular in distributions. Testing the Duhamel identity for u_m^T against a compactly supported smooth vector field and using the heat-kernel/Leray-projection duality gives convergence of the linear heat term and of the bilinear Duhamel term. The bounded mild duality formulation used in the setup paper then extends the identity from compactly supported tests to the standard bounded mild class, yielding (10.1) for u^T .

Centered time translations and covariant translations are exact symmetries of (1.2); in physical variables they are compositions of translations, scalings, and finite time shifts, which preserve the Duhamel identity. Retained tail limits, concentrating descendants, and recurrent-core limits are locally smooth limits of such normalized representatives, so the preceding limit argument applies. The affine normalization either removes a spatially homogeneous parasitic component or assigns the profile to $\mathcal{C}_{\text{aff}}$; outside $\mathcal{C}_{\text{aff}}$ no homogeneous parasitic forcing remains, and the same Duhamel identity holds for the canonical representative. $\square$

Theorem 10.7 (Parasitic-free mildness inheritance for normalized profiles). *Let (U, Π) be a retained terminal profile, retained covariant tail descendant, concentrating descendant limit, or recurrent-core limit represented in the normalized residual profile class generated by the Seregin extraction in the setup paper. Then exactly one of the following alternatives applies.*

- (i) (U, Π) belongs to the affine/parasitic lower stratum of Theorem 8.16;
- (ii) after the canonical affine normalization already built into the residual representative, the physical pullback

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0,$$

is parasitic-free and has the following property: for every $T < 0$, the finite-terminal shift

$$u^T(x, s) := u(x, T + s), \quad s < 0,$$

is a bounded mild ancient Navier–Stokes solution in the Albritton–Barker sense on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. Affine-constant centered profiles are precisely the homogeneous parasitic profiles removed by [Theorem 8.16](#); they are assigned to alternative (i). We therefore consider a representative outside that lower stratum.

The setup paper constructs normalized Seregin limits as centered bounded mild ancient representatives, after removal of the spatially homogeneous parasitic mode. The stability statement [Theorem 10.6](#) shows that the limiting operations used in this paper preserve the bounded mild formulation: exact centered symmetries preserve the Duhamel identity, local limits preserve it by bounded mild duality, and affine normalization either removes the homogeneous parasitic mode or places the profile in $\mathcal{C}_{\text{aff}}$. Thus retained terminal profiles, covariant tail descendants, concentrating descendant limits, and recurrent-core limits remain parasitic-free mild unless they have fallen into the affine/parasitic lower stratum.

Finally fix $T < 0$. On $\mathbb{R}^3 \times (-\infty, T]$, the physical pullback

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right)$$

is smooth and bounded by $(-T)^{-1/2} \|U\|_{L^\infty}$, and it solves the physical Navier–Stokes equations by the direct self-similar change of variables. By the inherited parasitic-free mild formulation, for $-\infty < s < t \leq T$,

$$u(t) = e^{(t-s)\Delta} u(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(\sigma) d\sigma.$$

After translating the finite terminal time T to 0, this is the mild ancient formulation for u^T on every finite forward interval $(-\infty, 0)$. The bound $\|u^T\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))} \leq (-T)^{-1/2} \|U\|_{L^\infty}$ is finite. Thus u^T is a genuine bounded mild ancient solution in the sense required by the Albritton–Barker endpoint theorem. $\square$

Theorem 10.8 (Albritton–Barker endpoint Liouville theorem). *Let $u \in L^\infty(\mathbb{R}^3 \times (-\infty, 0))$ be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists $t_k \downarrow -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem in the sequence- L^3 form proved by Albritton and Barker [\[1\]](#). $\square$

Lemma 10.9 (No retained indecomposable concentration profile). *No retained terminally indecomposable concentration profile exists.*

Proof. Let U have retained local concentration and be terminally indecomposable. By [Theorem 10.5](#), choose $\tau_k \rightarrow -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let u be the physical pullback and set $t_k = -e^{-\tau_k}$. Since $\tau_k \rightarrow -\infty$, the sequence satisfies $t_k \downarrow -\infty$, and critical scaling gives

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By [Theorem 10.7](#), either U is in the affine/parasitic lower stratum, which is not part of the normalized residual class, or else for each $T < 0$

$$u^T(x, s) := u(x, T + s), \quad s < 0,$$

is a bounded mild ancient solution. Choose $T < 0$ and discard finitely many indices so that $t_k < T$. Then $s_k := t_k - T \downarrow -\infty$, and

$$\|u^T(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} = \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} \leq \sup_j \|U(\cdot, \tau_j)\|_{L^3(\mathbb{R}^3)} < \infty.$$

[Theorem 10.8](#) gives $u^T \equiv 0$. Since $T < 0$ was arbitrary, the physical pullback vanishes on $\mathbb{R}^3 \times (-\infty, 0)$, and therefore $U \equiv 0$. This contradicts retained local concentration. $\square$

Lemma 10.10 (Diagonal lifting of retained concentrating tail limits). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate, represented by separated recentering sequences $g_{j,n} = (y_{j,n}, \tau_{j,n})$. Let (U_j, Π_j) be the terminal profile obtained in the j -th recentering. If (U_j, Π_j) has a retained concentrating tail limit (W, Π_W) , then there is a combined parabolic-covariant descendant recentering sequence $\mathfrak{d} = (d_n)$, in the sense of [Theorem 9.2](#), for the original terminal sequence such that:*

- (i) $\mathfrak{d}$ is asymptotically separated from g_j ;
- (ii) the recentered original sequence at $\mathfrak{d}$ converges locally to (W, Π_W) , after admissible pressure gauges;
- (iii) $\mathfrak{d}$ has retained local concentration.

Proof. By definition of retained concentrating tail limit, there are relative covariant tail recenterings $\zeta_m = (\xi_m, \sigma_m)$, with

$$|\xi_m| + |\sigma_m|^{1/2} \rightarrow \infty,$$

such that

$$\mathcal{T}_{\zeta_m}(U_j, \Pi_j) \rightarrow (W, \Pi_W)$$

locally in the terminal topology, and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Fix m . Since $V_n^{g_j} \rightarrow U_j$ locally smoothly and $(P_n^{g_j}) \rightarrow \Pi_j$ locally in $L^{3/2}$ modulo gauges, the convergence also holds on the fixed covariant tube

$$\{(y + e^{s/2}\xi_m, s + \sigma_m) : (y, s) \in Q_m(0, 0)\}.$$

Therefore one can choose $n(m)$ so large that the covariantly recentered original fields

$$V_{n(m)}(y + y_{j,n(m)} + e^{s/2}\xi_m, s + \tau_{j,n(m)} + \sigma_m)$$

and the corresponding pressure gauges are within m^{-1} of the $\mathcal{T}_{\zeta_m}$ -recentered fields of (U_j, Π_j) on $Q_m(0, 0)$.

Set

$$d_m := (g_{j,n(m)}; \xi_m, \sigma_m).$$

Relabeling the subsequence $n(m)$ as the terminal index gives a combined parabolic-covariant descendant recentering sequence $\mathfrak{d} = (d_m)$ for the original terminal sequence. Since the covariant parabolic distance from d_m to $g_{j,n(m)}$ is $|\xi_m| + |\sigma_m|^{1/2}$, the sequence $\mathfrak{d}$ is separated from the j -th recentering sequence. The diagonal choice gives local convergence of the original sequence at $\mathfrak{d}$ to (W, Π_W) . Let $\mu_{n(m)}^\mathfrak{d}$ denote the local defect measure of the original terminal sequence pulled back by this combined parabolic-covariant recentering and by the pressure gauges used in the diagonal convergence. The local CKN quantity is continuous under the strong local velocity convergence and weak lower semicontinuity of the pressure term after these fixed gauges, so

$$\liminf_{m \rightarrow \infty} \mu_{n(m)}^\mathfrak{d}(Q_1(0, 0)) \geq \mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Thus $\mathfrak{d}$ has retained local concentration. $\square$

Theorem 10.11 (Heredity of retained descendants). *Let (U, Π) be a normalized retained residual profile in $\mathfrak{X}_M$, and let (W, Π_W) be a retained covariant tail descendant of (U, Π) . Then $(W, \Pi_W) \in \mathfrak{X}_M$, it carries retained local concentration at the descendant threshold fixed in [Theorem 3.4](#), and it has the same parasitic-free normalization as a terminal profile generated directly by the Seregin extraction in the setup paper. Consequently exactly one of the following ordered outcomes occurs:*

- (i) (W, Π_W) belongs to one of the previously closed lower strata, in which case that descendant is already excluded;
- (ii) (W, Π_W) belongs to the normalized residual terminal profile class, so the concentrating descendant and terminal-indivisibility arguments may be iterated from W .

Proof. The descendant is obtained as a limit of covariant tail recenterings $\mathcal{T}_{\zeta_n}(U, \Pi)$. By [Theorem 8.9](#), the recentered sequence is locally precompact in $\mathfrak{X}_M$, and the limit remains a bounded smooth locally suitable centered solution with the same L^∞ bound and compatible pressure gauges. Retained local concentration is part of the definition of retained descendant; when the descendant is produced through a limit passage, [Theorem 9.4](#) fixes the threshold loss once and for all through [Theorem 3.4](#). The affine/parasitic normalization is canonical for the residual representative and is preserved by T_a , Θ_s , and their composition $\mathcal{T}_{\zeta_n}$; if a limit is affine-parasitic, it is assigned to the affine-constant lower stratum of [Theorem 8.16](#). Otherwise it is represented in the same normalized residual profile class. The ordered residual decomposition then places the descendant either in a previously closed lower class or in the generic residual profile class. This is a local profile reduction and does not use a global critical norm or global tail budget. $\square$

Lemma 10.12 (Escaping recenterings in a separated family). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. For each component $\mathfrak{s}_j$, the following ordered alternatives exhaust the possible successor profiles:*

- (i) the profile (U_j, Π_j) is terminally indecomposable;
- (ii) (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness;
- (iii) (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.

Proof. This is the definition of terminal indecomposability plus the diagonal lifting lemma, with no disjointness assertion. If (U_j, Π_j) is terminally indecomposable, (i) holds. If it is not terminally indecomposable, at least one of the three excluded alternatives in [Theorem 10.1](#) occurs: a retained concentrating tail limit, diffuse exterior concentration, or an escaping recentering sequence with loss of local compactness. The diffuse exterior and local-compactness-failure cases give (ii). In the retained concentrating tail-limit case, [Theorem 10.10](#) lifts the limit to a recentering sequence of the original terminal sequence with retained local concentration, and [Theorem 10.11](#) places the descendant either in a closed lower stratum or back in the normalized residual profile class, giving (iii) in the iterative residual case. If several successor profiles coexist, they are all recorded; the next lemma removes the nonconcentrating ones. $\square$

Lemma 10.13 (No diffuse or non-compact escaping recentering from a retained family). *In the setting of [Theorem 10.12](#), alternative (ii) cannot occur.*

Proof. Suppose that (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness. By the same diagonal procedure used in [Theorem 10.10](#), this tail profile is realized by a terminal extraction of the original refined residual candidate, after passing to a subsequence and choosing the local pressure gauges supplied by the setup paper `proof_setup.tex`. The fact that the diffuse or noncompact profile is not the sole source of the

original compact concentration does not exclude coexistence with the compact core (U_j, Π_j) . The paired exclusion [Theorem 9.69](#) rules out exactly this coexistence in the refined residual class. Therefore alternative (ii) cannot occur for a refined residual family with retained local concentration. $\square$

Definition 10.14 (Closed retained concentration successor relation). For a fixed threshold $\eta > 0$, let

$$\mathfrak{A}_\eta := \mathfrak{A}_\eta^\# = \{U \in \mathfrak{X}_M^\# : \Phi(U) \geq \eta\}$$

be the compact non-affine oscillation set of [Theorem 8.20](#). We write

$$U \mathcal{R}_\eta V$$

if $U, V \in \mathfrak{A}_\eta$ and V is a retained concentrating covariant tail descendant of U .

Lemma 10.15 (Closedness of the retained concentration successor relation). *The relation*

$$\mathcal{R}_\eta \subset \mathfrak{A}_\eta \times \mathfrak{A}_\eta$$

is closed in the local $\mathfrak{X}_M$ topology.

Proof. Let $U_m \rightarrow U$, $V_m \rightarrow V$ in $\mathfrak{X}_M^\#$, with $U_m \mathcal{R}_\eta V_m$. For each m , choose a retained covariant tail observer sequence $\zeta_{m,\ell}$ for U_m whose recentered fields converge to V_m on $Q_m(0,0)$ to accuracy m^{-1} , after admissible pressure gauges. A diagonal choice in (m, ℓ) , using [Theorems 8.9](#) and [9.4](#), gives a single escaping covariant tail sequence for U converging locally to V . Closedness of $\mathfrak{A}_\eta$ follows from [Theorem 8.20](#), and lower semicontinuity of the retained local CKN quantity after the fixed gauges preserves the descendant threshold. Hence $U \mathcal{R}_\eta V$. $\square$

Theorem 10.16 (Recurrence for retained concentration chains). *Assume there exists an infinite chain of retained concentrating descendants*

$$U_0 \mathcal{R}_\eta U_1 \mathcal{R}_\eta U_2 \mathcal{R}_\eta \cdots$$

Then the path space

$$\mathcal{P}_\eta := \left\{ (W_j)_{j \geq 0} \in \mathfrak{A}_\eta^\mathbb{N} : W_j \mathcal{R}_\eta W_{j+1} \text{ for all } j \right\}$$

is nonempty and compact, is invariant under the left shift

$$S(W_0, W_1, W_2, \dots) = (W_1, W_2, \dots),$$

and carries an S -invariant Borel probability measure $\mathbb{P}$. Its projected stationary profile measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported entirely on $\mathfrak{A}_\eta$.

Proof. The set $\mathfrak{A}_\eta = \mathfrak{A}_\eta^\#$ is compact by [Theorem 8.20](#). Hence $\mathfrak{A}_\eta^\mathbb{N}$ is compact and metrizable. By [Theorem 10.15](#),

$$\mathcal{P}_\eta = \bigcap_{j=0}^{\infty} \{(W_k)_{k \geq 0} : (W_j, W_{j+1}) \in \mathcal{R}_\eta\}$$

is closed, hence compact. The given infinite chain is an element of $\mathcal{P}_\eta$, so the path space is nonempty, and it is plainly invariant under the left shift.

For the path $p = (U_0, U_1, \dots)$, define empirical measures

$$\mathbb{P}_N := \frac{1}{N} \sum_{k=0}^{N-1} \delta_{S^k p}.$$

Weak-* compactness of probability measures on compact $\mathcal{P}_\eta$ gives a convergent subsequence $\mathbb{P}_{N_\ell} \rightharpoonup \mathbb{P}$. For each continuous F on $\mathcal{P}_\eta$,

$$\int F \circ S d\mathbb{P}_N - \int F d\mathbb{P}_N = \frac{F(S^N p) - F(p)}{N} \rightarrow 0.$$

Thus $\mathbb{P}$ is shift-invariant. Since every coordinate of every path in $\mathcal{P}_\eta$ belongs to $\mathfrak{A}_\eta$, the projected measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported in $\mathfrak{A}_\eta$. $\square$

Lemma 10.17 (Concatenation of retained concentrating descendants). *Let*

$$U_0 \mathcal{R}_\eta U_1 \mathcal{R}_\eta \cdots \mathcal{R}_\eta U_N$$

be a finite chain of retained concentrating descendants. Then U_N is obtained from U_0 by a single diagonal covariant recentering sequence, after admissible pressure gauges, and it remains in $\mathfrak{A}_\eta$. The pressure gauges in the intermediate limits can be chosen compatibly with the final diagonal limit.

Proof. For $N = 1$ this is the definition of $\mathcal{R}_\eta$. Assume it holds up to $N - 1$. The relation $U_{N-1} \mathcal{R}_\eta U_N$ gives escaping covariant observers for U_{N-1} which converge locally to U_N . Compose these observers with the diagonal observers which realize U_{N-1} from U_0 . The semidirect covariance identity in [Theorem 8.5](#) shows that the composition is again a covariant observer sequence for U_0 . A diagonal choice on $Q_m(0, 0)$, together with the pressure-gauge compatibility in [Theorem 9.4](#), gives local convergence to U_N . Since each link lies in $\mathfrak{A}_\eta$, the endpoint does too. $\square$

Lemma 10.18 (Recurrent path produces a compact retained concentration core). *Let $\mathbb{P}$ be a shift-invariant probability measure on the retained concentration path space $\mathcal{P}_\eta$. For $\mathbb{P}$ -a.e. recurrent path $(W_j)_{j \geq 0}$, the orbit closure of its coordinates, together with all covariant tail limits generated by the returning blocks, contains a nonempty compact invariant set $\mathcal{M}_\eta \subset \mathfrak{A}_\eta$. Every element of $\mathcal{M}_\eta$ is a retained descendant of the original profile realized from the original blow-up sequence.*

Proof. Poincaré recurrence for the left shift gives a full-measure set of paths which return arbitrarily close to themselves in the product topology. Fix such a path. For each returning block, use [Theorem 10.17](#) to compose the finitely many concentrating successor observers into one covariant recentering of the initial profile. The endpoints of the returning blocks remain in the compact set $\mathfrak{A}_\eta$. Taking the closure of all such endpoints and their covariant tail limits inside the compact normalized profile class $\mathfrak{X}_M^\#$ gives a nonempty compact set. Closedness of $\mathcal{R}_\eta$ and compactness of $\mathfrak{A}_\eta$ keep the closure inside $\mathfrak{A}_\eta$. Intersecting over all closed invariant subsets of this compact closure and applying Zorn's lemma gives a compact invariant minimal core $\mathcal{M}_\eta$. The diagonal construction in the concatenation lemma preserves realization from the original sequence and retained concentration, so every element of $\mathcal{M}_\eta$ is realized from the original blow-up sequence. $\square$

Lemma 10.19 (Retention and gauges on the recurrent concentration core). *The compact core $\mathcal{M}_\eta$ of [Theorem 10.18](#) carries the same retained oscillation threshold:*

$$\Phi(U) \geq \eta \quad (U, \Pi) \in \mathcal{M}_\eta,$$

and its pressure gauges are compatible with the invariant averaging used in [Theorem 8.28](#).

Proof. The inequality $\Phi \geq \eta$ is closed under the local topology by [Theorem 8.20](#), and all approximating profiles lie in $\mathfrak{A}_\eta$. Hence every element of the compact core lies in $\mathfrak{A}_\eta$. Gauge compatibility follows from [Theorem 9.4](#): on each compact cylinder the pressure gauges are chosen by a fixed normalization, and diagonal limits preserve the normalized local $L^{3/2}$ pressure topology.

Therefore the invariant measures on $\mathcal{M}_\eta$ used in the recurrent-core rigidity theorem average genuine normalized profiles with retained oscillation, not untracked pressure representatives. $\square$

Theorem 10.20 (No infinite retained concentration chain). *No infinite chain of retained concentrating descendants can occur in the refined residual class.*

Proof. Suppose an infinite chain of retained concentrating descendants exists. If $\limsup_j \Phi(U_j) = 0$, local compactness gives a subsequential limit $U_{j_k} \rightarrow U_\infty$ with $\Phi(U_\infty) = 0$. The diagonal lifting used in [Theorem 10.10](#) shows that this limit is still a retained tail descendant of the original residual profile. By [Theorem 8.21](#), U_∞ belongs to the affine/parasitic lower stratum, contradicting descendant heredity in the refined residual class. Hence $\limsup_j \Phi(U_j) > 0$. Choose $\eta > 0$ and an infinite ordered subchain contained in $\mathfrak{A}_\eta$. By transitivity of retained tail descendants through [Theorem 10.10](#), this ordered subchain is an $\mathcal{R}_\eta$ -chain.

[Theorem 10.16](#) gives a compact shift-invariant path space and a stationary path measure supported on $\mathfrak{A}_\eta$. [Theorems 10.17](#) to [10.19](#) turn a recurrent path into a compact tail-minimal recurrent core for (1.2) whose profiles remain in $\mathfrak{A}_\eta$ with compatible pressure gauges. The recurrent-core rigidity theorem [Theorem 8.28](#) forces this core to be affine-constant after the canonical affine normalization. But affine/parasitic profiles have $\Phi = 0$, whereas $\mathfrak{A}_\eta = \{\Phi \geq \eta\}$. This contradiction rules out the infinite chain of retained concentrating descendants. $\square$

Lemma 10.21 (Finite separated families produce indecomposable profiles or concentration chains). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. Then either one of the terminal profiles (U_j, Π_j) is terminally indecomposable, or the family generates an infinite chain of retained concentrating descendants. A finite recurrent loop is counted as an infinite chain by periodic repetition.*

Proof. Assume no (U_j, Π_j) is terminally indecomposable. By [Theorems 10.12](#) and [10.13](#), each (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.

Starting from A , enlarge the separated family whenever a lifted retained tail descendant is separated from every profile already selected. If this enlargement can be carried out indefinitely, the successive lifted descendants already form an infinite chain of retained concentrating descendants. Otherwise the procedure stops after finitely many enlargements at a finite separated family A' which is maximal for this descendant-lifting procedure.

Form a finite directed graph whose vertices are the retained profiles in A' . Draw an edge $i \rightarrow j$ if a retained tail descendant of U_i , lifted by [Theorem 10.10](#), is equivalent to U_j . There is no escape edge by the construction of A' : an escape edge would be exactly another separated retained unit concentrating cylinder in the original sequence, so the enlargement procedure would not have stopped. Every decomposable vertex has an outgoing edge by the preceding paragraph.

A finite directed graph in which every vertex has an outgoing edge contains a directed cycle. Therefore, if no terminally indecomposable profile appears, the directed recentering relation contains a finite loop

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \dots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}.$$

Repeating the loop periodically gives an infinite chain of retained concentrating descendants. The nonconcentrating successor alternatives at each vertex are excluded by [Theorem 10.13](#), while coexistence of a compact core with a diffuse or noncompact remainder is handled separately by [Theorem 9.69](#). This proof is graph/recurrence based and does not use a global counting or mass budget. $\square$

Lemma 10.22 (Finite recurrent loops of retained tail limits are impossible). *No refined residual terminal extraction contains a finite recurrent loop of retained concentrating tail limits.*

Proof. Let

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \cdots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}$$

be such a loop, and write (U_ℓ, Π_ℓ) for the corresponding retained concentration terminal profiles. Each arrow means that $U_{\ell+1}$ is a retained concentrating tail limit of U_ℓ . Thus, for each ℓ , there are escaping tail recentering sequences whose recentered fields converge locally to $U_{\ell+1}$.

Let $\mathcal{K}$ be the closure, in the local smooth velocity topology and local $L^{3/2}$ pressure topology modulo gauges, of all profiles obtained from the finite set $\{U_0, \dots, U_{q-1}\}$ by time translations and covariant terminal translations. The boundedness inherited from the Type I Seregin extraction theorem and the local pressure compactness from `proof_setup.tex` give compactness of $\mathcal{K}$ on every fixed terminal cylinder; a diagonal argument gives compactness of $\mathcal{K}$. The set is invariant under the same time and covariant translations by construction. The recurrent-loop relations imply that each U_ℓ returns to $\mathcal{K}$ through tail limits, so the closed invariant subset of $\mathcal{K}$ generated by the loop is nonempty, compact, invariant, and contains the retained concentration profiles.

Repeating the loop periodically gives an infinite chain of retained concentrating descendants. This contradicts [Theorem 10.20](#). Therefore no finite recurrent loop of retained concentrating tail limits exists. $\square$

Theorem 10.23 (Finite separated retained concentration profile families are impossible). *No refined residual candidate admits a finite separated family of retained concentration profiles.*

Proof. Suppose, for contradiction, that a refined residual candidate admits a finite retained concentration separated family $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$. By [Theorem 10.21](#), either some component of the family is terminally indecomposable or the family generates an infinite chain of retained concentrating descendants. The first alternative contradicts [Theorem 10.9](#). The second contradicts [Theorem 10.20](#). Hence the finite retained concentration separated family cannot exist. $\square$

Theorem 10.24 (Terminal stratification theorem). *Let (V_n, P_n) be a terminal residual sequence realized from the original blow-up sequence and satisfying the imported setup admissibility hypotheses, the pressure atlas hypotheses, and a retained unit CKN concentration lower bound*

$$\liminf_{n \rightarrow \infty} \mathcal{E}_1(V_n, P_n; z_n) \geq \eta_* > 0.$$

Assume all setup-closed lower classes have been removed. Then the paired terminal extraction of [Theorem 9.11](#) gives the following ordered alternatives:

- (1) *a compact concentrating descendant occurs;*
- (2) *a finite separated family of retained concentration profiles occurs;*
- (3) *an infinite chain of retained concentrating descendants occurs;*
- (4) *a diffuse exterior defect occurs after concentration-neighborhood removal;*
- (5) *a critical-tail defect occurs;*
- (6) *an affine/parasitic quotient profile occurs;*
- (7) *an indecomposable atomic concentration profile occurs and satisfies a sequence- L^3 bound.*

Alternative (1) is a reduction alternative: by descendant heredity it is iterated inside the same normalized profile class until one reaches (2), (3), a mixed diffuse/noncompact/critical-tail branch, or the atomic case (7). Alternatives (2)–(6) are excluded by the preceding branch closures and local terminal arguments. Alternative (7) is excluded by the endpoint bounded-mild ancient Liouville theorem after finite terminal shift. Hence no terminal residual profile satisfying the retained unit CKN concentration lower bound remains.

Proof. This is the assembly of the local results and explicit profile-class arguments in [Sections 9](#) and [10](#). The local concentration decomposition is the paired concentration-set/remainder

decomposition [Theorem 9.11](#). The pure radiative alternatives enter only as sole-source exclusions, and are used only in that form through [Theorem 9.14](#). Coexistence of a compact concentration core with diffuse or noncompact tail remainder is separately ruled out by [Theorem 9.69](#), using realized critical-tail rigidity [Theorems 9.60](#) and [9.68](#). Infinite concentration chains and finite recurrent loops are ruled out by the retained-chain recurrence theorem [Theorem 10.20](#), while finite separated concentration families are ruled out by [Theorem 10.23](#). Therefore any retained concentration terminal profile that survives the ordered lower-class removals is single and has no retained concentrating descendant or diffuse/noncompact tail remainder. By [Theorem 10.4](#) it is terminally indecomposable, and then [Theorems 10.5](#), [10.7](#) and [10.9](#) give the sequence- L^3 , parasitic-free finite-shift mildness, and endpoint contradiction. All estimates used in this assembly are localized to fixed terminal cylinders, except for the final sequence- L^3 conclusion, which is derived by contradiction from terminal indecomposability and is not assumed as a global bound. $\square$

Remark 10.25 (No global budget in the terminal theorem). The terminal stratification theorem assumes no global L^3 , global CKN budget, finite energy bound on expanding regions, or global counting measure for concentration profiles. Compactness and recurrence are local, normalized, or realized from the original blow-up sequence.

Corollary 10.26 (R2 closure). *The ordered rotational relative-equilibrium residual class is empty:*

$$\mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. By [Theorem 5.8](#), every R2 residual either belongs to a previously closed lower class or produces a terminal concentration profile realized from the original blow-up sequence with retained unit CKN concentration. The lower classes are empty by the imported setup theorem, and the terminal concentration case is empty by [Theorem 10.24](#). Hence no R2 residual remains. $\square$

Corollary 10.27 (R3 closure). *For every admissible stationary phase space $\mathfrak{B}$,*

$$\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Proof. By [Theorems 6.12](#) and [7.6](#), every R3 stationary-hull residual either lies in a setup-closed lower class or produces a stationary terminal concentration profile realized from the original blow-up sequence. The lower classes are empty by the imported setup theorem. The stationary terminal profile satisfies the retained unit CKN concentration hypotheses of [Theorem 10.24](#), so it is reduced to the atomic alternative only after all separated, diffuse, noncompact, critical-tail, and affine alternatives have been removed. In that atomic case [Theorems 10.5](#), [10.7](#) and [10.9](#) give the sequence- L^3 bound, finite-shift bounded mildness, and the endpoint Albritton–Barker contradiction. Hence no stationary terminal concentration profile remains. This proves the deferred stationary concentration and degenerate stationary-family occurrence statements [Theorems 6.10](#), [7.4](#) and [7.5](#), and therefore the R3 class is empty. $\square$

11. CLOSURE OF THE GENERIC TERMINAL RESIDUAL

Definition 11.1 (Generic residual profile). A generic residual profile is a smooth bounded normalized Seregian ancient solution of [\(1.2\)](#) which carries persistent compact velocity L^3 -mass and belongs to none of the previously closed alternatives, none of $\mathcal{C}_{\text{ax}}$, $\mathcal{C}_{\text{rot}}$, $\mathcal{C}_{\text{stat}}^{\text{ret}}$, and none of the structured or tight classes removed earlier, including the ordered critical-tail lower strata $\mathcal{C}_{\text{log diff}}$, $\mathcal{C}_{\text{Young}}$, $\mathcal{C}_{\text{homcrit}}$, $\mathcal{C}_{\text{logper}}$, $\mathcal{C}_{\text{ap}}$.

Theorem 11.2 (Closure of the generic residual class by local concentration analysis). *One has*

$$\mathcal{C}_{\text{gen}} = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{gen}}$. By the tail geometry developed above, constant tails may be removed by the exact affine symmetry and nonconstant recurrent profile tail cores collapse to affine constants. Thus the remaining representative may be taken to be tail-normalized without leaving the centered equation or the residual decomposition. If this normalization eliminates non-affine oscillation, then [Theorem 8.21](#) places the profile in the affine/parasitic lower stratum already removed from $\mathcal{C}_{\text{gen}}$. Thus, for the residual representative still under consideration, a positive non-affine oscillation threshold survives.

Apply the paired terminal local concentration decomposition to a terminal extraction of this tail-normalized candidate. The distinguished compact velocity mass from the setup paper still forces the terminal concentration set to be nonempty by [Theorem 9.12](#); after affine normalization the surviving residual oscillation is measured by Φ . The pure radiative alternatives cannot be the sole location of the compact mass by [Theorem 9.14](#), and mixed compact–diffuse or compact–noncompact remainders are excluded by [Theorem 9.69](#). Infinite chains of retained concentrating descendants are excluded by [Theorem 10.20](#). Thus the concentration set reduces, in the refined residual class, either to a finite separated family or to a single retained profile with no surviving tail remainder.

The finite separated-family case contradicts [Theorem 10.23](#). In the single-profile case, [Theorem 10.4](#) makes the terminal profile terminally indecomposable. Then [Theorem 10.9](#) excludes it. All paired profile possibilities are impossible. Hence no generic residual profile exists. $\square$

12. FINAL ASSEMBLY

The class closures above form the following complete dependency chain. The rotational relative-equilibrium arrow is the localized Coriolis-flux argument; the terminal concentration-set compactness, no finite separated-family, and indecomposability arrows are proved in this paper from local estimates:

$$\begin{aligned}
\mathcal{C}_{\text{ax}} &\xrightarrow{\text{r-Liouville}} \emptyset, \\
\mathcal{C}_{\text{rot}} &\xrightarrow{\text{local flux}} \emptyset, \\
\mathcal{C}_{\text{stat}}^{\text{ret}} &\xrightarrow{\text{terminal concentration}} \emptyset, \\
\mathcal{C}_{\text{aff}} &\xrightarrow{\text{affine/parasitic normalization}} \text{no retained normalized representative}, \\
\mathcal{C}_{\text{log diff}} &\xrightarrow{\text{log-window descendants + hidden-scale closure}} \emptyset, \\
\mathcal{C}_{\text{Young}} &\xrightarrow{\text{minimal mesoscopic-scale reduction}} \emptyset, \\
\mathcal{C}_{\text{homcrit}} \cup \mathcal{C}_{\text{logper}} \cup \mathcal{C}_{\text{ap}} &\xrightarrow{\text{bounded-origin realization}} \emptyset, \\
\mathcal{C}_{\text{gen}} &\xrightarrow{\text{critical-tail closure, finite separation, sequence-}L^3} \emptyset.
\end{aligned}$$

The positive local velocity mass inherited by every normalized Seregin limit, after affine normalization through the non-affine oscillation functional Φ , is the common contradiction target in the complete ordered decomposition.

Equivalently, the final exclusion is the composition of six steps:

- (1) *Entry step:* The setup paper supplies a positive local velocity concentration sequence; the endpoint weak Serrin local Type I bounds [Theorem 1.23](#) gives the terminal local energy/no-escape estimate in [Theorem 1.24](#). Thus the sequence is admissible in the sense

of [Theorem 1.16](#), and [Theorem 1.5](#) turns it into a normalized Seregin profile with retained compact velocity concentration.

- (2) *Stratification step*: the ordered residual decomposition places every remaining retained profile in the complete list of alternatives

$$\mathcal{C}_{\text{ax}} \cup \mathcal{C}_{\text{rot}} \cup \mathcal{C}_{\text{stat}}^{\text{ret}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{log diff}} \cup \mathcal{C}_{\text{Young}} \cup \mathcal{C}_{\text{homcrit}} \cup \mathcal{C}_{\text{logper}} \cup \mathcal{C}_{\text{ap}} \cup \mathcal{C}_{\text{gen}}.$$

- (3) *Closed-strata step*: the small, stationary L^3 , tight, closed structured/decay, axisymmetric, rotational, and stationary-hull strata are excluded by the results cited above.
- (4) *Lower-stratum step*: no retained normalized representative remains in $\mathcal{C}_{\text{aff}}$, by oscillatory entry normalization. The alternative $\mathcal{C}_{\text{log diff}}$ is excluded by log-window descendant selection plus hidden-scale compactness; see [Theorem 9.30](#). The alternative $\mathcal{C}_{\text{Young}}$ is excluded by the same hidden-scale compactness; see [Theorem 9.62](#). The alternatives $\mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}$ are excluded by bounded-origin realization of coherent critical tails.
- (5) *Residual terminal step*: the generic residual is reduced by the paired concentration-set decomposition, diffuse-defect compactness, diffuse-defect trichotomy, critical-tail compactification, realized critical-tail rigidity, descendant heredity, recurrence for concentration chains, finite separated-family exclusion, and terminal indecomposability.
- (6) *Endpoint Liouville step*: a terminally indecomposable retained profile either belongs to the affine/parasitic lower stratum or has a parasitic-free bounded mild physical pullback with a backward L^3 -bounded sequence. In the latter case it is zero by Albritton–Barker.

Proof of [Theorem 1.13](#), revisited. Assume, for contradiction, that a retained admissible residual profile exists. By the refined ordered decomposition, after extraction it belongs to one of

$$\mathcal{C}_{\text{ax}}, \mathcal{C}_{\text{rot}}, \mathcal{C}_{\text{stat}}^{\text{ret}}, \mathcal{C}_{\text{aff}}, \mathcal{C}_{\text{log diff}}, \mathcal{C}_{\text{Young}}, \mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}, \mathcal{C}_{\text{gen}}.$$

The axisymmetric class is empty by [Theorem 4.3](#). The rotational class is reduced to the terminal concentration theorem by [Theorem 5.8](#) and is empty by [Theorem 10.26](#). The stationary-hull class is reduced to the terminal concentration theorem by [Theorems 6.12](#) and [7.6](#) and is empty by [Theorem 10.27](#). No retained normalized representative remains in $\mathcal{C}_{\text{aff}}$ by [Theorem 9.61](#). The log-diffuse and Young branches are empty by [Theorems 9.30](#) and [9.62](#). The coherent critical-tail classes are empty by [Theorem 9.67](#). The generic terminal class is empty by [Theorems 10.24](#) and [11.2](#).

Every refined residual alternative is impossible, contradicting the ordered decomposition. Hence $\mathcal{R}^\#(\mathcal{S}; I, J) = \emptyset$, which is NoResidual. $\square$

Corollary 12.1 (Verification of the setup residual hypothesis). *The residual-class hypothesis `p1:hyp:no-remainder` of the setup paper holds.*

Proof. By [Theorem 1.3](#), the setup residual hypothesis is exactly the assertion that the retained residual class $\mathcal{R}^\#(\mathcal{S}; I, J)$ is empty after the lower setup classes are removed. This is [Theorem 1.13](#). $\square$

Corollary 12.2 (Local pointwise Type-I exclusion from the two-paper chain). *By the setup paper’s conditional final assembly `p1:thm:final-assembly` and [Theorem 12.1](#), no local pointwise Type I singularity exists under the stated admissibility hypotheses.*

Proof. This is exactly the conditional final assembly imported in [Theorem 1.3](#), with the residual hypothesis verified by [Theorem 12.1](#). $\square$

Remark 12.3 (Inherited inputs and new arguments). The paper inherits the base setup, including local concentration, the local Type I Type II dichotomy, the Type I Seregin extraction framework, and the initial structural decomposition, from the setup paper `proof_setup.tex`. All remaining case-closure steps are proved in this paper using local concentration and terminal dynamics:

- terminal concentration-set extraction;
- compactness of descendant/tail families and pressure controls;
- recurrence for chains of retained concentration profiles;
- finite separated-family exclusion;
- oscillatory-entry normalization;
- hidden-scale compactness reduction for critical tails;
- bounded-origin realization and critical-tail alternative exclusion;
- and the compactness and alternative exclusions culminating in [Theorem 11.2](#).

Remark 12.4 (No hidden global Liouville or critical-norm input). The residual closure does not invoke any global critical-norm bound on the original residual class, any global Coriolis–Leray Liouville theorem for bounded centered profiles, or any global Liouville theorem for bounded ancient solutions of the centered equation. The only external Liouville input used after residual closure is the Albritton–Barker endpoint Liouville theorem [Theorem 10.8](#), applied in exactly two places. First, the local weak-Serrin Type I estimate [Theorems 1.23](#) and [1.24](#) upgrades the local pointwise Type I envelope to terminal velocity no-escape; this is an entry-bridge use, optional inside Paper IV in the sense of [Section 1.2](#). Second, in [Theorem 10.9](#), the endpoint ancient Liouville theorem is applied to a parasitic-free bounded mild physical pullback that carries a backward L^3 -bounded sequence supplied by indecomposable sequence- L^3 completeness [Theorem 10.5](#). No other global Liouville theorem and no global critical-norm estimate is used in any of the closure steps for $\mathcal{C}_{\text{ax}}, \mathcal{C}_{\text{rot}}, \mathcal{C}_{\text{stat}}^{\text{ret}}, \mathcal{C}_{\text{aff}}, \mathcal{C}_{\text{log diff}}, \mathcal{C}_{\text{Young}}, \mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}, \mathcal{C}_{\text{gen}}$. In particular, the rotational closure [Theorem 10.26](#) proceeds through the localized Coriolis-flux identity and terminal local concentration, and the variance/critical-tail closures proceed through the minimal mesoscopic-scale reduction theorem [Theorem 9.54](#) together with bounded-origin realization [Theorem 9.65](#).

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